FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Design of an Asynchronous Flash Analog-to-Digital Converter

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Resumo

A aquisição de sinal de baixo consumo é uma necessidade crescente na Internet das Coisas e na deteção distribuída, onde muitos dos sinais de interesse são não periódicos. Surgem como acontecimentos ocasionais e de variação lenta, como uma leitura de temperatura que se altera ao longo do tempo, separados por longos intervalos de inatividade. Um conversor analógico-digital síncrono convencional adapta-se mal a este tipo de entrada. O seu relógio global mantém o banco de comparadores e a rede de distribuição de relógio a comutar a uma taxa fixa, pelo que o conversor dissipa potência dinâmica quer a entrada traga ou não informação nova.

Esta dissertação desenvolve um conversor Flash assíncrono que consome potência apenas quando a entrada efetivamente muda. O relógio global é removido e a conversão é conduzida pelo próprio sinal, de forma orientada a eventos (level-crossing), pelo que a potência dinâmica acompanha a atividade da entrada e cai para um nível estático quando a entrada está em repouso. Remover o relógio elimina também a rede de distribuição de alta velocidade que, num Flash síncrono, representa uma grande parte do orçamento de potência e reduz a latência de conversão.

O conversor é projetado num processo CMOS de 16 nm com alimentação de 0.8 V, resolução de 4 bits e um banco de quinze slicers de comparação. Um amplificador operacional de transcondutância complementar de cinco transístores, configurado como seguidor de tensão de ganho unitário, reproduz a entrada em quase toda a gama de alimentação e alimenta o banco de slicers. Cada slicer é um comparador de espelho de corrente cujo nível de decisão é fixado apenas pela razão das larguras dos seus transístores, pelo que os quinze limiares são definidos por dimensionamento e não por uma escada de referência. Como esses limiares dependentes do dimensionamento variam com o processo de fabrico, uma estratégia de calibração offline por polarização do substrato (body bias) ajusta as tensões de bulk de cada slicer para repor os pontos de disparo ao longo dos cantos de processo, sem acrescentar atividade comandada por relógio nem atrasar a conversão Flash.

Este trabalho foi realizado em ambiente empresarial, na Synopsys Portugal. Engloba a análise teórica, o projeto ao nível esquemático e a validação do sistema através de co-simulação analógica e de sinal misto, e não a implementação física. O conversor proposto é comparado com uma referência Flash síncrona da mesma resolução e tecnologia, caracterizada nas mesmas condições. Com a mesma carga de conversões por segundo, o conversor assíncrono consome cerca de um nono da potência da referência síncrona, e a poupança aumenta quando a entrada está quase sempre inativa, à medida que o conversor se aproxima do seu nível de repouso.

Abstract

Low-power signal acquisition is a growing need in the Internet of Things and in distributed sensing, where many of the signals of interest are non-periodic. They appear as occasional, slowly changing events, such as a temperature reading that drifts over time, separated by long, quiet intervals. A conventional synchronous analog-to-digital converter is a poor match for this kind of input. Its global clock keeps the comparator bank and the clock distribution network switching at a fixed rate, so the converter dissipates dynamic power whether or not the input carries any new information.

This dissertation develops an asynchronous Flash ADC that spends power only when the input actually changes. The global clock is removed, and the conversion is driven by the signal itself, in a level-crossing manner, so the dynamic power follows the activity of the input and falls towards a static floor when the input is idle. Removing the clock also takes out the high-speed distribution network that, in a synchronous Flash, accounts for a large part of the power budget, and it shortens the conversion latency.

The converter is designed in a 16 nm CMOS process at a 0.8 V supply, with 4-bit resolution and a bank of fifteen comparison slicers. A complementary five-transistor operational transconductance amplifier, configured as a unity-gain voltage follower, reproduces the input across almost the whole supply range and drives the slicer bank. Each slicer is a current-mirror comparator whose decision level is fixed purely by the ratio of its transistor widths, so the fifteen thresholds are set by sizing rather than by a reference ladder. Those sizing-defined thresholds drift with process variation, so an offline body bias trimming strategy adjusts the bulk voltages of each slicer to bring the trip points back into place across the process corners, without adding any clocked activity or slowing the Flash conversion.

This work was carried out in an enterprise environment at Synopsys Portugal. It covers the theoretical analysis, the schematic-level design and the validation of the system through analog and mixed-signal co-simulation. The proposed converter is benchmarked against a synchronous Flash reference of the same resolution and technology, characterised under the same conditions. At a matched conversions-per-second workload, the asynchronous converter draws about a ninth of the power of the synchronous reference, and the saving grows when the input is quiet for most of the time, as the converter settles towards its idle floor.

Keywords: Analog-to-Digital Converter, Asynchronous Design, Flash ADC, Level-Crossing Sampling, Offset Trimming, Low Power, IoT.

UN Sustainable Development Goals

The United Nations Sustainable Development Goals (SDGs) provide a global framework to achieve a better and more sustainable future for all. It includes 17 goals to address the world's most pressing challenges, including poverty, inequality, climate change, environmental degradation, peace, and justice.

Electronic systems play a pivotal role in modern society, acting as the backbone for smart infrastructure, healthcare monitoring, and environmental sensing. However, the massive deployment of battery-operated devices poses a significant challenge regarding energy consumption and electronic waste. This dissertation contributes directly to the efficiency and sustainability of these systems.

The specific Sustainable Development Goals addressed by this work are:

SDG 7 Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable and modern energy for all. By optimising the power consumption of data converters, we extend the battery life of devices and reduce the overall energy footprint of IoT networks.

SDG 9 Industry, Innovation and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialisation, and foster innovation. This work advances the state-of-the-art in microelectronics by proposing novel asynchronous architectures that enable smarter and more efficient industrial sensors.

SDG	**Target**	**Contribution**	**Performance Indicators and Metrics**
7	7.3	By double the global rate of improvement in energy efficiency, this work reduces the power waste in standby modes of electronic sensors.	Reduction in Energy per Conversion (fJ/conv) compared to synchronous architectures.
9	9.5	Enhance scientific research, upgrade the technological capabilities of industrial sectors, encouraging innovation.	Successful validation of the proposed calibration algorithm and circuit topology.

Agradecimentos

Aos meus pais, por todos os sacrifícios que fizeram para que eu pudesse chegar até aqui. Nada neste trabalho teria sido possível sem eles, e devo-lhes muito mais do que estas poucas linhas alguma vez conseguirão exprimir.

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Por último, à Ana, por ter estado sempre lá e pelo seu apoio incondicional.

"Get busy living, or get busy dying"

Andy Dufresne

Declaration of honour

I declare that this dissertation is my original work and has not previously been submitted or assessed in any other course or curricular unit, whether at this or any other institution.

References to other authors (statements, ideas, thoughts), as well as the use of published works fully complies with the rules of attribution and are appropriately cited in the text and bibliography, in accordance with reference standards. I am aware that plagiarism and self-plagiarism constitute academic misconduct.

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Gonçalo Pinto Monteiro

Contents

List of Figures

List of Tables

List of Acronyms

ADC	Analog-to-Digital Converter
CC-II	Second-Generation Current Conveyor
CMF-ADC	Current-Mirror Flash Analog-to-Digital Converter
CMOS	Complementary Metal-Oxide-Semiconductor
DAC	Digital-to-Analog Converter
DNL	Differential Non-Linearity
ENOB	Effective Number of Bits
FBB	Forward Body Bias
FF	Fast-Fast (process corner)
FFT	Fast Fourier Transform
FoM	Figure of Merit
FS	Fast-NMOS, Slow-PMOS (process corner)
GS/s	Giga-Samples per Second
INL	Integral Non-Linearity
IoT	Internet of Things
LCS	Level-Crossing Sampling
LSB	Least Significant Bit
MS/s	Mega-Samples per Second
MSB	Most Significant Bit
OTA	Operational Transconductance Amplifier
PVT	Process, Voltage and Temperature
RBB	Reverse Body Bias
SAR	Successive Approximation Register
SF	Slow-NMOS, Fast-PMOS (process corner)
SFDR	Spurious-Free Dynamic Range
SHA	Sample-and-Hold Amplifier
SINAD	Signal-to-Noise-and-Distortion Ratio
SNDR	Signal-to-Noise-and-Distortion Ratio
SNR	Signal-to-Noise Ratio
SQNR	Signal-to-Quantisation-Noise Ratio
SS	Slow-Slow (process corner)
THD	Total Harmonic Distortion
TT	Typical-Typical (process corner)

Chapter 1

Introduction

This chapter introduces the topic of the dissertation. It presents the context, the research question and the motivation for the work, states the objectives and the scope, and ends with the structure of the document.

Smart devices are now everywhere, a development usually grouped under the name Internet of Things (IoT). The processing and storage of information are digital and have benefited from the scaling described by Moore's law, but the physical world is analog, temperature, sound and pressure are continuous in both time and amplitude. The Analog-to-Digital Converter (ADC) is the interface between the two domains, and every digital system that measures the real world depends on one [71].

Energy efficiency has moved from a secondary performance metric to a primary design constraint. Many sensors run on small batteries or on harvested energy, so every joule dissipated shortens the autonomy of the system. The data conversion block is frequently the largest consumer in these interfaces, particularly when the digital transmission is duty-cycled and the radio spends most of its time off.

Many real-world signals are also sparse and non-periodic. Voice activity, machine vibration and environmental monitoring data arrive in bursts, short periods of high information content followed by long intervals in which nothing changes.

Traditional data conversion, based on uniform sampling at the rate dictated by the Nyquist-Shannon theorem [124], treats these silent periods exactly like the active ones. The system dissipates power to generate redundant samples that carry no new information. Exploiting this sparsity is the most direct route to ultra-low-power conversion interfaces.

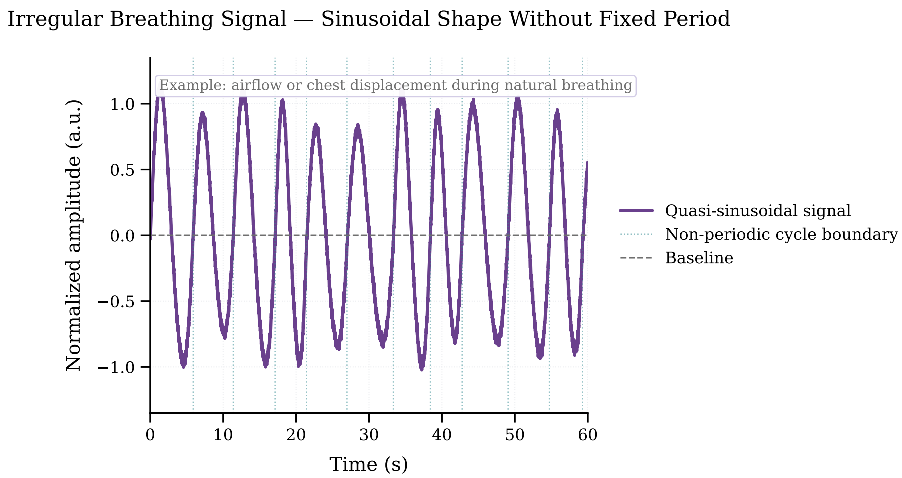


Figure 1.1: Example of a non-periodic signal, a quasi-sinusoidal waveform whose cycles vary in duration and never settle into a fixed period, here illustrated by the airflow or chest displacement of natural breathing. The information is carried by the slow, irregular variation of the waveform rather than by a fixed repetition rate.

Figure 1.1 shows an example of such a signal, a quasi-sinusoidal waveform whose cycles vary in length and never settle into a fixed period. Its information lies in these slow, irregular variations rather than at a fixed rate, so a converter that reacts to the signal's own changes matches it far better than one sampling uniformly at the Nyquist rate.

1.1 Motivation and Problem Statement

The energy wasted in large IoT deployments, together with the poor match between synchronous architectures and sparse data, motivates converters that activate only when the input actually produces an event.

1.1.1 Inefficiency of Synchronous Sampling

Traditional ADCs sample at a fixed rate set by a global clock (f_s). The Nyquist-Shannon sampling theorem [124] ties this rate to the maximum bandwidth of the signal, $f_s \geq 2B$. In many real applications, however, the signal is active for only a fraction of the time, so its true information content is far below what the bandwidth alone suggests.

The cost of this mismatch follows from the expression for dynamic power consumption in CMOS circuits [9]:

$$P_{dynamic} = \alpha \cdot f \cdot C_L \cdot V_{DD}^2 \tag{1.1}$$

where:

- $P_{dynamic}$ is the dynamic power dissipated.
- α is the activity factor (or switching factor).
- f is the clock frequency.
- C_L is the load capacitance being switched (e.g., in the clock tree).
- V_{DD} is the supply voltage.

In a uniformly clocked ADC, f is fixed by the Nyquist criterion and the clock network toggles every cycle, so α stays high even when the input is almost static. The converter keeps charging and discharging the clock and comparator capacitances at every period to produce samples that carry no new information. If a sensor signal is active only 10% of the time and the ADC samples continuously, 90% of the switching events are spent on intervals where the input is essentially constant.

1.2 Objectives and Scope of the Dissertation

Given the limitations of synchronous data conversion for time-sparse sensor signals, the primary objective of this work is to design and implement an ADC that exploits signal inactivity to reduce power consumption.

This overall goal breaks down into three specific objectives:

1. Circuit Proposal and Typical Validation. To implement a valid asynchronous Flash ADC circuit proposal and validate it under typical operating conditions.

2. Trimming Implementation. To implement a trimming approach for the proposed schematic, which may be based on a hardware-description-language model.

3. Complete System Validation. To validate the complete ADC system, the schematic together with the trimming, through analog and mixed-signal co-simulation.

1.3 Document Structure

This dissertation is organised into five chapters. Chapter 1 introduces the topic, presents the motivation and problem statement, and defines the objectives and scope of the work. Chapter 2 reviews the relevant background on analog to digital conversion, sampling paradigms, synchronous and asynchronous architectures, clockless event generation, calibration and trimming techniques, and presents the synchronous reference Flash ADC used as the benchmark throughout the work. Chapter 3 develops the proposed asynchronous Flash ADC through its successive architectures, presenting for each one the circuit, the supporting calculations and the simulation results that motivated the step to the next, and then develops the body bias trimming strategy adopted to correct the trip-point deviations of the comparison stage across process corners. Chapter 4 presents the final results of the trimmed converter and the comparison against the synchronous reference, including the power measurement methodology. Finally, Chapter 5 summarises the main conclusions and outlines directions for future work.

Chapter 2

Literature Review

Chapter 1 defined the goal. A converter that spends power only when the input produces events. This chapter assembles the background needed to design one. It presents the state-of-the-art in ADCs, reviewing theoretical foundations and analysing survey data to justify the proposed architecture.

Subsequent sections focus on asynchronous and level-crossing sampling strategies, including prior work on event-driven converters and their advantages for time-sparse signals. The chapter closes with a review of calibration and trimming techniques for comparator non-idealities, which is the reference framework for the bulk-driven body bias trimming mechanism proposed later in this work.

2.1 Fundamentals of Analog-to-Digital Conversion

This section provides the theoretical background required to understand the operation and performance characterisation of data converters. It covers the fundamental steps of the conversion process such as sampling, quantisation, and coding.

2.1.1 The Data Conversion Process

2.1.1.1 Ideal Data Conversion

Analog-to-digital conversion bridges the continuous physical world and the discrete digital domain. It involves two distinct operations, discretisation in time (sampling) and discretisation in amplitude (quantisation). Ideally, the process is instantaneous and lossless within the signal bandwidth of interest [71].

2.1.1.2 The Sampling Operation

Sampling converts a continuous-time signal $x(t)$ into a discrete-time sequence $x[n]$ [71].

According to the Nyquist-Shannon sampling theorem [124], a band-limited signal with maximum frequency B can be perfectly reconstructed if the sampling frequency f_s satisfies:

$$f_s \geq 2B \tag{2.1}$$

Violating this condition results in aliasing, high-frequency spectral components fold into the baseband and become indistinguishable from the original signal.

2.1.1.3 The Quantisation Operation

While sampling discretises the time axis, quantisation maps the continuous amplitude of each sample to one of a finite number of levels [71]. An N-bit ADC divides the input range (V_{ref}) into 2^N discrete levels. The step size between adjacent levels is the Least Significant Bit (LSB):

$$V_{LSB} = \frac{V_{ref}}{2^N} \tag{2.2}$$

Unlike sampling, quantisation is not reversible and introduces a deterministic error. This error, the quantisation noise, is usually modelled as additive white noise with a uniform probability distribution. For an ideal quantiser, the Signal-to-Quantisation-Noise Ratio (SQNR) is:

$$SQNR_{dB} \approx 6.02N + 1.76 \tag{2.3}$$

2.1.1.4 Coding

The final stage encodes the quantised level into a binary format. The most common schemes are the straight binary and the Gray code. The choice depends on the system requirements for data processing and transmission and does not affect the analog performance [9].

2.1.2 Classical Performance Metrics

A standard set of metrics, divided into dynamic and static parameters, allows data converters to be evaluated and compared objectively.

2.1.2.1 Resolution and Sampling Rate

Resolution defines the theoretical dynamic range, while the sampling rate determines the maximum signal bandwidth. Both are nominal values. The actual performance is limited by noise and non-linearities [71].

2.1.2.2 Signal-to-Noise-Distortion Ratio (SNDR)

The signal-to-noise-and-distortion ratio (SNDR) is the primary dynamic metric. It is the ratio of the signal power to the total power of all noise and harmonic distortion components. From SNDR, the Effective Number of Bits, ENOB, is derived [71]:

$$ENOB = \frac{SNDR_{dB} - 1.76}{6.02} \tag{2.4}$$

This value represents the true resolution of the converter at a specific input frequency.

2.1.2.3 Differential and Integral Non-Linearity (DNL and INL)

Static linearity is characterised by measuring the deviation of code transition levels from their ideal positions [71]. The DNL measures the deviation of a single step width from the ideal 1 LSB. A DNL less than -1 LSB implies a missing code in the transfer function.

The INL is the cumulative sum of DNL errors, representing the deviation of the transfer curve from a straight line. Specific INL patterns like saw-tooth shapes can reveal systematic errors in the architecture, such as gain mismatch or non-linear biasing [109]. Figure 2.1 illustrates both errors on a staircase distorted by device mismatch, the deviation in each code width being the DNL and the displacement of each transition from its ideal position being the INL.

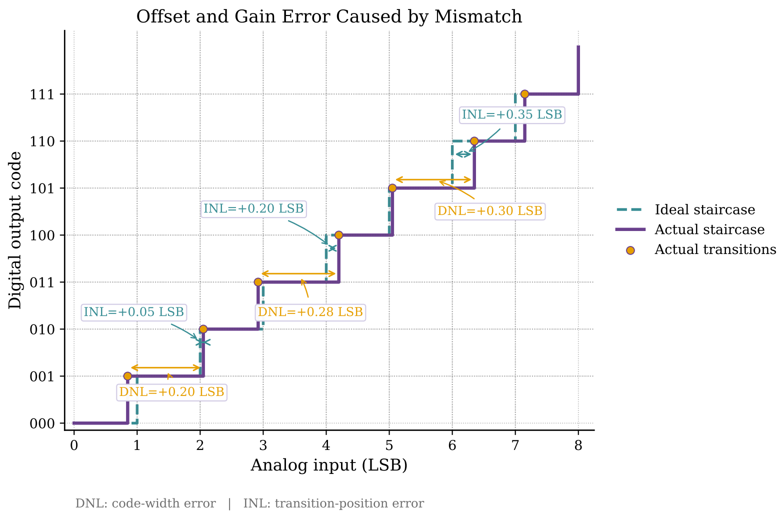


Figure 2.1: Effect of device mismatch on the ADC transfer characteristic. The actual staircase departs from the ideal one, the deviation in each code width is the differential non-linearity (DNL) and the displacement of each transition from its ideal position is the integral non-linearity (INL).

2.1.2.4 Offset and Gain Error

These are linear errors, offset is a constant shift of the transfer characteristic, while gain error is a deviation in its slope [71]. Unlike non-linearity, both preserve the signal shape and can usually be calibrated out with little effort. A constant DC shift moves the whole curve up or down (offset error), while a slope deviation alters the full-scale span (gain error), and both can be compensated by simple digital calibration.

2.1.2.5 Figures of Merit

Figures of merit (FoMs) provide normalised metrics to compare ADC performance across different architectures and technologies. The Walden FoM quantifies energy efficiency per conversion step:

$$\text{FoM}_W = \frac{\text{Power}}{2^{\text{ENOB}} \cdot f_s} \tag{2.5}$$

where lower values indicate better performance. This is the figure of merit used throughout this work.

2.1.3 Metrics Adaptation for Asynchronous Systems

The Walden FoM was developed for Nyquist converters, but it still applies to event-driven converters, provided one point is kept in mind, in a level-crossing architecture, it is the power, not the sampling rate, that depends on the input signal. The denominator thereforem stays fixed at twice the signal bandwidth, as for any converter of that bandwidth, while the power term carries the signal dependence. The efficiency is consequently reported as a range rather than a single number.

Level-crossing quantisers, as Weltin-Wu and Tsividis demonstrated, generate a sample only when the input crosses a quantisation threshold [150]. When the input is sparse, few crossings occur and little power is spent. When the input is active, the crossing rate and the power both rise. The absence of a uniform sampling clock also changes the noise spectrum, the output is free of aliasing and of the usual quantisation noise floor, and under some signal conditions, the SNDR can exceed the theoretical limit of a conventional converter with the same nominal number of bits.

The discrete-time level-crossing ADC of Mou et al. shows how this is reported in practice [104]. With its 8 kHz bandwidth and 10.1-bit ENOB held fixed in the Walden denominator, its power consumption of 95 nW for a quasi-static input and 250 nW for a full-scale Nyquist-rate sinusoid maps directly onto a FoM range of 5.3 to 14.2 fJ per conversion step. The converter is most efficient on the sparse inputs it is designed for, and the figure of merit captures that advantage entirely through the power term.

The power range itself has a physical origin. The amplitude change between successive samples, ΔV_{in}, determines how much work the converter does per event. Slow events, with steps between one and two LSB, require only a linear ± 1 LSB update and a few comparisons. Abrupt variations beyond two LSB trigger a binary search similar to a SAR conversion, with more comparisons and more energy. Sparse signals produce mostly slow events, so the average power, and with it the FoM, falls.

Throughout this work, the standard Walden FoM is used, with the rate taken as twice the signal bandwidth, exactly as in the level-crossing literature above. The efficiency of the proposed converter is reported at its full-scale operating point, and the drop in power for sparse inputs is discussed separately, since that is where the event-driven advantage actually appears.

2.2 Sampling Paradigms: Synchronous and Asynchronous

Data conversion systems have traditionally relied on periodic sampling. An alternative is to let the dynamics of the input signal dictate when conversion happens. This choice separates synchronous from asynchronous operation.

2.2.1 Uniform Sampling and Nyquist Rate Systems

Uniform sampling follows the Nyquist-Shannon theorem. A constant sampling frequency of at least twice the highest frequency component of the signal. A clock distribution network times the sample-and-hold circuits and the quantisation stages that follow. The equally spaced temporal grid gives predictable throughput and simplifies digital signal processing. The drawback is that the converter spends dynamic power continuously, whether or not the input contains new information, Section 2.8 quantifies this overhead.

2.2.2 Non-Uniform Sampling and Event-Driven Systems

Non-uniform sampling removes the periodic clock altogether. Conversion is governed by amplitude variations rather than by fixed time intervals. Level-crossing ADCs are the clearest example. A sample is generated only when the input crosses one of a set of predefined thresholds, so the effective sampling rate follows the instantaneous slope of the signal. Weltin-Wu and Tsividis [150] showed that the sample rate then becomes proportional to input activity, and power consumption follows it. The active components stay quiescent during low activity and dissipate dynamic power only when a conversion event occurs. The resulting representation of the continuous-time signal is also free of aliasing [150]. Discrete-time level-crossing architectures, such as the implementation by Mou et al. [104], add an adaptive search that switches between linear and binary tracking according to whether the comparator delay classifies each crossing as slow or fast.

2.2.3 Signal Sparsity and Theoretical Energy Efficiency Benefits

Signal sparsity is intrinsic to many physical phenomena, particularly in non-periodic signals, whose waveforms alternate long quiet intervals with sporadic bursts of high-frequency information. A uniform-sampling system must run at a Nyquist rate high enough to capture the bursts, and then wastes that effort during the silences. Event-driven architectures adapt the conversion rate to the information rate of the source, when the input is static, conversion activity drops to zero. Power dissipation is thus coupled directly to signal activity, which is what makes the nanowatt-level consumption reported by Weltin-Wu and Tsividis [150] and Mou et al. [104] possible.

2.3 Synchronous Architectures

Synchronous ADCs are the necessary starting point for any discussion of asynchronous ones. In these converters, a global clock dictates the sampling instants and synchronises the internal operations. The architectures are mature and widely used, but they face power-efficiency trade-offs at high frequency and when the data is sparse [71].

2.3.1 Synchronous Flash ADC

The Flash ADC is conceptually the simplest and fastest architecture, performing a complete conversion in a single clock cycle [9]. It uses a resistive ladder to generate $2^N - 1$ reference voltages, which are compared simultaneously to the input signal by a large bank of comparators. This parallel comparison produces a thermometer code that a digital logic block then converts into a standard binary output. Its primary limitation is the exponential increase in hardware complexity, area, and power consumption as resolution increases.

The thermometer term refers to the parallel output from the Flash ADC comparators, where '1's run up to the input voltage level like mercury in a thermometer, indicating signal strength before being converted to standard binary by an encoder.

2.3.1.1 The Power Bottleneck

The primary drawback of the synchronous Flash architecture is its exponential scaling. Survey data clearly illustrate that high speed translates into high power in these designs. For example, a 6-bit 500 MS/s CMOS Flash ADC reported in 1999 already consumed 400 mW [132].

More extreme cases, such as a 22 GS/s 5-bit design, can consume up to 1.2 W [119].

The continuous clocking of a massive comparator bank creates a significant energy dissipation even when the input signal is static. This lack of adaptivity makes synchronous Flash architectures increasingly unsuitable for battery-powered or energy-harvesting applications.

2.3.2 SAR (Successive Approximation Register)

The SAR ADC operates using a binary search algorithm [109], for each sample, the internal logic tests one bit at a time, starting from the most significant bit (MSB). A single comparator compares the input to the output of an internal digital-to-analog converter (DAC), if the input is higher, the bit is set to '1' otherwise, it is set to '0'. This process repeats for N cycles. Because it uses very few active components, it is the most energy-efficient choice for low speeds.

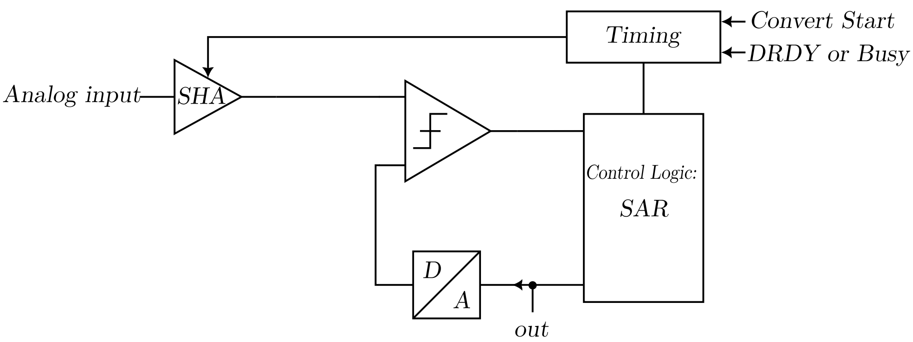


Figure 2.2: Block diagram of the SAR ADC architecture.

The binary search operation of a SAR ADC is implemented by successively testing each bit through the interaction of the sample-and-hold amplifier (SHA), DAC, comparator, and control logic, as shown in Fig. 2.2. At every conversion step, the SAR logic updates the DAC output and the comparator decides whether the input is above or below the trial voltage, gradually converging from the MSB to the LSB towards the final digital code.

2.3.3 Pipeline

The Pipeline ADC breaks the conversion into several sequential stages [71]. Each stage resolves a few bits, quantises them, and passes the amplified remaining error (the residue) to the next stage. Because several samples are processed concurrently, the architecture reaches high throughput at high resolution. The price is the processing latency.

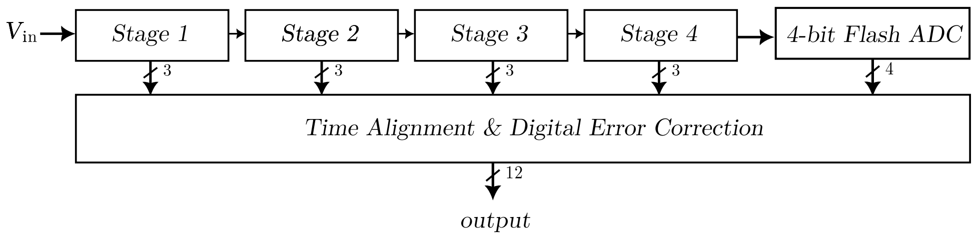


Figure 2.3: Block diagram of the pipeline ADC architecture.

The pipelined converter processes multiple samples simultaneously by distributing the resolution across several cascaded stages, as shown in Fig. 2.3. Each stage resolves a subset of bits, generates a partial digital output, and amplifies the residue for the next stage, while a digital error-correction block aligns the timing of all stage outputs to produce the final high-throughput conversion result.

2.3.4 Sigma-Delta (ΣΔ)

The ΣΔ architecture relies on oversampling, sampling much faster than the Nyquist rate [16], and noise-modulation. A modulator integrates the difference between the input signal and a feedback version of the quantised output. This process pushes the quantisation noise into higher frequencies, outside the signal band of interest a digital filter then removes the high-frequency noise, resulting in high resolution and precision [118].

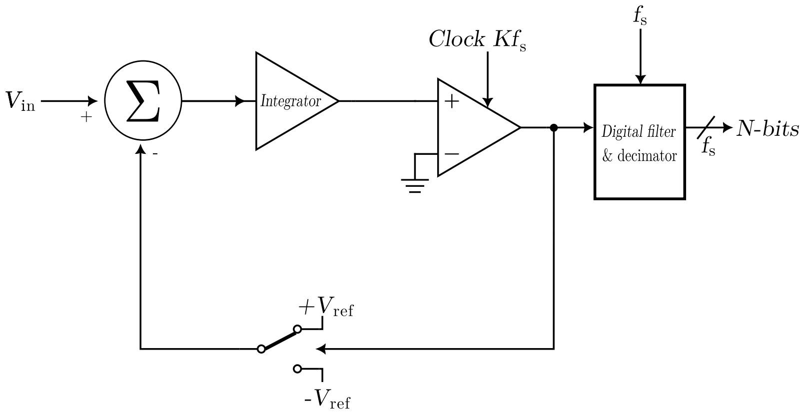


Figure 2.4: Block diagram of the first-order sigma-delta ADC architecture.

Oversampling and noise shaping are achieved by the feedback loop formed by summer, integrator, 1-bit quantiser, and digital decimation filter in the first-order sigma-delta architecture, as shown in Fig. 2.4. The modulator continuously integrates the difference between the analog input and the 1-bit DAC feedback, pushing most of the quantisation noise to high frequencies, which are later removed by the digital filter.

2.3.5 Dual-slope

This integrating architecture [115] measures the time a capacitor takes to charge and discharge. In the first phase the capacitor is charged by the input voltage for a fixed period. In the second, it is discharged by a known reference voltage. The time needed to return to zero is proportional to the average value of the input. The technique is accurate and immune to high-frequency noise, but much slower than the other architectures.

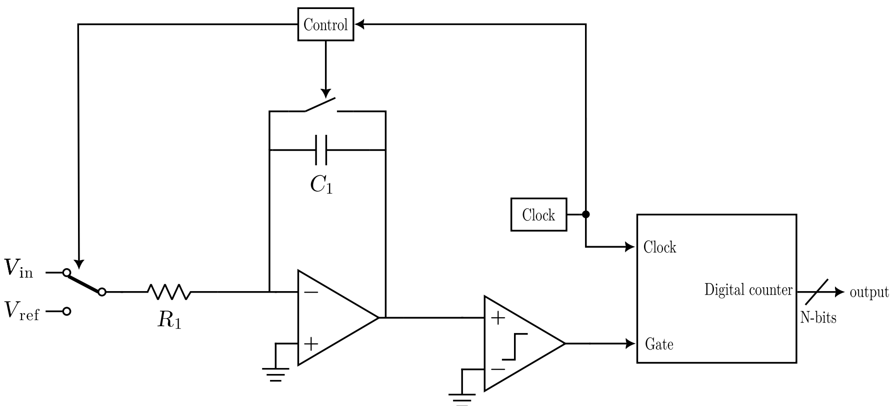


Figure 2.5: Block diagram of the dual-slope ADC architecture.

The dual-slope architecture converts the input voltage into time by integrating it during a fixed interval and then discharging the capacitor with a known reference, as shown in Fig. 2.5. The digital counter measures the discharge interval controlled by the reference voltage, so the resulting count is proportional to the average value of the input signal and inherently rejects high-frequency noise.

2.4 Synchronous Reference Flash ADC

This work benchmarks the proposed asynchronous converter against a synchronous Flash ADC of the same resolution and technology. This section introduces that reference, a synchronous 4-bit Flash ADC re-implemented in the same technology and supply conditions, and defines the role it plays in the rest of the dissertation. It explains why this particular design was selected, and describes its architecture, its StrongARM comparators and its clock generation. Its measured results appear in Chapter 4, side by side with those of the asynchronous converter proposed in Chapter 3.

2.4.1 Motivation for Architectural Selection

Benchmarking a novel converter architecture requires a reference point that is close enough structurally to make the comparison genuinely informative. The synchronous reference used in this work is a 4-bit Flash ADC with rail-to-rail comparators, originally presented in [38] and designed in a 65-nm CMOS process. For the purpose of this comparison, the architecture was re-implemented and simulated in a 16-nm technology node, under the same conditions applied to the proposed asynchronous converter throughout this dissertation.

The choice of this particular design was not arbitrary. Several properties directly align with those of the asynchronous architecture proposed in Chapter 3, and it is precisely this alignment that makes the comparison meaningful. Both converters use the Flash topology, in which the input is compared in parallel against a set of fixed decision thresholds and the result is produced as a thermometer code, so the fundamental conversion principle is shared. Both are designed for 4-bit resolution with $2^4 - 1 = 15$ parallel comparison elements, so the complexity of the comparison stage is the same on both sides. Both were implemented in the same 16-nm technology node and operate from the same 0.8 V supply, eliminating technology and supply voltage as sources of asymmetry. Finally, both architectures are explicitly oriented towards low-power operation, which is the design space where the asynchronous approach is expected to have the most significant impact.

One difference between the designs must nevertheless be stated explicitly so that the comparison remains honest, the input ranges are close but not identical. The StrongARM comparators of the reference resolve inputs all the way to the rails, while the complementary follower of the asynchronous converter covers almost the entire supply range but loses functionality within a small margin of each rail, as Chapter 3 will show. This asymmetry is accounted for by characterising both converters over their respective effective input ranges, and the energy comparison is in any case performed through resolution-normalised figures of merit, in which the effective number of bits of each converter appears explicitly.

The fundamental difference, and the one this comparison is intended to illuminate, is the presence of a global clock. The synchronous reference depends on a periodic clock to coordinate its comparators and define valid output instants, while the asynchronous architecture eliminates the clock entirely. Removing the clock distribution network is one of the primary motivations for adopting asynchronous operation, since the clock tree contributes a non-trivial fraction of total power in high-speed designs. By holding the remaining parameters constant and changing only this aspect, the comparison gives a direct, controlled view of what asynchronous operation actually costs, or saves, in power and performance.

2.4.2 Architecture Overview

A Flash ADC resolves an analogue input voltage in a single conversion step by comparing it simultaneously against $2^N - 1$ evenly spaced reference levels, where N is the number of output bits. For a 4-bit converter, this means 15 comparators operate in parallel at every active clock edge, each producing a binary decision about whether the input exceeds its assigned threshold. The outputs of all 15 comparators form a thermometer code (a sequence of ones followed by zeros, with no transition other than the boundary between them), which an encoder then maps to the final 4-bit binary word, as detailed in Section 2.4.4.

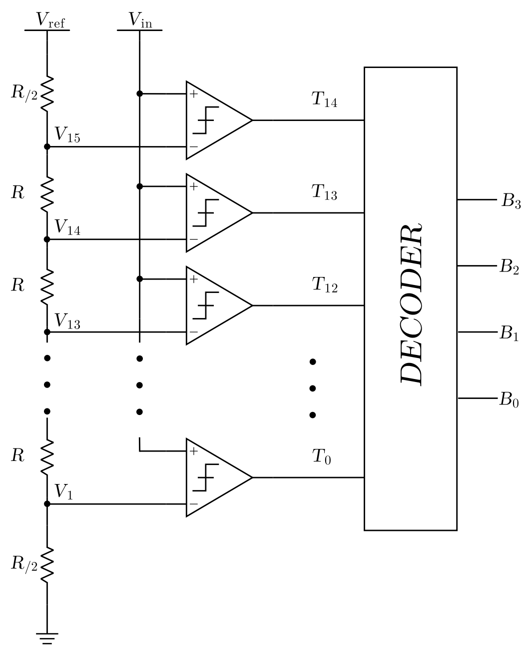


Figure 2.6: Top-level block diagram of the synchronous 4-bit Flash ADC [38].

The reference voltages fed to each comparator are generated by a resistor ladder connected between V_{REF+} and V_{REF-}, which in this rail-to-rail implementation are tied directly to the supply rails V_{DD} and V_{SS}. The ladder divides the full input range into 15 equally spaced steps of one LSB each, so the k-th comparator from the bottom receives a reference of

$$V_{REF,k} = V_{SS} + k \cdot \frac{V_{DD} - V_{SS}}{2^N}, \qquad k = 1, 2, \ldots, 2^N - 1. \tag{2.6}$$

where $V_{REF,k}$ is the reference voltage of the k-th comparator, V_{SS} and V_{DD} are the supply rails and N is the number of bits.

Since all comparators are driven by the same clock, every decision is made within a single clock period and the output is valid only at well-defined time instants. This is the defining characteristic of a synchronous converter and stands in direct contrast to the asynchronous operation discussed in Chapter 3.

2.4.3 The StrongARM Latch Comparator

2.4.3.1 Circuit Description and Operating Principle

The comparator used in this architecture is the StrongARM latch, a dynamic topology that has become something of a standard reference in both high-speed receiver design and data converter circuits [116]. It is widely adopted for good reasons, high regeneration speed, good noise performance and essentially zero static power dissipation, all of which matter in a Flash ADC where 15 comparators operate in parallel.

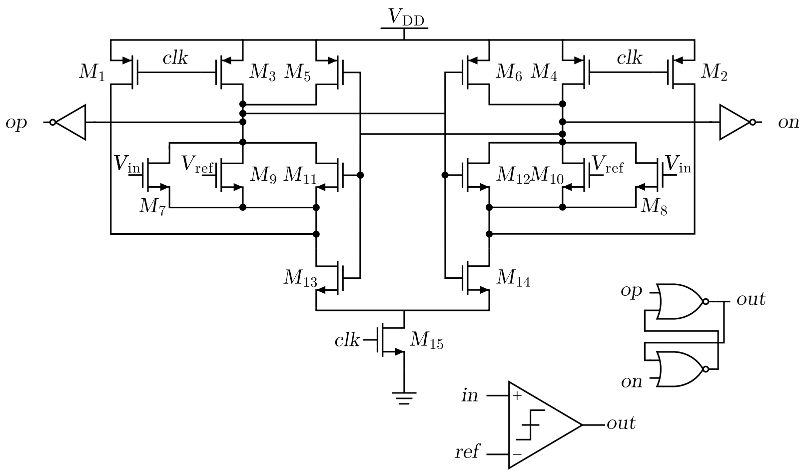


Figure 2.7: Schematic of the StrongARM latch comparator [38].

The circuit operates in two alternating phases controlled by the clock signal, a reset phase and an evaluation phase.

The reset phase occurs when the clock is low, the tail transistor is turned off and a pair of reset switches pulls both output nodes up to V_{DD}. This precharges the outputs to a known state before every comparison, so that each evaluation begins from the same initial condition regardless of the previous result. Because the tail transistor is off and no current path exists through the core of the circuit, static power dissipation is identically zero during this phase.

The evaluation phase occurs when the clock rises, the tail transistor turns on and current begins to flow through the differential input pair. The input transistors steer this current according to the voltage difference $\Delta V = V_{INP} - V_{INN}$: the side with the higher input voltage conducts more, causing that output node to discharge faster than the other. Once a sufficient imbalance develops across the internal nodes, the cross-coupled inverter pair at the output enters regeneration, a positive feedback mechanism that amplifies the initial imbalance exponentially until the outputs settle at valid logic levels [116]. The time needed to complete the regeneration is inversely related to the input overdrive, larger differences between the input voltage and the reference threshold resolve faster, while inputs very close to the threshold require more time. This input-dependent latency is a fundamental property of dynamic comparators and is one of the key differences exploited by the asynchronous architecture.

Since current only flows during the evaluation phase, and only briefly at that, the average power consumption is directly proportional to the clock frequency. This makes the StrongARM latch well suited to low-power converter design, reducing the sampling rate reduces the power linearly.

2.4.3.2 Rail-to-Rail Input Range

A key property of this comparator implementation is its ability to correctly resolve input voltages across the full supply range, from V_{SS} to V_{DD}. In a Flash ADC, the comparators assigned to the top and bottom reference levels must operate very close to the supply rails, and a comparator that loses sensitivity or gain in those regions would introduce nonlinearity at the extremes of the transfer characteristic, degrading the integral linearity of the converter. The rail-to-rail design ensures this does not happen, maintaining consistent performance across the entire input range.

The asynchronous converter will pursue the same goal with a complementary input stage, but, as Chapter 3 shows, its effective window stops slightly short of both rails due to the overdrive lost by the input pairs near the supply limits. The benchmarking methodology described in Chapter 4 therefore, evaluates each converter over its own effective input range.

2.4.4 Thermometer-to-Binary Encoder

The 15 comparator outputs form a thermometer code that has to be reduced to the final 4-bit binary word. A direct thermometer-to-binary mapping is sensitive to bubble errors, isolated zeros within the run of ones, or ones within the run of zeros, caused by comparator metastability or mismatch near a decision threshold. In a direct encoder such a single misplaced bit can translate into a large output error. To make the encoding more robust, the conversion is split into two stages, the thermometer code is first converted to a Gray code and then from Gray to binary.

2.4.4.1 Thermometer-to-Gray Conversion

In the first stage, each Gray bit is formed by combining selected thermometer outputs through XOR gates, so that the code depends only on the position of the boundary between the ones and the zeros rather than on the absolute pattern. Because a Gray code changes only a single bit between adjacent levels, a lone bubble shifts the encoded value by at most one LSB instead of producing a large jump, which is the property that gives this two-stage scheme its tolerance to comparator errors. Figure 2.8 shows the thermometer-to-Gray conversion logic.

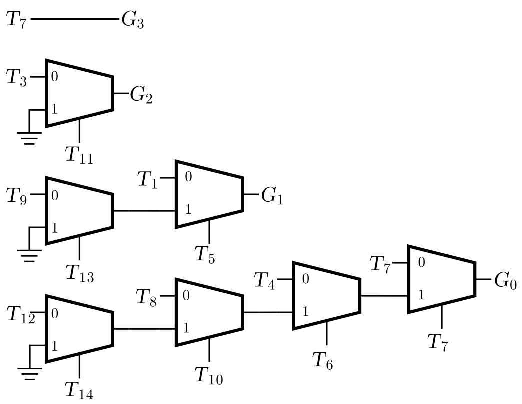


Figure 2.8: Thermometer-to-Gray conversion logic of the synchronous reference encoder.

2.4.4.2 Gray-to-Binary Conversion

The second stage converts the Gray code into the final natural binary word. The most significant binary bit equals the most significant Gray bit, and each lower binary bit is obtained as the XOR of the corresponding Gray bit with the binary bit immediately above it, so the conversion reduces to a short chain of XOR gates. Figure 2.9 shows this stage.

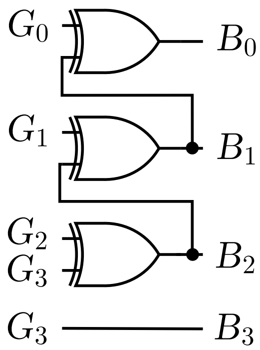
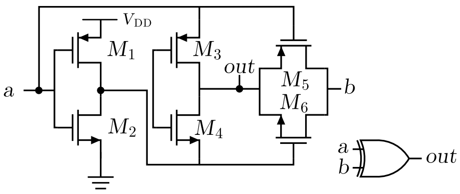


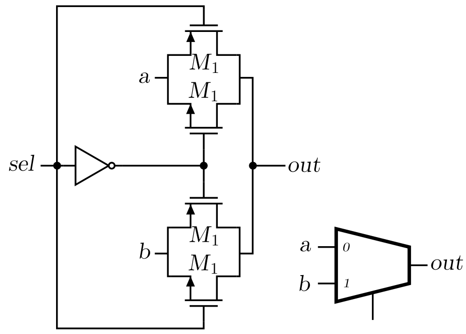
Figure 2.9: Gray-to-binary conversion logic of the synchronous reference encoder.

2.4.4.3 Logic Primitives

Both encoding stages are assembled from two standard cells, the two-input XOR gate that performs the modulo-2 additions of the conversion and the 2:1 multiplexer that routes the bits along the encoding path. Their gate-level realisations are shown in Figure 2.10.



(a) Two-input XOR gate



(b) 2:1 multiplexer

Figure 2.10: Logic primitives used to build the thermometer-to-binary encoder: (a) the two-input XOR gate and (b) the 2:1 multiplexer.

Once the binary word is available, the encoder output is registered, completing the single-cycle Flash conversion.

2.4.5 Clock Generation

Since the architecture is synchronous, a stable clock signal must drive the comparator array and determine the conversion instants. For simulation purposes, the clock was generated using a ring oscillator, which provides a compact and realistic on-chip clock source without requiring an ideal external generator.

Rather than driving the comparators directly from the oscillator output, the signal was routed through a tapered inverter buffer chain designed to model the loading and parasitics of a realistic clock distribution network. The chain consists of a sequence of inverters with progressively increasing drive strength, with each stage sized at approximately twice the fanout of the previous one. The final stage of the chain is a $\times 24$ fanout inverter, whose size reflects the capacitive load that a clock signal meets when distributed through the metal routing to the ADC core, the interconnect wires, vias and comparator gate inputs that the clock must charge and discharge on every cycle.

A $\times 24$ final buffer is a reasonable and representative choice for modelling the routing from a clock source to the ADC, as it captures the kind of loading one would expect in a physically compact but realistic clock distribution. Neglecting this stage would result in a power estimate that is overly optimistic, in a real implementation, the clock tree is rarely a negligible contributor to total power, and any fair assessment of the synchronous architecture must account for it.

The complete clock path used in simulation is therefore the ring oscillator, followed by the buffer chain ($\times 2$ fanout per stage), the $\times 24$ output buffer and, finally, the comparator array.

2.4.6 Summary

This section fixed the reference against which the proposed converter will be judged, a 4-bit synchronous Flash ADC from [38], re-implemented in 16 nm at 0.8 V. The reference shares the

Flash topology, the resolution (fifteen comparison elements) and the low-power orientation of the asynchronous design, and differs from it in the property under study, the global clock, plus one declared asymmetry in the input range. Its comparators are StrongARM latches, clocked through a ring oscillator and a tapered buffer chain whose $\times 24$ output stage models a realistic clock route. The power measurement methodology applied to this reference, together with its simulated performance figures, is presented alongside the results of the proposed asynchronous design in Chapter 4, so that the two converters are characterised and compared within a single framework. The asynchronous converter it benchmarks is developed in Chapter 3.

2.5 Asynchronous Architectures

Asynchronous, or event-driven, architectures are a fundamentally different approach to data conversion [88]. Unlike synchronous ADCs, which are bound by a global clock and the Nyquist-Shannon sampling theorem, asynchronous ADCs operate based on the signal's activity. This approach is a more efficient alternative for processing sparse signals.

2.5.1 Level-Crossing Sampling (LCS)

The core principle behind many asynchronous ADCs is Level-Crossing Sampling (LCS) [88]. In traditional uniform sampling, the signal is captured at fixed time intervals (T_s), and the amplitude is quantised. In LCS, the process is inverted. The amplitude levels are fixed (quantisation thresholds), and the ADC records the exact time at which the input signal crosses these thresholds.

This method is particularly powerful for signals that remain constant or change slowly over long periods. Instead of generating redundant samples that capture no new information, the LCS ADC remains idle, only producing a digital event when the signal effectively changes by more than the defined threshold.

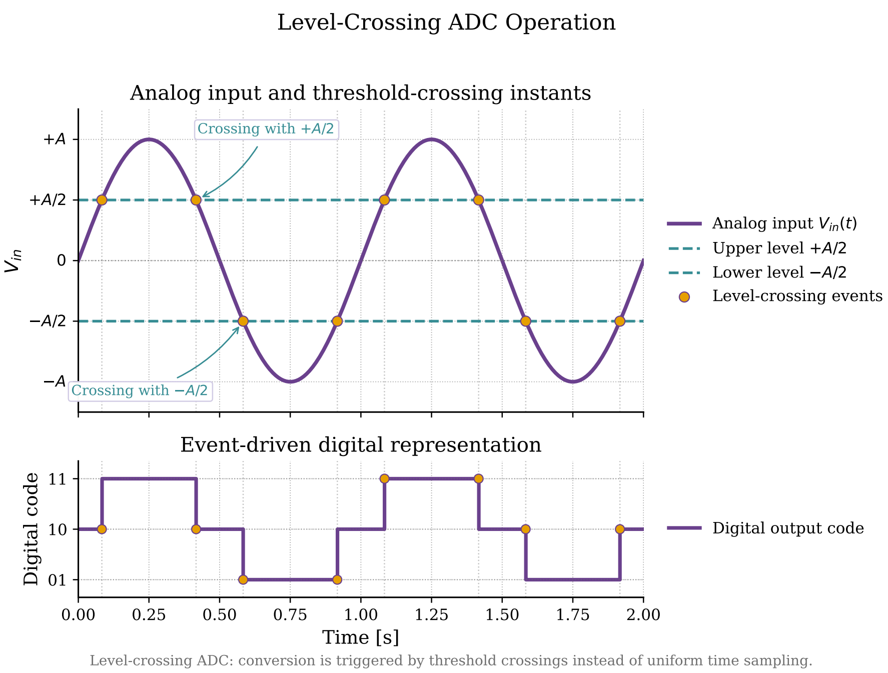


Figure 2.11: Operation of a level-crossing ADC. The analog input $V_{in}(t)$ (top) is compared against a set of fixed amplitude thresholds, here the levels $+A/2$ and $-A/2$, and a conversion event is produced only at the instants where the input crosses one of them (markers). The event-driven digital code (bottom) is updated at each crossing and held constant in between, so the output changes only when the input does rather than at a uniform sampling rate.

The core of the event-driven conversion scheme is illustrated in Fig. 2.11. The converter holds a set of fixed amplitude thresholds and continuously monitors the input against them. As long as V_{in} stays between two adjacent levels nothing happens and no conversion is performed. Each time the input crosses a threshold a single event is generated and the output code is updated by one step, upward for an upward crossing and downward for a downward crossing.

The signal is therefore represented as a sequence of events rather than as uniformly spaced samples, each event recording both the new amplitude level and the instant at which the crossing occurred. Conversion activity concentrates in the intervals where the input varies and disappears during the flat regions, so the converter's power consumption scales with the actual signal dynamics rather than with a fixed global sampling clock.

2.5.2 Flash ADC

The Asynchronous Flash ADC adapts the parallel structure of a standard Flash ADC but removes the sampling clock entirely. In this architecture, the comparators are not latched by a clock, they operate in continuous time, constantly monitoring the input voltage V_{in} against the reference ladder. The event generation happens when the input signal crosses a threshold, the corresponding comparator changes its output state immediately, this transition triggers an asynchronous logic to generate a digital output pulse.

2.6 Clockless Event Conversion

This section reviews current-mode techniques for generating conversion events without a clock, voltage-to-current conversion and current comparator topologies. A caveat applies to all of this material. These techniques are the conceptual basis of the first design phase in Chapter 3, the current-mirror Flash configuration, which the final architecture later abandoned because the voltage headroom demanded by current-mode structures did not fit within the 0.8 V supply. The review therefore documents the starting point of the architectural exploration and the criteria by which the current-mode candidates were judged and, in the end, rejected.

2.6.1 The Voltage to Current Conversion Concept in Event Detection

A central challenge in asynchronous Analog to Digital Converters is the generation of conversion events without a global clock. In a synchronous architecture, periodic sampling imposes the temporal discretisation from outside. An asynchronous system instead needs a mechanism that detects meaningful changes directly in the analog input. Voltage-to-current conversion is one efficient way to build such a mechanism.

The idea is to translate variations of the input voltage into proportional current variations that current-domain circuits can monitor continuously. Current-based processing often allows faster operation, which suits event-driven architectures where response time and energy efficiency both matter.

In this context, the analog input signal is applied to a transconductance stage that converts the input voltage into a current proportional to its amplitude. This generated current is then compared against a reference current or a set of threshold currents that represent predefined quantisation levels. Whenever the difference exceeds a decision threshold, an event is generated and propagated to the digital control logic.

This mechanism turns changes in signal amplitude into discrete temporal events. The converter stays inactive while the signal varies little and activates only on relevant transitions, which ties power consumption directly to input activity.

2.6.2 Current Comparator Operating Principles

Current comparators are the fundamental decision elements used in event driven architectures operating in the current domain. Their purpose is to determine whether an input current is greater or smaller than a reference current and to generate a corresponding digital output that signals the occurrence of an event.

Unlike conventional voltage comparators, which rely on charging and discharging internal nodes until a voltage threshold is reached, current comparators process information directly through current differences. This characteristic enables reduced voltage excursions, lower parasitic charging requirements, and potentially faster response times.

In a typical implementation, two input branches receive the signal current and a reference current. The comparator evaluates the difference between these currents and amplifies the resulting imbalance until one of the output states becomes dominant. Depending on the adopted topology, this amplification may occur through current mirrors, regenerative feedback structures, resistive loads, or current conveyor stages.

The performance of a current comparator is usually evaluated according to several parameters including propagation delay, resolution, sensitivity, power consumption, and input impedance. In asynchronous event generation applications, propagation delay becomes particularly relevant since it directly determines the temporal precision of the generated conversion events.

An ideal current comparator should therefore combine low input impedance, fast regeneration, low static power consumption, and robust operation under process and temperature variations.

2.6.3 Topologies for Event Detection in the Literature

Several event detection topologies have been proposed in the literature to enable continuous monitoring of analog signals and asynchronous generation of digital events, differing mainly in the method used to compare currents and regenerate the output transition. The six representative structures detailed below span the trade-off space between speed, power consumption, implementation complexity, and robustness. For asynchronous Flash Analog to Digital Converters, the preferred solution should preserve continuous operation while minimising propagation delay and avoiding unnecessary static power dissipation.

2.6.3.1 Current-Mirror-Based Current Comparator

The current-mirror-based comparator represents the simplest class of current-mode comparators. In this architecture, the input current and a reference current are applied to two branches of a mirror network that generates a difference current, which is then converted into a voltage across a high-resistance node. This voltage variation is sensed by a subsequent amplifier or inverter, producing a logic output that indicates whether the input is larger or smaller than the reference. The small-signal behaviour is governed by the output resistance of the mirror devices. A high intrinsic resistance allows small current differences to produce significant voltage variations, which improves sensitivity.

The main advantage of this structure is its simplicity, very small silicon area and low design effort, since it relies on basic current mirrors and standard logic inverters. It can also offer good static power characteristics if the bias currents are kept small, and the absence of complex feedback paths simplifies stability considerations. The speed of the structure is set by the load it drives. The high output resistance that improves sensitivity has to charge whatever capacitance hangs on the sensing node, so a large capacitive load, or a buffered rail-to-rail output added through extra inverter stages, lengthens the propagation delay and makes it sensitive to process and temperature. Driving only a light internal node at low current, however, the same cell settles quickly, which is the regime it operates in inside the current-mode Flash ADC of this work.

A small-signal view of the canonical mirror comparator of Freitas and Current makes the trade-off explicit [41]. The difference current $I_{in} - I_{ref}$ is forced onto a single sensing node whose resistance is the parallel output resistance of the mirror devices, $R_n = r_{op} \parallel r_{on}$. The voltage developed there is $\Delta V = (I_{in} - I_{ref})R_n$, so a high R_n turns a small current difference into a large voltage and gives the good sensitivity noted above. The same resistance sets the dominant time constant $\tau = R_n C_n$ with the node capacitance C_n, so the response time is governed by the load, a large C_n makes the cell slow and strongly process-dependent, whereas a light internal node keeps τ short. Sensitivity and load capacitance therefore set the speed jointly, which is why the bare mirror comparator can be either the slow or the fast option depending on the node it has to drive.

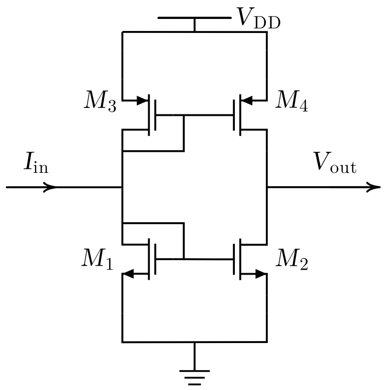


Figure 2.12: Transistor-level schematic of the current-mirror current comparator.

2.6.3.2 Traff Current Comparator

The Traff comparator introduces a high-speed continuous-time current comparison scheme based on a source-follower input stage and a CMOS inverter with positive feedback [136]. The input current is first converted into a voltage at a low-impedance node through a pair of source followers, this node then drives an inverting stage that regenerates the decision. The positive feedback loop keeps the input node from slewing rail-to-rail, so its voltage excursion is limited around a quiescent point and the effective input impedance remains low, which shortens the time required to develop the voltage difference that triggers the output transition. When the differential input current exceeds the resolution threshold, the latch-like action of the feedback rapidly pulls the output to one of the rails, providing a logic-compatible decision.

The small-signal input resistance set by the push-pull source followers is $r_{in} \approx 1/(g_{mn} + g_{mp})$, where g_{mn} and g_{mp} are the transconductances of the two follower devices. Because this resistance is low the sensing node barely moves, the input current is steered into the inverter quickly, and the propagation delay is short. The weakness lies in the same expression. For $|I_{in}|$ small, both followers approach cut-off, g_{mn} and g_{mp} collapse, r_{in} rises, and a dead zone appears, which is exactly the regime of small currents an asynchronous converter has to resolve.

Among its main advantages, this architecture achieves a high speed, with reduced propagation delay for a given input current step and relatively low power consumption for high-speed operation. Furthermore, the absence of large high-resistance nodes improves the frequency response and makes the design attractive for high-bandwidth current-mode ADCs. On the other hand, the Traff comparator exhibits a deadband region for very small input currents in which both input transistors operate near cut-off, temporarily increasing input resistance and degrading response time, making it not suitable for this application.

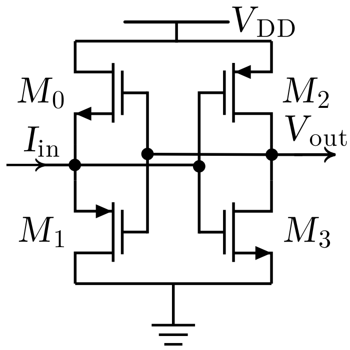


Figure 2.13: Transistor-level schematic of the Traff current comparator.

2.6.3.3 Lu Chen Current Comparator

The continuous-time comparator proposed by Lu Chen employs a CMOS complementary amplifier with resistive feedback, followed by two resistive-load amplifiers and CMOS inverters [24]. The core stage uses a pair of complementary transistors operating in saturation, with a MOS transistor in the linear region acting as a feedback resistor between input and output nodes. Small-signal analysis of this structure shows that, for typical device sizing, both input and output resistances of the complementary amplifier are approximately $1/(g_{mn}+g_{mp})$, the inverse of the sum of the transconductances of the input devices, since the feedback divides the open-loop resistance by the loop gain. The internal swing for a given input current is then $\Delta V \approx I_{in}/(g_{mn}+g_{mp})$, small enough that the stage settles fast. The following resistive-load amplifiers provide additional gain with relatively low static current, while the final inverters restore rail-to-rail logic levels.

This design achieves a favourable speed-to-power ratio because the reduced internal voltage swings translate into shorter response time for a given input current, while the resistive-load stages limit power consumption, particularly when the input current increases. Compared with other high-speed comparators, simulations reported by the authors show that the average delay for small currents is comparable to more complex solutions, but with lower or similar power, and without requiring external bias currents or voltages, improving process robustness. As a drawback, the achievable resolution remains dependent on the sizing of the feedback device and the effective transconductance of the input pair, so aggressive scaling of bias currents to save power can degrade sensitivity.

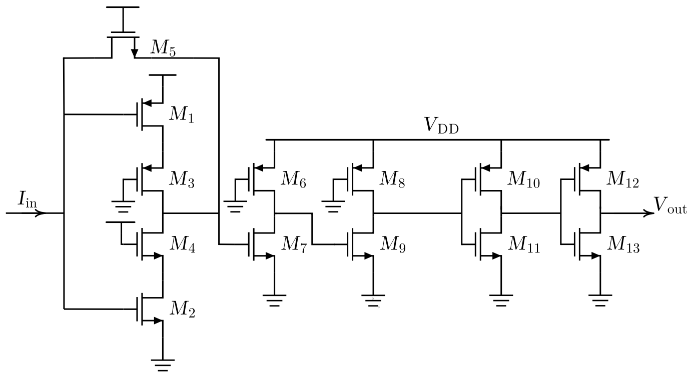


Figure 2.14: Transistor-level schematic of the Lu Chen current comparator.

2.6.3.4 Hysteresis Current Comparator

The hysteresis current comparator studied by Sridhar et al. introduces a current gain stage with non-linear feedback and a CMOS inverter output [129]. The gain stage consists of cascaded resistive amplifiers and an inverter enclosed by a pair of transistors that implement non-linear feedback, whose gates are driven by the output node while their sources sense the difference-node voltage. For small $|I_{\text{diff}}|$, both feedback transistors are off and the node behaves as a capacitive load around a quiescent operating point, whereas for larger positive or negative differences one of the feedback devices turns on, enhancing the loop gain around that point and accelerating the transition.

In small-signal terms, the added feedback transistor contributes a transconductance g_{mf} only once it conducts, so the loop gain, and with it the regeneration speed, rises with $|I_{\text{diff}}|$ instead of staying constant. The width of the hysteresis is the input current of opposite polarity needed to turn that device back off, which is the price paid in residual offset for immunity to noise-induced chatter near the decision boundary.

This non-linear feedback mechanism creates an effective hysteresis, once the output has switched, a finite current difference of opposite polarity is required to bring the internal node back to the decision boundary, which improves noise immunity and metastability behaviour. The use of multiple gain stages and feedback devices increases circuit complexity and area compared with simpler comparators, and the hysteretic behaviour must be carefully dimensioned so that the added memory effect does not introduce excessive offset or distort the transfer characteristic in applications where strictly monotonic behaviour is required.

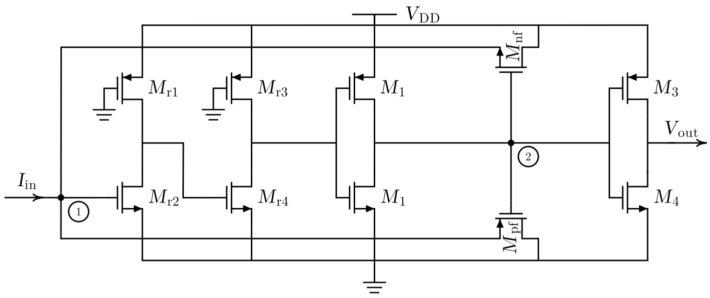


Figure 2.15: Transistor-level schematic of the hysteresis current comparator.

2.6.3.5 Min and Kim Current Comparator

The Min and Kim comparator employs a resistive feedback network around a current-source inverting amplifier to reduce input and output impedances and improve dynamic response [102]. The structure consists of several current-source amplifiers connected in cascade, with the first stage incorporating a resistor R_f that feeds back a portion of the output voltage to the input node. With an open-loop gain A in that first stage, the closed-loop input resistance falls to $r_{in} \approx R_f/(1+A)$ and the output resistance drops by the same factor, so the swing needed to represent a given current difference is small and the decision is reached quickly. A final CMOS inverter converts the amplified voltage into a digital logic level.

Simulation results reported in the literature indicate that this architecture achieves low propagation delay and good process immunity, with the resistive feedback helping to stabilise the operating point and reduce sensitivity to parameter variations. Additionally, the reduced node swings make the comparator suitable for relatively high-speed applications as long as the bias currents are chosen appropriately. A key limitation, however, is that both speed and resolution are strongly dependent on the bias current of the current-source amplifiers. Achieving very high resolution requires higher bias currents, which directly increases static power dissipation and degrades energy efficiency. Moreover, because the design relies on current-source loads rather than resistive loads, its power consumption does not decrease when the input current increases, unlike other comparators where dynamic behaviour can partially trade off with static consumption.

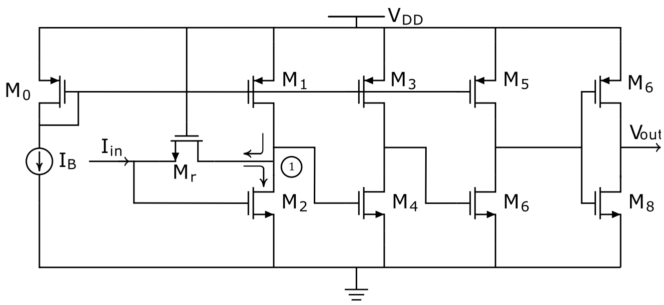


Figure 2.16: Transistor-level schematic of the Min and Kim current comparator.

2.6.3.6 CC-II-Based Current Comparator

The CC-II-based comparator proposed by Chavoshisani and Hashemipour uses a current subtractor followed by a current conveyor of the second generation and a multi-inverter output stage [22]. The current subtractor is implemented with a p-type current mirror that produces the difference between the input current and a reference current across a resistor, effectively converting the difference into a proportional voltage. This voltage is then applied to the low-impedance X-terminal of a CC-II, whose behaviour is characterised by a unity voltage gain from Y to X and current conveyance from X to Z, so that the current difference is sensed at X and reproduced at the high-impedance Z terminal. The inverter chain at the output converts the conveyor's analog voltage into full-swing logic levels suitable for digital processing.

The speed of this structure follows from the X-terminal resistance of the conveyor, $r_X \approx 1/g_m$, set by the transconductance of the conveyor input device. This low resistance holds the swing at the sensing node small for a given current difference, which is the source of the nanosecond delays and sub-5 µA resolution reported for the topology. The positive feedback from the output inverter to the Y terminal raises the effective gain around the decision point and sharpens the transition, at the cost of the compensation needed to keep the loop from oscillating.

A key feature of this architecture is the use of positive feedback from one of the inverters back to the Y terminal of the conveyor, so that small variations in the X-node voltage induced by the input current difference are reinforced through the CC-II action. This positive feedback shortens the response time and increases sensitivity to small current steps, allowing the comparator to achieve sub-5 µA resolution with nanosecond-level propagation delay in simulations. The number of devices is also moderate, and the CC-II realisation provides inherently low input impedance at X and high output impedance at Z. On the downside, the presence of the op-amp or amplifier block used to realise the CC-II, together with the feedback network, increases design complexity and power compared with very simple mirror-based comparators. Furthermore, the performance is sensitive to the bandwidth and stability of the conveyor implementation, which must be carefully compensated to prevent oscillations due to the positive feedback.

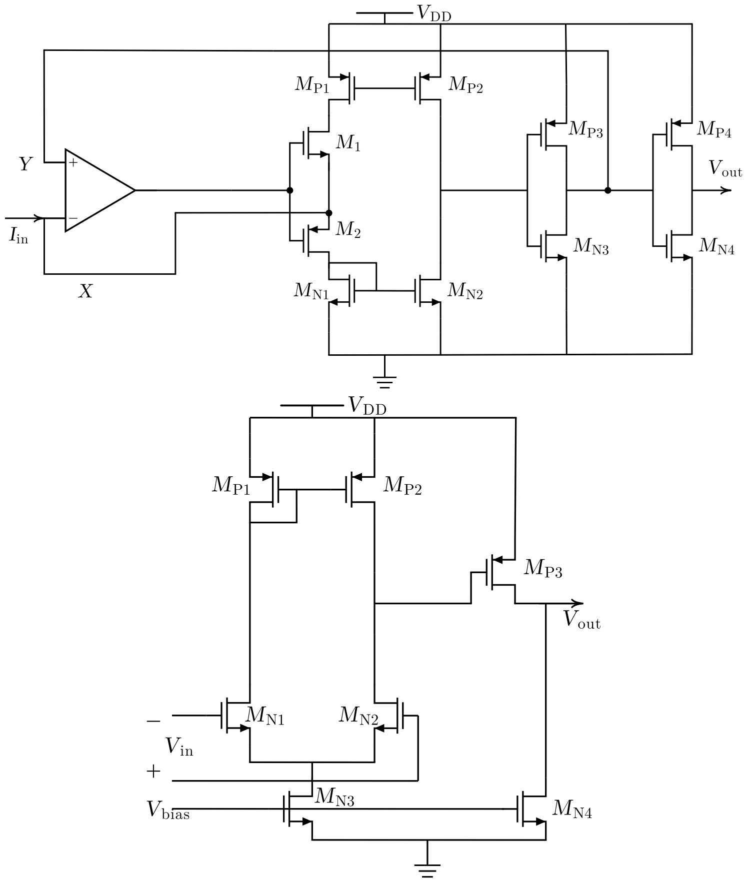


Figure 2.17: Transistor-level schematic of the CC-II-based current comparator, with its internal operational amplifier subcircuit shown below.

2.7 Calibration and Trimming Techniques

In high-speed Flash ADCs, the accuracy of the system is fundamentally limited by the precision of the comparators. While the architecture fundamentals were established in the previous sections, practical implementations must address the non-idealities of the fabrication process, specifically the Input Offset Voltage (V_{os}).

2.7.1 The Component Mismatch Problem

In deep sub-micron CMOS technologies, transistors that are drawn with identical dimensions on the layout will exhibit slight differences in their electrical parameters after fabrication. This phenomenon, known as mismatch, affects the threshold voltage (V_{th}) and the current gain factor (β) of the differential pair in a comparator [9].

According to Pelgrom's law [108], the standard deviation of the threshold voltage mismatch (σ_{Vth}) is inversely proportional to the square root of the transistor area ($W \cdot L$):

$$\sigma_{V_{th}} = \frac{A_{V_{th}}}{\sqrt{W \cdot L}} \tag{2.7}$$

where $A_{V_{th}}$ is a technology-dependent constant. This creates a trade-off. To keep the offset small without calibration, the transistors must be large, which increases parasitic capacitance and degrades the speed and power efficiency of the ADC. In practice, small transistors are used for speed and calibration corrects the resulting offset.

2.7.2 Calibration Classifications

Calibration techniques can be classified by timing and by domain [9]. In timing, foreground calibration interrupts normal operation to measure and correct errors, while background calibration runs continuously at the cost of considerable extra complexity. In domain, digital calibration corrects the output code mathematically, whereas analog calibration adjusts the circuit biasing or load conditions to cancel the offset at its source.

2.7.3 Trimming Techniques

2.7.3.1 Internal Loading

The internal resistive loading method involves placing a variable resistive network in parallel or in series with the output loads of the comparator stage. By digitally switching small resistors or MOS switches operating in the triode region in parallel with the load branch, the effective resistance R_L is modulated. If the ADC has an offset, the trimming network is adjusted to introduce an equal and opposite value to effectively zero the error. This technique is typically implemented using a binary-weighted bank of PMOS transistors as the variable resistance, where the digital control code is determined at startup and stored in a register.

2.7.3.2 Reference Ladder Trimming

The reference ladder trimming technique corrects comparator offset by using the main reference ladder as a calibration source [154]. In this architecture, a switching network selects specific voltages from the resistor ladder and applies them to the bulk terminals of the input transistors M1 and M2. The body effect then shifts the threshold voltage (V_{th}) of the differential pair and cancels the input-referred offset caused by process mismatch.

Simulation results reported in [154] show that the threshold voltage of transistors M1 and M2 increases linearly from approximately 356.5 mV to 372 mV as the calibration voltage is adjusted. In contrast, the threshold voltage of transistors M3 and M4 remains constant at approximately 356.5 mV because their bulk terminals are not connected to the tuning network. This targeted adjustment of the bulk potential for the input differential pair provides a tuning range sufficient to compensate for random process variations and ensure the accuracy of the comparator trip points. As the calibration voltage increases, the threshold of M1 and M2 shifts linearly while M3 and M4 remain fixed, providing the tuning range required to compensate comparator offset caused by random process variations [154].

2.7.4 Advantages of Static Bulk-Driven Trimming for Asynchronous ADCs

Dynamic offset-cancellation techniques such as auto-zeroing rely on accurately timed clock phases ϕ_1 and ϕ_2 and on storage capacitors that must be charged and discharged periodically [9]. Static trimming techniques, whether implemented with resistive networks or with bulk-driven threshold adjustment as in [154], suit an event-driven architecture far better. Once the correction values are found during a foreground calibration phase, the trimming network stays static, no further switching or clocking is required, and the comparator bias conditions are held by DC conduction paths alone. The trimming circuitry therefore does not disturb the event-driven operation of the ADC, which should remain idle whenever the input is not crossing a quantisation threshold.

The bulk-driven variant has a second advantage. The correction acts on terminals outside the signal path. Driving the bulk of the comparison devices shifts their threshold voltage through the body effect without adding capacitance to the high-speed nodes, so the small-signal bandwidth and regeneration speed are preserved. Dynamic calibration schemes, by contrast, place sampling capacitors and switch parasitics at sensitive nodes, which slows the comparator response and limits the maximum sampling rate in high-speed Flash architectures [9]. These properties motivate the static, bulk-driven trimming strategy adopted in Section 3.7, where the body bias of the comparison stage corrects deterministic trip-point deviations across process corners.

2.8 Asynchronous vs Synchronous

2.8.1 Advantage

The primary motivation for adopting asynchronous architectures is the direct relationship between signal activity and power consumption [88].

In a synchronous system, the clock tree and the comparators switch at every clock cycle, regardless of whether the input is changing, the power consumption is dominated by the fixed clock frequency [9]:

$$P_{sync} \approx f_{clk} \cdot C_L \cdot V_{DD}^2 \tag{2.8}$$

In an asynchronous ADC, the switching frequency is replaced by the event rate. If the signal is constant or slowly varying, the comparators and digital logic remain static, leading to the following proportionality:

$$P_{async} \propto Activity \times V_{DD}^2 \tag{2.9}$$

The energy consumed is therefore proportional to the information content of the signal. For sparse signals this is a large advantage. During periods of inactivity the converter draws close to no dynamic power.

2.8.2 Comparative Evaluation

State-of-the-art data show that asynchronous ADCs are among the most energy-efficient converters over a wide range of sampling rates, particularly at low and medium bandwidths [88].

The plots and tables below draw on a set of 140 published designs. All entries come from ISSCC and JSSC, where the reporting of data converter performance is consistent and the peer review is strict. Within these sources, only papers reporting both the supply voltage (V_{DD}) and the signal-to-noise-and-distortion ratio (SNDR) were retained, so that every point in the performance envelopes rests on documented electrical and dynamic measurements. To ensure transparency and reproducibility, the complete list of references used to generate these performance envelopes, covering Flash, SAR, Pipeline, and Sigma-Delta architectures, is provided in [1–8, 11–15, 17–21, 23, 26–28, 30–32, 34–37, 39, 40, 42–46, 48–66, 68, 69, 72, 73, 75–85, 87, 89–98, 100, 101, 103, 106, 107, 110–113, 120–123, 125–128, 130–134, 138–149, 151–153, 155–174].

In each plot, the entry marked [1] is the Flash ADC with the best performance among all surveyed works. It marks the limit of the performance envelope reached by Flash converters in that figure and is the benchmark against which the SAR, Pipeline and Sigma-Delta entries are compared. Because each plot ranks a different metric, the specific design occupying this position is not necessarily the same from one figure to the next.

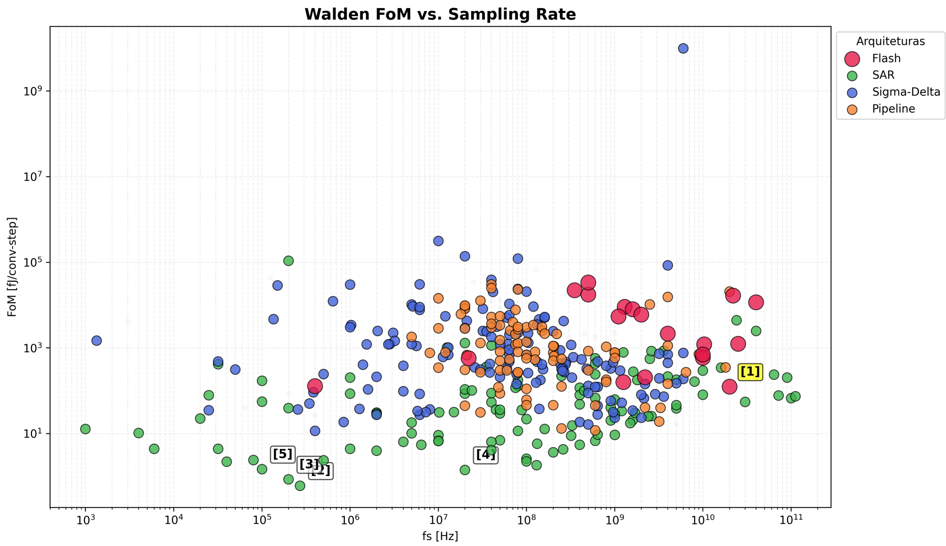


Figure 2.18: Walden FoM vs Sampling Frequency (Flash reference [1] highlighted)

The evolution of the Walden Figure of Merit as a function of sampling rate for different ADC architectures is summarised in Fig. 2.18. This plot allows identifying the performance envelope across a wide range of f_s and highlights how specific implementations approach the energy-efficiency limit at their target bandwidth.

Table 2.1: Walden FoM vs Sampling Frequency (Flash reference [1] highlighted)

Ref.	Architecture	Publication	f_s [Hz]
[1]	**Flash**	**Lukas [99]**	**20G**
[2]	SAR	Jang [67]	270k
[3]	SAR	Ginsburg [47]	200k
[4]	SAR	Cano [33]	20M
[5]	SAR	Muller [105]	200

While Synchronous Flash ADCs are designed for peak performance at a specific frequency and suffer from a much higher power consumption caused by the continuous clocking, asynchronous designs scale their power consumption linearly with the input activity.

Table 2.2: Comparison of Power Consumption between Synchronous and Asynchronous Flash ADCs. Simulation results from ISSCC conference showing the power advantage of asynchronous operation for sparse signals.

Topology	**Average power (P [W])**
Synchronous	0.1016
Asynchronous	0.0314

Figure 2.19 shows how this architectural difference plays out over time. The plot tracks the energy per conversion (P/f_s) over the last two decades. Both paradigms trend downwards with process scaling, but the asynchronous designs sit consistently at the lower bound of the energy envelope, which suggests that removing the global clock is one of the few remaining ways to push past the present efficiency limits.

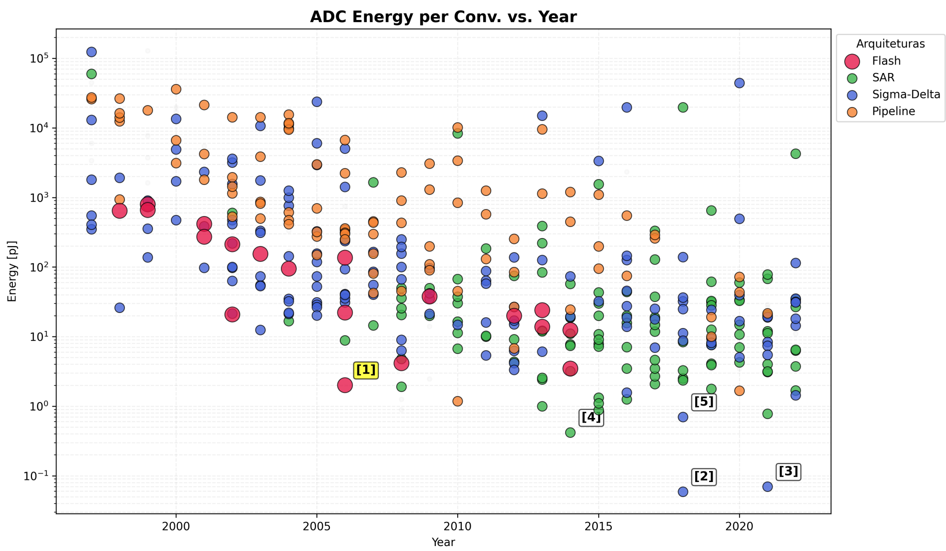


Figure 2.19: Energy per Conversion vs Year (Flash reference [1] highlighted)

Each point in Fig. 2.19 corresponds to a published design, so the plot shows how the different ADC architectures fill the energy-versus-year plane and which solutions are the most efficient. The conclusion of this analysis is straightforward. Synchronous architectures waste power on

Table 2.3: Energy per Conversion vs Year (Flash reference [1] highlighted)

Ref.	Architecture	Publication	Energy [pJ]
[1]	**Flash**	**Chen [25]**	**2.02**
[2]	Sigma-Delta	Jiang [70]	0.059
[3]	Sigma-Delta	Veldhoven [137]	0.070
[4]	SAR	Ginsburg [47]	0.420
[5]	Sigma-Delta	Kim [74]	0.703

time-sparse signals because the global clock keeps the switching activity, and hence the dynamic power, tied to the clock frequency even when the input carries nothing new. Event-driven schemes tie power to the rate of relevant signal transitions instead. The survey data adds the second half of the argument, Flash ADCs define the speed frontier but pay for it with a poor Walden FoM, while the more efficient SAR and sigma-delta architectures are confined to lower bandwidths. An asynchronous Flash topology sits at the intersection of the two observations. By removing the sampling clock and letting the comparators operate in a level-crossing manner, it keeps the low latency of the Flash architecture and uses signal sparsity to close the efficiency gap with the best synchronous designs in the targeted operating region.

Event-driven operation is not exclusive to Flash. The discrete-time level-crossing converter of Mou et al. [104] consumes between 95 and 250 nW around a SAR-like structure, but it resolves each event through successive comparisons in a feedback loop, whereas a Flash bank resolves it in a single parallel step. For the low-latency, event-driven operation targeted here, that single-step conversion was the deciding argument in favour of Flash.

2.8.3 Chapter Summary and Design Directives

Asynchronous Flash ADCs combine a parallel comparison stage with event-driven operation to exploit the time-sparse nature of many sensor signals. In these architectures, the conversion is triggered only when the signal crosses predefined decision levels, avoiding unnecessary activity during idle periods. Compared with conventional synchronous converters, this enables a significant reduction in dynamic power, since large capacitive nodes associated with sampling clocks and comparator arrays are only switched when relevant information is present in the input waveform.

From an architectural point of view, the proposed converter follows a Flash topology, where a bank of comparison elements observes different decision levels in parallel and produces a thermometer-like code that is subsequently encoded into binary. As in a conventional Flash ADC, this guarantees single-cycle conversion and very low latency, but here the global clock is removed and the decision activity is instead driven by input events. This choice transfers the power bottleneck from the sampling network to the design of the comparison elements themselves, since their intrinsic delay, input range and power consumption directly determine the achievable resolution, maximum event rate and overall energy efficiency.

For the comparison elements themselves, the current-mode techniques reviewed in Section 2.6 were the starting point of the design, processing the input as a current promises fast, clockless decisions, and the study of current comparator topologies shaped the first architectural phase in Chapter 3. That exploration will show, however, that the cumulative voltage headroom demanded by the current-mode structures did not fit the 0.8 V supply of the target technology. The final architecture operates in voltage mode instead, combining a complementary unity-gain voltage follower with a bank of ratioed CMOS slicers whose switching thresholds are set by transistor sizing. The literature in this chapter is therefore used in two ways, the current-mode material records where the first design phase came from and why it was rejected, and the trimming techniques of Section 2.7 are the basis for the body bias correction strategy of Section 3.7.

Chapter 3

Proposed Asynchronous Flash ADC Design

The previous chapter reviewed the field and fixed the benchmark, this one develops the converter that will be measured against it. The presentation follows the order the work took, a sequence of architectures, each shown with its circuit, the calculations that support it and the simulation results that motivated the move to the next one. The chapter then characterises the final architecture at the typical corner and across process corners, and develops the body bias trimming strategy (Section 3.7) that those corner results make necessary.

3.1 Design Specifications and Terminology

Before describing the architectural exploration, the specifications that constrained every design decision in this chapter are stated explicitly. The converter is designed in a 16 nm CMOS technology node, operating from a nominal supply voltage of $V_{DD} = 0.8$ V. The target application is the acquisition of time-sparse signals, as motivated in Chapter 1, which dictates the central architectural requirement, fully clockless, event-driven operation, with dynamic power dissipated only when the input crosses a decision level.

A point of terminology must also be fixed, since it is a recurring source of ambiguity in Flash converter descriptions. A Flash quantiser with a binary resolution of N bits requires $2^N - 1$ parallel comparison elements, whose collective outputs form a thermometer code of $2^N - 1$ levels. The number of thermometer outputs is therefore always distinct from, and larger than, the number of binary resolution bits. Throughout this chapter, the term thermometer outputs refers to the individual comparator or slicer outputs, while resolution bits refers exclusively to the binary word produced after encoding.

The converter implemented in this work has a resolution of 4 bits. A bank of $2^4 - 1 = 15$ slicers produces a fifteen-level thermometer code that the encoder maps to sixteen quantisation levels. This matches the resolution of the synchronous reference converter used for benchmarking (Section 2.4), which also uses fifteen comparators, so the two designs are compared at equal resolution. A presentational convention applies throughout the chapter. Whenever the operating principle of the comparison stage is explained step by step, a reduced bank of five slicers (outputs out_4 to out_0) stands in for the full fifteen, purely to keep the figures and the code examples readable. The five-slicer bank is an explanatory device, not the implemented circuit. Everything said about it extends directly to the fifteen-slicer bank of the actual design.

3.2 Architectural Evolution and Methodology

The design of the voltage comparator responsible for generating the digital output code of the analog-to-digital converter was not arrived at directly. It evolved through a sequence of candidate topologies, each evaluated against the constraints imposed by the target supply voltage, the required conversion accuracy, the effective input dynamic range, and the overall demands on circuit compactness. The following sections document this progression, presenting each topology that was considered and explaining the physical reasoning that either led to its adoption or prompted its replacement by a subsequent candidate.

3.3 First Architecture: The Baseline Current-Mirror OTA Flash Configuration

The first topology investigated was the current-mirror flash ADC (CMF-ADC) introduced by Lee et al. [86], originally proposed as the sensing stage of a coarse-fine dual-loop digital low-dropout regulator. Although the application context of that work differs substantially from the present design, the CMF-ADC converts a voltage directly into a thermometer code, which is exactly what a multi-bit comparison stage needs, so it was a natural starting point for the architectural exploration. It is also the architecture that puts into practice the current-mode event-detection concepts reviewed in Section 2.6.

3.3.1 Operating Principle and Core Topology

The CMF-ADC topology is structured around two cascaded functional stages, a transconductance cell, which converts a differential input voltage into a proportional error current, and a current-to-code converter, which maps that current into a parallel thermometer-coded output without requiring a sampling clock. The complete circuit, annotated with the current-flow conventions used throughout this analysis, is shown in Figure 3.1.

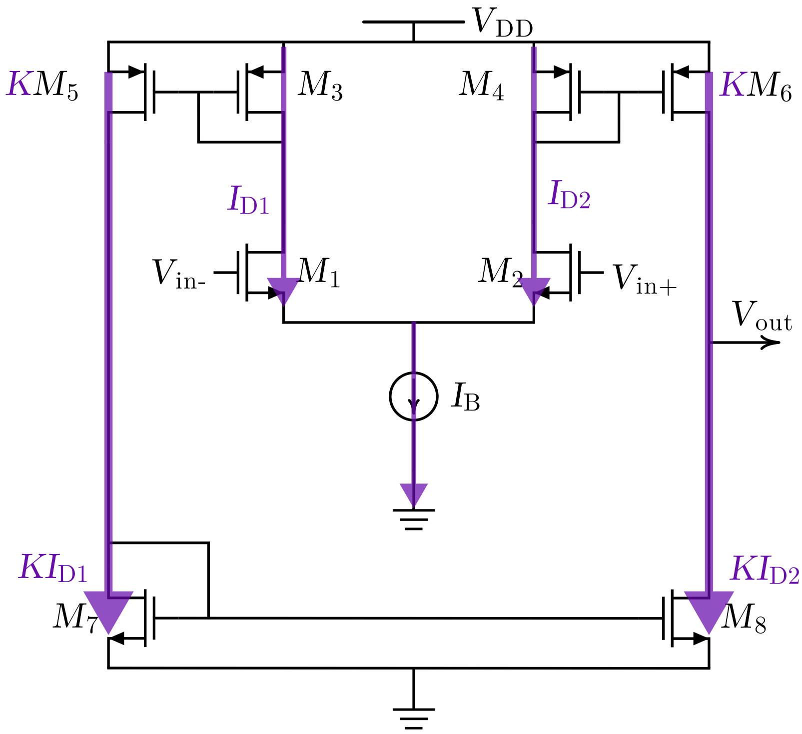


Figure 3.1: Operating principle of the current-mirror Flash ADC (CMF-ADC): (a) equilibrium operation, (b) signal operation, and (c) current-to-code conversion with asymmetric mirror ratios.

3.3.1.1 Transconductance Cell

The transconductance cell is a differential pair formed by transistors M_1 and M_2, both biased by a common tail current source of magnitude I_B. The non-inverting input of the pair receives the analog signal V_{in}, while the inverting input is driven by an adjustable reference voltage $V_{ref,c}$. An active current mirror, composed of diode-connected M_3 and mirroring transistor M_4, constitutes the load of the differential pair and steers the drain current of M_1 toward the output. An additional NMOS mirror generates a third current branch I_3 as a replica of I_1, which is then distributed to the conversion stage.

Equilibrium operation. See Figure 3.1(a) for the operation at equilibrium. When $V_{in} = V_{ref,c}$, M_1 and M_2 conduct equal quiescent currents of magnitude $I_B/2$. Because M_3 is diode-connected, its gate-to-source voltage adjusts to carry exactly $I_B/2$, and M_4, sharing this gate voltage, mirrors the same current to the output branch. The mirrored current from M_4 and the drain current of M_2 are therefore equal in magnitude, so their contributions cancel at the output node of the transconductance cell and the net error current is zero: $I_{err} = 0$. Under this condition, as indicated in Figure 3.1(a), both branches carry identical currents in the same direction, resulting in no differential current available to drive the downstream converter.

Signal operation. See Figure 3.1(b) for the differential signal analysis. When a differential voltage $v_{id} = V_{in} - V_{ref,c}$ is applied at the input, the symmetry of the pair is broken. Assuming matched transconductances $g_m = g_{m1} = g_{m2}$, the two transistors steer the tail current unequally, producing branch currents:

$$I_1 = \frac{I_B}{2} - I_{err}, \qquad I_2 = \frac{I_B}{2} + I_{err}, \tag{3.1}$$

where $I_{err} = g_m \cdot v_{id}/2$ is the error current. As annotated in Figure 3.1(b), when $V_{in} > V_{ref,c}$, M_1 carries the excess current (indicated by the thickened arrow on the M_1 branch), while the current in the M_2 branch is correspondingly reduced (indicated by the crossed symbol). The mirror formed by M_3 and M_4 copies I_1 toward the output, and the resulting imbalance between this mirrored current and I_2 constitutes the differential output signal $2I_{err}$ that drives the code converter. The effective transconductance of the cell is therefore:

$$G_m = g_{m1} = g_{m2} = g_m. \tag{3.2}$$

3.3.1.2 Current-to-Code Converter and Threshold Generation

Rather than producing a single differential output voltage, the current-to-code converter distributes I_2 and I_3 across N parallel comparison nodes, one per thermometer output. Each node is loaded simultaneously by a PMOS current mirror, which replicates I_2 with a multiplication factor α_k, and an NMOS current mirror, which replicates I_3 with a complementary factor β_k. The two factors are chosen such that α_k decreases monotonically with the node index k while β_k increases correspondingly, as illustrated schematically in Figure 3.1(c).

The binary output at node k is resolved by a simple current competition. If the PMOS sourcing current $\alpha_k I_2$ exceeds the NMOS sinking current $\beta_k I_3$, the node is pulled to a logic high, otherwise the NMOS pull-down dominates and the node settles to logic low. The quantisation threshold associated with node k is the specific input voltage at which the two competing currents are in equilibrium:

$$\alpha_k I_2 = \beta_k I_3 \Rightarrow v_{id}^{(k)} = \frac{I_B}{g_m} \cdot \frac{\alpha_k - \beta_k}{\alpha_k + \beta_k}. \tag{3.3}$$

Because the PMOS ratio α_k decreases monotonically with k, the thresholds $v_{id}^{(k)}$ also decrease monotonically, guaranteeing a valid thermometer code by construction. Nodes with a dominant PMOS mirror transition to high first as V_{in} rises above $V_{ref,c}$, followed progressively by nodes with higher NMOS ratios as the input continues to increase.

At the reference condition $V_{in} = V_{ref,c}$, the error current vanishes, $I_2 = I_3 = I_B/2$, and the state of each node reduces to comparing α_k against β_k directly. For an odd number of output bits with symmetric, evenly spaced ratio assignments, the central node satisfies $\alpha_k = \beta_k$ and is therefore at its decision boundary, producing the thermometer pattern `11···1?0···00` that is characteristic of a comparator array operating precisely at its reference level.

3.3.1.3 Adaptation to an Asynchronous Flash ADC

The circuit as described operates entirely in continuous time. No clock is needed to trigger a comparison, and the thermometer output updates immediately whenever V_{in} crosses one of the thresholds defined by Equation (3.3). This property is the primary motivation for adopting the CMF-ADC topology as the comparison stage of an asynchronous Flash ADC.

In this context, each crossing event of V_{in} through a threshold level generates a spontaneous transition on the corresponding output bit without the intervention of any external timing reference. The ensemble of N output bits forms a thermometer code that tracks the input amplitude in real time. As V_{in} increases past each threshold, one additional bit transitions from low to high, and the reverse process occurs as V_{in} decreases. The asynchronous control logic described in Section 3.6.5 then detects these transitions and encodes them into the final binary output.

A further advantage in the ADC context is that monotonicity of the thermometer code is enforced at the topology level by the ordered mirror ratios, rather than relying on post-processing error correction. Bubble errors (isolated low codes within an otherwise high thermometer sequence) cannot arise as long as the PMOS ratios are strictly decreasing and the NMOS ratios are strictly increasing across the output nodes, since any such ordering guarantees that the threshold voltages $v_{id}^{(k)}$ form a strictly monotone sequence as required by (3.3).

The main design parameters that control the ADC's resolution and input range are the mirror ratios $\{\alpha_k, \beta_k\}$, the tail current I_B, and the transconductance g_m of the input pair. These relationships, along with the specific ratio assignments used in the implementation explored in this work, are discussed in Section 3.3.2.

3.3.2 Reference Implementation and Mirror Ratio Assignment

The reference implementation described in [86] instantiates the operating principle of the previous section with five comparison nodes, i.e., five thermometer outputs (not five bits of binary resolution, see Section 3.1). The PMOS mirror ratios decrease monotonically across the five output nodes ($\times 14$, $\times 12$, $\times 10$, $\times 8$ and $\times 6$ for thermometer outputs $C_{\mathrm{LPT}}[4]$ through $C_{\mathrm{LPT}}[0]$), while the NMOS mirror ratios increase correspondingly from $\times 6$ to $\times 14$. The odd number of output nodes provides a well-defined, symmetric midpoint. The central node $C_{\mathrm{LPT}}[2]$ carries equal PMOS and NMOS ratios ($\times 10$), so its decision threshold aligns exactly with the condition $V_{\mathrm{in}} = V_{\mathrm{ref,c}}$. Figure 3.2 shows the transistor-level schematic of this five-output reference implementation, with the PMOS and NMOS mirror ratios annotated on each branch.

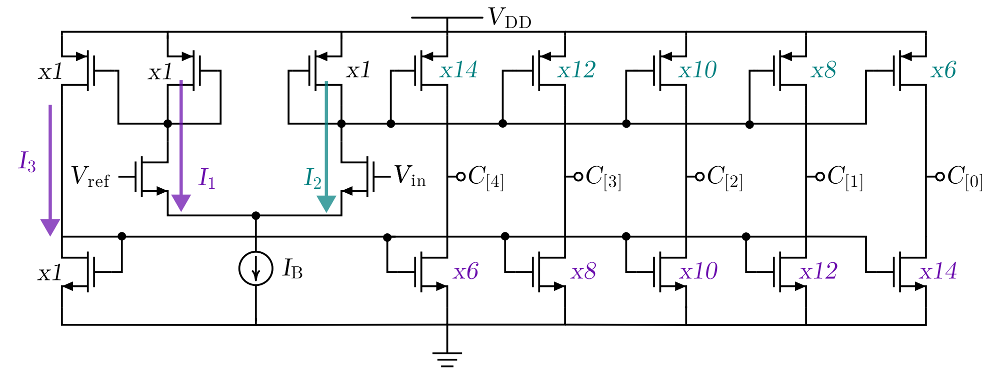


Figure 3.2: Transistor-level schematic of the first architecture, the current-mirror Flash ADC, showing the differential input pair driven by V_{ref} and V_{in}, the tail current source I_B and the five current-mirror output branches whose PMOS and NMOS width ratios set the thermometer thresholds $C[4]$ down to $C[0]$.

Applying Equation (3.3) to these ratio assignments confirms the expected behaviour at the reference condition, with $V_{\text{in}} = V_{\text{ref,c}}$, the two outermost node pairs resolve to high ($C_{\text{LPT}}[4]$, $C_{\text{LPT}}[3]$) and low ($C_{\text{LPT}}[1]$, $C_{\text{LPT}}[0]$) respectively, while the balanced central node sits at its decision boundary, yielding the thermometer code 11?00, where the question mark denotes the indeterminate state of a comparator operating precisely at its decision threshold.

The thresholds governing the transition of each output are fully determined by the mirror ratios, as established analytically in [86], and result in a monotonic, symmetric thermometer-coded response across the input voltage range. This architecture combines inherently fast conversion speed stemming from the absence of a conventional error amplifier, whose finite bandwidth would otherwise limit the response time of the control loop [86] with a straightforward, hardware-efficient structure and an explicitly quantifiable input-output characteristic. It was therefore adopted as the first candidate for the present comparator design, and its operating principle informed all subsequent topological choices.

3.3.3 Small-Signal Behaviour of the Current-Mirror OTA

The transconductance cell and the current-to-code converter together form a current-mirror operational transconductance amplifier (OTA), a single-stage topology whose distinguishing feature is that the input current is replicated to the output through current mirrors sized with a multiplication factor K (the ratios α_k introduced above). A short small-signal analysis, following the treatment of single-stage OTAs in [29], explains why this structure was attractive as a fast, current-domain input stage.

When the input pair is matched with transconductance g_m, each input device delivers a signal current $g_m v_{id}/2$. The output mirrors copy this current with the factor K and merge the two halves at the output node, so the effective transconductance of the cell is

$$G_m = K\,g_m, \tag{3.4}$$

K times larger than that of a plain differential pair. The output node sees two device output resistances in parallel,

$$r_{out} = r_{oK} \parallel r_o \approx \frac{r_o}{2}, \tag{3.5}$$

where r_o is the output resistance of a single transistor at the operating bias. The low-frequency gain and the single-pole frequency response then follow from the product of these two quantities and the load capacitance C_L:

$$A_0 = -G_m r_{out} = -\frac{K}{2} g_m r_o, \qquad \omega_{3\mathrm{dB}} = \frac{1}{r_{out} C_L}, \qquad \omega_{GB} = |A_0| \, \omega_{3\mathrm{dB}} = \frac{K g_m}{C_L}. \tag{3.6}$$

The gain-bandwidth product is therefore K times that of a single differential pair, which is the source of the speed advantage of the current-mode approach. The same K amplification acts on the large-signal behaviour. The current available to charge the output node is the mirrored tail current, so the slew rate

$$SR = \frac{K I_{tail}}{C_L} \tag{3.7}$$

is not bounded by the input bias current, unlike in a simple OTA. Finally, because only two transistors stack at the output node, the output can swing between V_{ov} and $V_{DD} - V_{ov}$, the widest output range of the single-stage OTA family and effectively rail-to-rail at this resolution.

High G_m, wide bandwidth, fast slewing and a wide output swing all suit an event-driven comparison stage, and they justified the choice of the current-mirror configuration for the first approach. The same analysis also shows the costs that later ruled the current-mode line out. The K extra current paths raise the power consumption, the mirror devices add noise that is amplified to the output, and, as the next architecture shows, every stacked mirror spends voltage headroom the 0.8 V supply cannot afford.

3.3.4 Current Comparator Selection

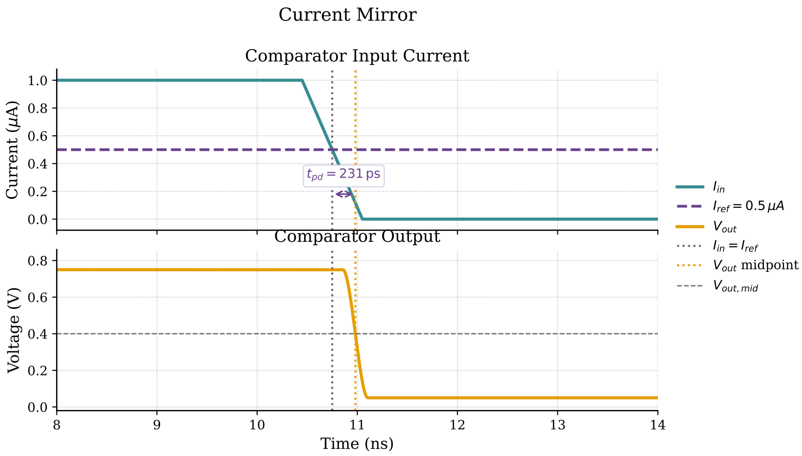
The current-to-code converter resolves each thermometer output through a current comparison, so the decision element of this first approach is a current comparator, and its choice sets the speed and the power of the whole stage. This choice was not made from the literature alone. All six topologies reviewed in Section 2.6, the current-mirror, Traff, Lu Chen, hysteresis, Min and Kim and CC-II comparators, were redrawn at transistor level and simulated under identical conditions, and the current-mirror cell was selected on the basis of those simulations. This section documents the testbench, the small-signal behaviour and the measured results, so that the power and delay figures used later in the chapter rest on an explicit comparison rather than on an asserted choice.

The small-signal behaviour of each topology was analysed in Section 2.6 and is the starting point for the comparison. The transistor-level schematic of each comparator, with its input and output swing ranges annotated, was presented alongside that analysis in Section 2.6, the input swing being the range of input current (or the equivalent node voltage) over which the comparator resolves correctly, and the output swing the excursion of the decision node. These ranges, together with the small-signal input resistance of each topology, set the sensitivity and the speed that the transient measurements then quantify.

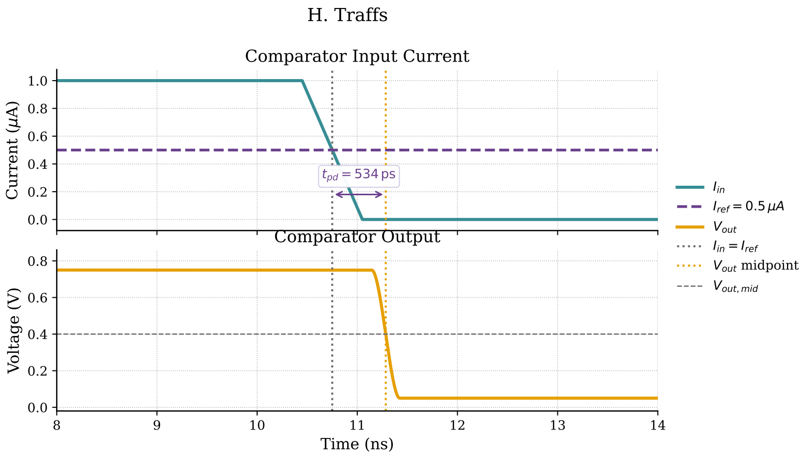
The six comparators were placed in a common testbench, the same supply, the same reference current, and the same input current step, so that the figures are directly comparable. From the step response, the propagation delay was measured as the time from the input crossing the decision threshold to the output reaching its logic level, and the maximum and average power were integrated over the switching event. Table 3.1 collects the three figures of merit and Figures 3.3 and 3.4 show the transient response of each comparator to the same input current step.

Table 3.1: Comparison of the simulated current comparator topologies.

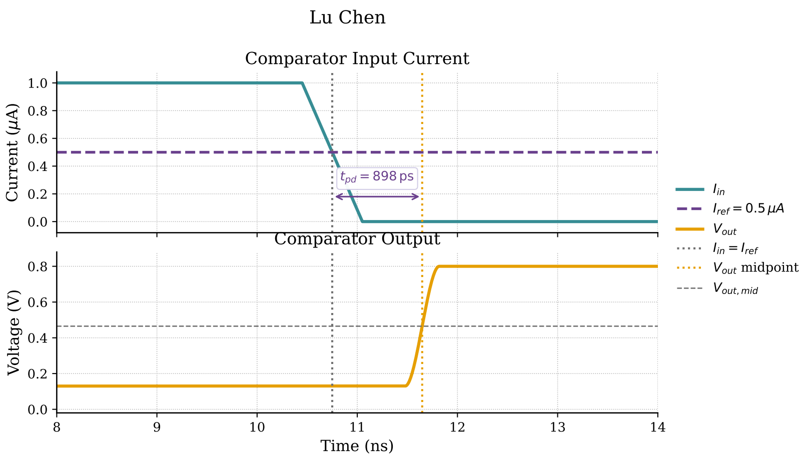
Topology	**Average power (μW)**	**Propagation delay (ps)**
Current-mirror	1.624	231
Traff	0.824	534
Lu Chen	108.8	898
Hysteresis	35.28	1230
Min and Kim	291.2	767
CC-II based	19.52	10030



(a) Current-mirror

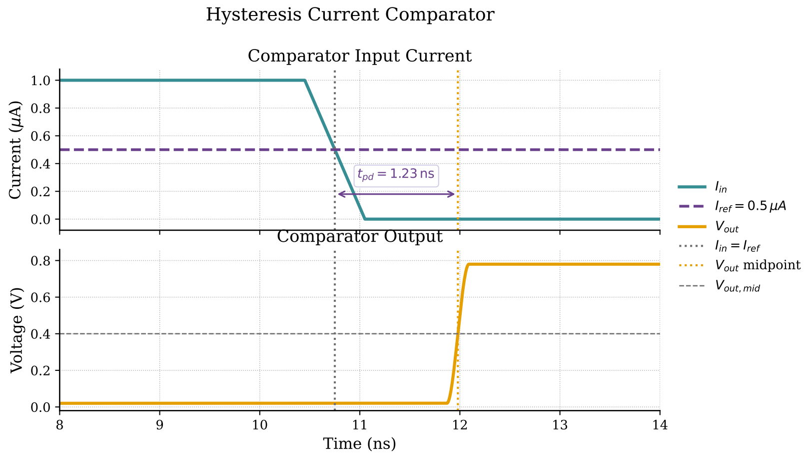


(b) Traff

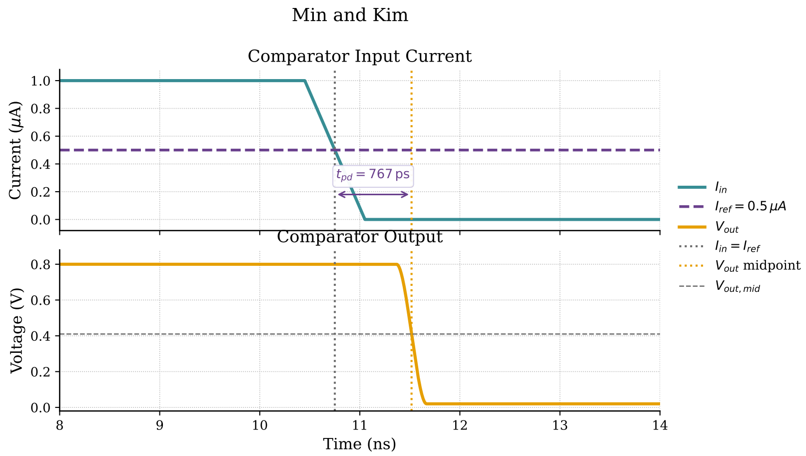


(c) Lu Chen

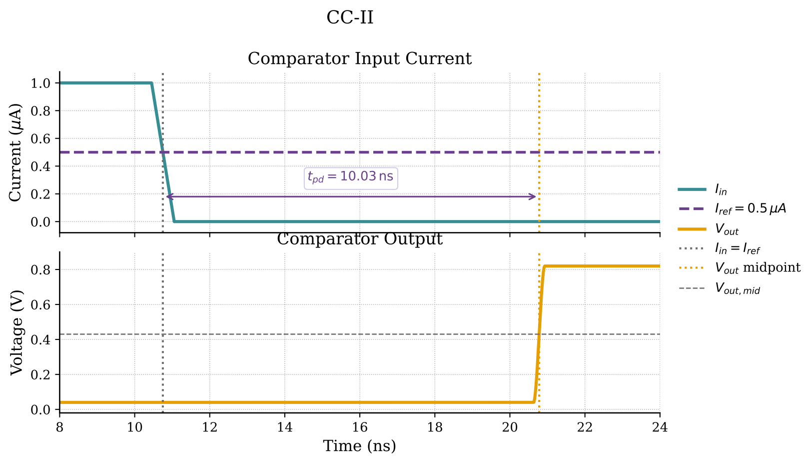
Figure 3.3: Transient response of the simulated current comparators to the same input current step, showing the output voltage together with the input and reference currents against time, from which the propagation delay is measured: (a) current-mirror, (b) Traff, (c) Lu Chen. The remaining three topologies are shown in Figure 3.4.



(a) Hysteresis



(b) Min and Kim



(c) CC-II based

Figure 3.4: Transient response of the simulated current comparators, continued from Figure 3.3: (d) hysteresis, (e) Min and Kim, (f) CC-II based.

With the lowest propagation delay (231 ps, about 2.3 times shorter than the next fastest) and a very competitive average power (1.624 µW, the second-lowest of the six), the current-mirror was the selected topology. Arguably, the Traff comparator could be an option, since it draws slightly less power (0.824 µW), but it is more than twice as slow (534 ps) and carries the deadband limitation described in Section 2.6, so it does not match the speed-and-power combination of the current-mirror cell. The choice also keeps the comparison stage structurally homogeneous with the rest of the CMF-ADC, which is itself built on current mirrors. Working as the decision node of the CMF-ADC, at low current and driving only a light internal load rather than a buffered rail-to-rail output, the cell avoids the large-capacitive-load penalty discussed in Section 2.6 and reaches the lowest delay of the six.

3.3.5 Results of the First Approach

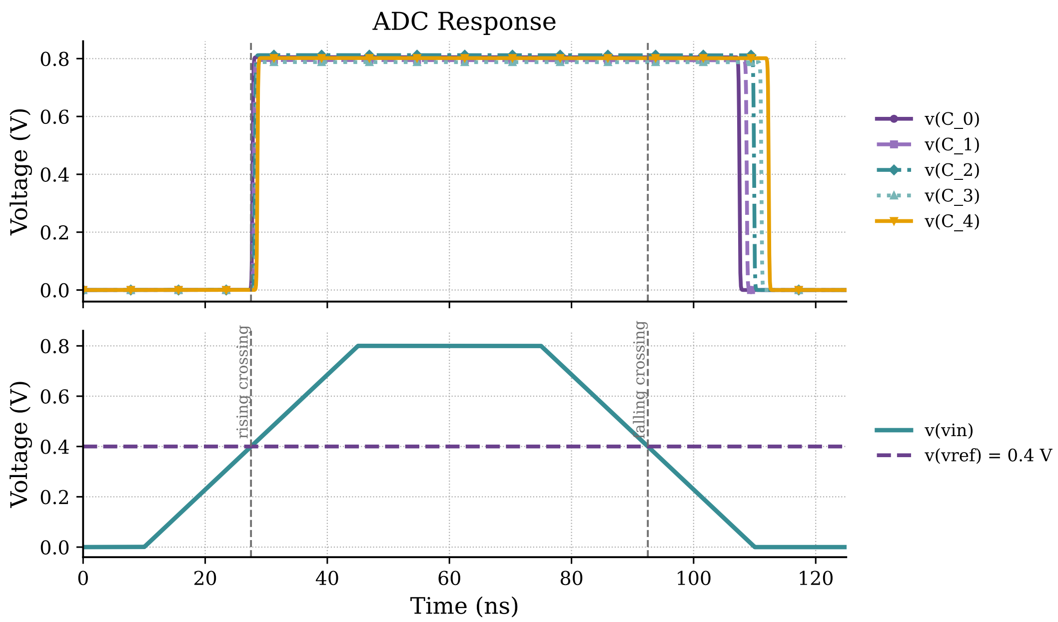


Figure 3.5: Simulated operation of the first architecture. The five comparator outputs (top) follow a trapezoidal input crossing the 0.4 V reference (bottom). The rising crossing is tracked sharply, but the falling transition lags well behind the input crossing, the slow-recovery asymmetry of the high-impedance current-mirror node.

Figure 3.5 shows the implemented CMF-ADC in operation, as the input crosses the reference level, the thermometer outputs switch in the expected monotonic order, confirming the operating principle of Section 3.3.2. The same simulations, however, exposed the limitation that ended this architecture. The usable input common-mode range is far from the supply rails, bounded by the headroom of the differential pair and its tail source. For the sensor-driven applications targeted in Chapter 1, that restriction is too costly, and the work moved to a rail-to-rail input stage.

3.4 Second Architecture: The Complementary Dual-Input Flash Configuration

3.4.1 Motivation for Wide Input Common-Mode Range Extension

A primary goal in developing the proposed comparator was to extend its input common-mode range as close as possible to the supply rails. A rail-to-rail input lets the converter process signals whose common-mode voltage falls anywhere between ground and V_{DD} without any added level-shifting or biasing stage. This matters most in sensor interfaces, where the common-mode level is set by the physical quantity being measured rather than by the input stage, so a restricted input range would either leave part of the sensor output unusable or force extra conditioning circuitry, with its associated power and area cost.

A wide input range also makes fuller use of the available swing. With the signal spanning a larger fraction of the supply, the dynamic range and the signal-to-noise ratio improve at no extra supply voltage, a benefit that grows in low-voltage CMOS where the shrinking V_{DD} already limits the analog headroom. Finally, it adds robustness, since small common-mode shifts from mismatch or supply variation are then less likely to push the input transistors out of their intended operating region, keeping the transconductance and the comparator behaviour predictable across PVT.

For these reasons, a complementary dual-input architecture capable of rail-to-rail operation was investigated as a second design candidate.

3.4.2 Rail-to-Rail Architecture Evaluation and Voltage Headroom Limitations

Although the proposed CMF-ADC satisfies the functional requirements of the event-driven conversion scheme, its input common-mode range remains inherently limited by the voltage headroom required by the differential input pair and the tail current source. This limitation becomes particularly critical in modern low-voltage CMOS technologies, where the continuous reduction of the supply voltage severely restricts the available analog signal swing. As a consequence, input signals whose common-mode voltage approaches either supply rail may drive the transconductance stage outside its intended operating region, degrading both linearity and sensitivity.

The complementary dual-input topology evaluated to overcome this limitation employs two transconductance stages operating simultaneously one based on an NMOS differential input pair and another based on a PMOS differential input pair. The rationale behind this arrangement is well established in rail-to-rail input amplifiers [114]. An NMOS differential pair operates correctly when the common-mode voltage is high enough above the negative rail to provide gate overdrive for the pair and saturation for its tail source, so it covers the upper part of the input range, a PMOS differential pair has the complementary behaviour and covers the region near the negative rail. With both stages operating simultaneously, at least one transconductor remains functional across almost the entire supply voltage range.

The implementation explored in this work did not simply combine the output currents of the two OTAs in parallel. Instead, one output branch of each OTA was connected to the PMOS current mirrors responsible for generating the individual bits of the thermometer code, preserving the current-comparison principle established in the original CMF-ADC architecture.

The decision thresholds in this architecture are set by the ratios of those current mirrors, in exactly the same way as in the first architecture. Each output branch uses a distinct PMOS-to-NMOS mirror ratio, and that ratio fixes the input voltage at which the corresponding thermometer bit switches, so the fifteen thresholds are defined by sizing rather than by any reference ladder. This is the same threshold-setting mechanism later adopted by the final architecture, where the ratioing is moved from the current mirrors to the PMOS-to-NMOS width ratio of the output slicers (Section 3.6.4). What changes between the architectures is only the input stage that drives these ratioed branches, not the ratioed quantisation principle itself. The remaining output branch of each OTA was then cross-coupled to the complementary transconductor, such that the residual output current of one stage modulated the output node of the other, with the same arrangement applied symmetrically in the opposite direction.

The objective of this bidirectional cross-coupling was to exploit the complementary transconductance of both stages so that each OTA could reinforce the operation of the other near the boundaries of its valid input common-mode range. In principle, this mechanism would extend the effective dynamic operating region of the converter while preserving the event-driven current-comparison methodology. Figure 3.6 shows the implemented circuit, with the role of each transistor annotated.

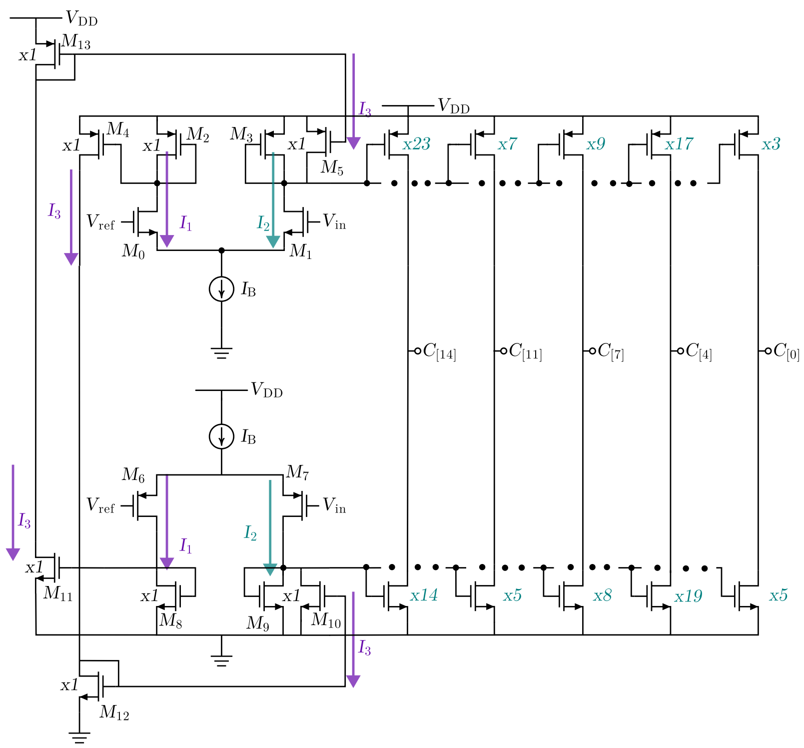


Figure 3.6: Complementary dual-input current-mirror OTA of the second approach, with the function of each transistor annotated: NMOS and PMOS input pairs, tail sources, mirror loads and cross-coupled output branches.

Despite the conceptual advantages of this architecture, its practical implementation proved incompatible with the target supply voltage of $V_{\mathrm{DD}} = 0.8$ V. A conventional five-transistor OTA already requires a minimum voltage headroom to accommodate the gate-to-source voltage of the differential pair, the saturation voltage of the tail current source, and the voltage drop across the active load. At a supply voltage of only 0.8 V, this headroom budget is already severely constrained.

The current-mirror OTA implementation did not fit this budget. In that topology, each load is itself a mirror that copies the branch current onward to the thermometer-code conversion stage. Along the critical path the tail current mirror and the input device each drop their own saturation voltage, and at the output node the cross-coupling places a second device in parallel with the mirror load, so the node must clear whichever of the two needs more headroom. The budget of the critical path is

$$V_{DD} \geq V_{ov,\mathrm{tail}} + V_{ov,\mathrm{in}} + \max(V_{ov,\mathrm{M9}}, V_{ov,\mathrm{M12}}), \tag{3.8}$$

where $V_{ov,\mathrm{tail}}$ is the saturation voltage of the tail current mirror, $V_{ov,\mathrm{in}}$ that of the input device, and the last term is the larger of the saturation voltages of the mirror load M_9, $V_{ov,\mathrm{M9}}$, and of the cross-coupling device M_{12} brought in from the complementary OTA, $V_{ov,\mathrm{M12}}$, since the two share the output node and the more demanding one sets the limit. Each of these saturation voltages is taken out of the supply before any output swing is left over. The basic mirror-OTA stack already spends two of them, and the cross-coupling adds the third through the maximum term, so at $V_{DD} = 0.8$ V the remaining headroom is not enough to hold the output device in saturation while the output swings to the rail. No operating point kept all active devices in saturation over the intended input range, and the architecture could not function correctly. This shows up directly in the DC behaviour of the stage. Figure 3.7 plots the simulated DC response, and the output cannot reach V_{DD}, saturating well short of the upper rail. That ceiling is the signature of equation (3.8). With the supply fixed at 0.8 V, the stacked saturation voltages, and in particular the extra one the cross-coupling adds through the maximum term, leave no room for the output to reach the rail regardless of how the transistors are sized. The argument needs no specific device parameters, only the count of stacked saturation voltages against the available 0.8 V.

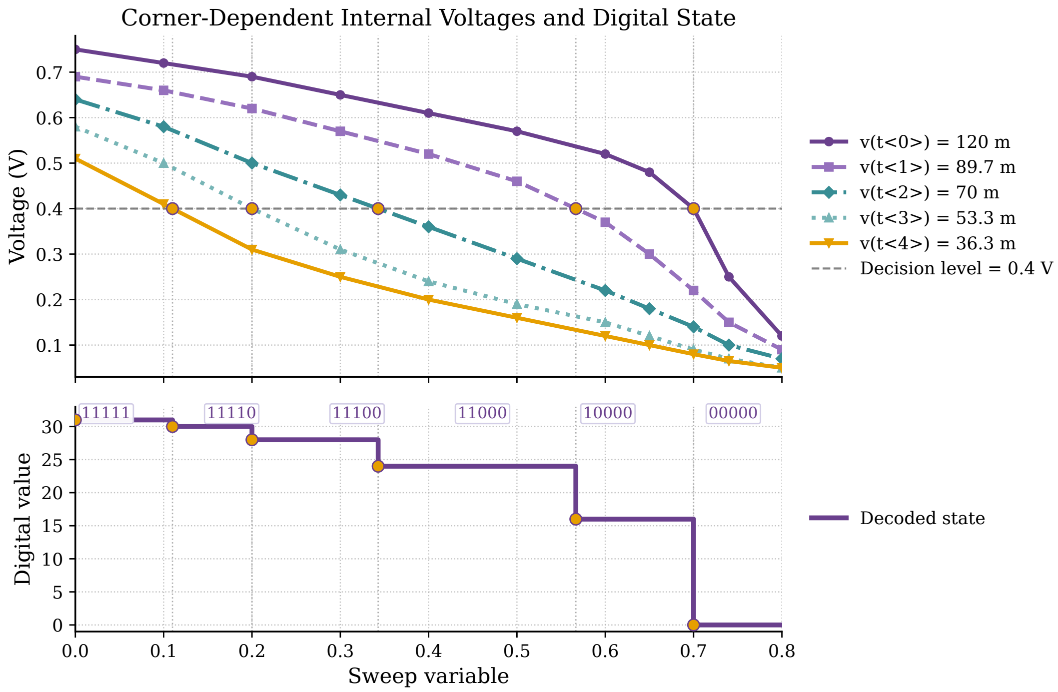


Figure 3.7: Simulated DC response of the second-architecture stage. The complementary outputs (out_nmos, out_pmos) and the internal node voltages never reach the supply rails and the transfer is visibly non-linear, the headroom shortfall predicted by equation (3.8) at the 0.8 V supply.

Part of the lost headroom could be recovered at the tail of the OTAs. The conventional tail current source was replaced with the low-voltage current mirror described in Razavi's book [114], which needs less voltage across the mirror branch because it removes one stacked gate-to-source drop from the path, at the price of a more complex bias network.

Figure 3.8 shows this mirror. The reference current I_1 flows into the left branch, where M_1 is the reference mirror device and M_0 sits above it as a cascode, while M_2 and M_3 form the output branch that carries I_{out}. The two mirror devices M_1 and M_2 share a gate voltage, so they copy the current faithfully only when their drain-source voltages match. Those drains are the nodes A and B, and the job of the cascode devices M_0 and M_3, both biased by V_b, is to hold $V_A = V_B$ no matter where the output node settles. With $V_A = V_B$ the two mirror devices see equal V_{DS} and the copy stays accurate. A plain cascode would set $V_b = V_{GS3} + V_{GS1}$, which raises A and B to a full V_{GS} above ground and makes the branch demand two overdrive voltages plus one threshold of headroom [114]. The low-voltage version sets V_b about one threshold lower, so A and B sit at roughly a single overdrive, $V_{DS} \approx V_{GS} - V_{TH}$, the smallest value that still keeps M_1 and M_2 in saturation. The accurate copy survives because A and B stay equal, while the headroom taken at the tail drops to about two overdrives and the threshold that a plain cascode wastes is recovered [114]. This is the saving the second architecture used to claw back part of its lost headroom.

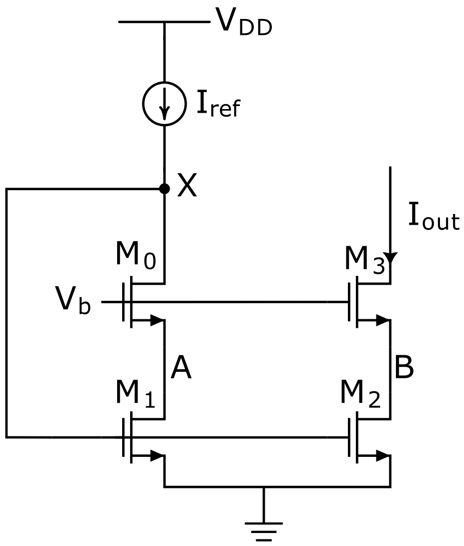


Figure 3.8: Low-voltage cascode current mirror used at the tail of the OTAs, after Razavi [114]. The reference current I_1 feeds the M_0–M_1 branch and the output branch M_2–M_3 carries I_{out}. The cascode devices M_0 and M_3, biased by V_b, hold the drains of the mirror devices, nodes A and B, at equal voltage, so the current copies accurately while each node sits at only about one overdrive above ground.

The change helped, but not enough. The combined voltage requirements of the differential input pair, the low-voltage tail mirror, the mirror loads and the cross-coupled output stage still exceeded the available supply.

No feasible bias point could be identified that simultaneously maintained all transistors within their intended operating regions while preserving the desired dynamic-range extension mechanism. What this architecture showed is that the current-mirror implementation of the rail-to-rail concept cost too much headroom at 0.8 V. The concept itself survived, along with the low-voltage tail mirror. The search for a rail-to-rail input stage therefore continued, first through a bulk-driven amplifier and then through the five-transistor OTA that was finally adopted.

3.5 A Bulk-Driven Alternative

The headroom problem of the second architecture comes from stacking gate-driven devices under a low supply. A bulk-driven input stage sidesteps that problem at the source. Driving the signal into the bulk terminal rather than the gate takes the threshold voltage out of the input common-mode budget, so a bulk-driven differential pair accepts an input that swings rail to rail even when the supply is below a single threshold. The amplifier evaluated here follows the single-stage bulk-driven OTA of Ballo, Grasso and Pennisi [10], in which a non-tailed bulk-driven pair is loaded by current mirrors and a cross-coupled load adds local positive feedback that boosts the otherwise weak bulk transconductance back to a useful level. Figure 3.9 shows the circuit, with the role of each transistor annotated.

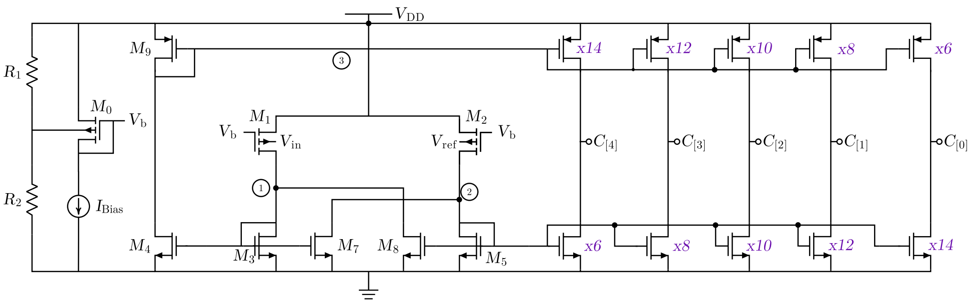


Figure 3.9: Bulk-driven OTA of the third architecture [10], with the function of each transistor annotated, the non-tailed bulk-driven input pair M_1, M_2, the current-mirror loads and the cross-coupled load M_7, M_8 that provides the local positive feedback. The circled points 1 and 2 mark the two differential output nodes and point 3 the supply-side node of the input pair.

The operation follows the bulk-driven OTA of Ballo, Grasso and Pennisi [10]. The bias network on the left, the divider R_1 and R_2 with M_0, the source I_{Bias} and the current device M_9, sets the operating point of the stage and the gate voltage V_b shared by the input devices. The input pair M_1 and M_2 is bulk-driven, the signals V_{in} and V_{ref} entering through their bulk terminals while the gates are held at V_b. Driving the bulk rather than the gate keeps the threshold voltage out of the input common-mode budget, which is what lets the pair track an input that swings close to both rails. Because the pair is non-tailed, the sources of M_1 and M_2 reach the supply at node 3 without a tail current source between them and the rail, so nothing constrains their common mode. The two drain currents develop the differential output at node 1 and node 2, where the input pair meets the NMOS load formed by M_3 to M_8. Within that load M_7 and M_8 are cross-coupled, each taking its gate from the opposite output node, so a rise at node 1 reinforces the current at node 2 and the reverse holds as well. This local positive feedback raises the otherwise weak bulk transconductance back to a useful level [10]. Node 1 and node 2 then drive the ratioed current-mirror output stage on the right, whose asymmetric PMOS-to-NMOS ratios, from $\times 14/\times 6$ down to $\times 6/\times 14$, generate the thermometer outputs $C[4]$ to $C[0]$.

The transconductance of the cell follows from the small-signal analysis of the paper [10]. With the cross-coupling ratio and the mirror-load ratio defined as

$$\alpha = \frac{(W/L)_7}{(W/L)_3} = \frac{(W/L)_8}{(W/L)_5}, \qquad \beta = \frac{(W/L)_4}{(W/L)_3} = \frac{(W/L)_6}{(W/L)_5}, \tag{3.9}$$

where α sets the strength of the positive feedback from the cross-coupled pair M_7, M_8 and β the gain of the mirror loads, the differential transconductance of the stage is

$$G_m = \frac{\beta}{1-\alpha}\, g_{mb1,2}, \tag{3.10}$$

where $g_{mb1,2}$ is the bulk transconductance of the input devices M_1 and M_2. The factor $1/(1-\alpha)$ is the boost provided by the local positive feedback, and it lifts the weak bulk transconductance, which is roughly 60 to 90% smaller than the gate transconductance for the same bias and sizing [10], back to a useful level. The feedback has to stay below unity, so $\alpha < 1$, and it is kept comfortably under it, around 0.9, to leave a margin against process mismatch and to keep the stage from latching.

Only the input stage changes here. The part that makes the decision, the bank of ratioed current comparators, stays the same. The bulk-driven amplifier is tried purely as a rail-to-rail replacement for the input follower, and the current comparison behind it is exactly the one used in the first architecture, the same ratioed current comparators with the same dimensions as those of Section 3.3.2. Keeping the current comparators fixed makes the comparison fair, since any change in behaviour is then due to the input stage alone.

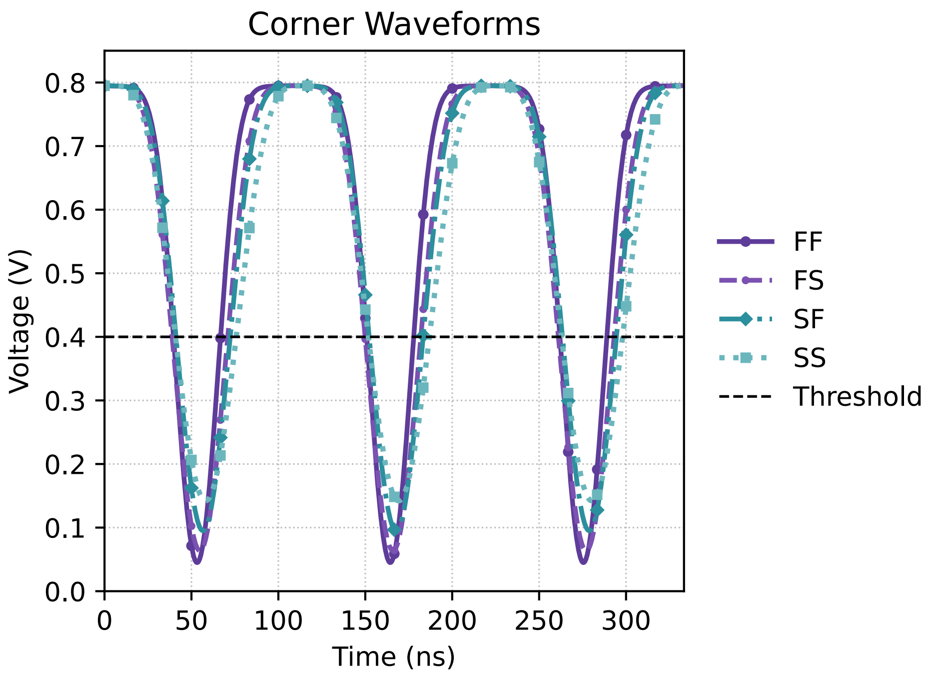


Figure 3.10: Response of the bulk-driven OTA follower across the four process corners (FF, FS, SF, SS). The output waveform diverges markedly between corners, most of all near the lower rail, the sensitivity that led to discarding the architecture.

At the nominal corner the cell behaved well. It reproduced a rail-to-rail input at 0.8 V with adequate gain and a clean response, and as a unity-gain follower it tracked the input across the full supply, which is the property the converter needs.

The difficulty appeared across process corners. The transconductance boost relies on a positive-feedback factor set by the ratio of the cross-coupled devices, and that factor multiplies not only the nominal transconductance but also its spread. A small mismatch in the feedback ratio, or a corner shift in the device transconductances, moves the effective gain and bandwidth by a large amount. Figure 3.10 shows this directly, the follower output diverging between the four corners, most of all near the lower rail. The same sensitivity is visible in the source design, whose reported corner data spread the DC gain from roughly 20 to 45 dB and the gain-bandwidth product by more than an order of magnitude [10]. There are certain applications where this divergence is tolerable, but the Flash converter needs fifteen consistent decision paths and a follower whose behaviour repeats from corner to corner. An input stage whose gain and bandwidth move that much with process would shift the slicer thresholds in a way the body-bias trimming of Section 3.7 could not reliably absorb.

The same corner sensitivity appears in the time domain. Figure 3.11 reports the input-referred hysteresis of the bulk-driven follower measured under the 9 MHz full-scale sinusoid at each process corner, and the worst case falls on the slow SS corner.

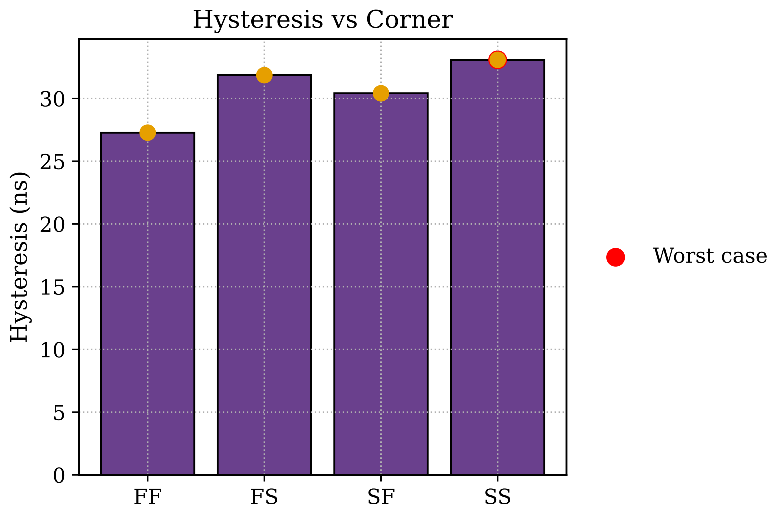


Figure 3.11: Input-referred hysteresis of the bulk-driven OTA follower at the four process corners, measured under the 9 MHz full-scale sinusoid, with the worst case marked at the slow SS corner.

For that reason the bulk-driven amplifier was set aside despite its attractive nominal behaviour, and the design moved to the more robust five-transistor OTA of the next section, trading some of the rail-to-rail reach of the bulk-driven input for predictable, corner-stable operation.

3.6 Final Architecture: The Complementary Five-Transistor OTA

3.6.1 Complementary Five-Transistor OTA Core and Optimised Transistor Sizing

This final architecture kept the rail-to-rail objective of the second architecture and rebuilt its implementation around the classical five-transistor OTA. The complementary input is preserved. Two five-transistor OTAs operate in parallel on the same input, one built around an NMOS differential pair and the other around a PMOS differential pair, so the usable input range is extended in the same way as before. The load structure is what changes. Each pair is loaded by a single active current mirror, in which one device operates in diode connection and sets the gate voltage of the mirrored branch. This is the conventional active load that defines the five-transistor OTA [114]. Figure 3.12 shows the circuit with the role of each transistor annotated.

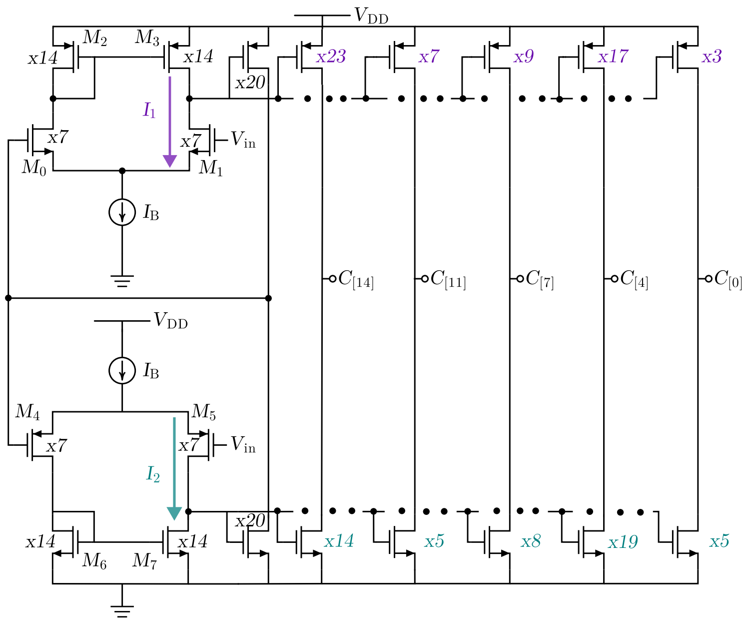


Figure 3.12: Final architecture, a complementary pair of five-transistor OTAs with the function of each transistor annotated, including the low-voltage tail mirrors and the current-mirror active loads.

The final architecture keeps a single current-mirror active load per pair, which removes the extra stacked devices of the second architecture. For its PMOS-input OTA the critical-path budget is

$$V_{DD} \geq V_{ov,\text{tail}} + V_{ov,\text{in}} + \max\left(V_{ov,\text{M7}}, V_{GS,20}\right), \tag{3.11}$$

where $V_{ov,\text{tail}}$ is the tail saturation voltage, $V_{ov,\text{in}}$ that of the input device, $V_{ov,\text{M7}}$ the saturation voltage of M_7, and $V_{GS,20}$ the gate-to-source voltage of the mirror device of multiplicity 20. This last device dominates the maximum and sets the output-node requirement. This budget and that of the second architecture in equation (3.8) share the tail and input-device terms and differ only in the device that sets the output-node voltage.

The tail bias carries over a result from the earlier architectures. Both OTAs draw their tail current through the low-voltage current mirror from Razavi's book [114], the same structure tested in the second architecture, because even this reduced topology operates close to the headroom limit in contrast to a conventional tail mirror which requires higher voltage headroom.

One limitation remains. Close to either supply rail neither input pair has enough overdrive to stay in saturation, so the effective input window of the complementary structure stops slightly short of the rails. The consequences for the test stimuli and for the idle power consumption are discussed in Section 3.8.2.

The initial implementation evaluated the complementary OTA under open loop conditions to maximise gain and improve sensitivity to small differential input voltages. Through transistor sizing optimisation, high voltage gain could be achieved and with it higher comparison sensitivity. Practical simulations, however, revealed a critical limitation of this approach. Small unintended differential inputs originating from offset, thermal noise or marginal overdrive caused the output node to saturate rapidly at one of the supply rails before the intended comparison could be completed.

Once saturation occurred, the recovery was limited by the bias current available to charge or discharge the high-impedance output node. This increased the settling time considerably and introduced variability between consecutive conversion cycles.

To overcome this limitation, the complementary OTA was reconfigured as a unity gain voltage follower by directly connecting the output node to the inverting input and establishing full negative feedback. Under this configuration, the differential input voltage remains close to zero in steady state and the amplifier operates continuously in its linear region.

As a consequence, the OTA acts as a low impedance replica of the input signal, reproducing V_{in} at its output while avoiding saturation effects. The resulting accuracy is determined primarily by the open loop gain and the input referred offset of the amplifier.

3.6.2 Small-Signal Analysis of the Five-Transistor OTA

The same small-signal framework applied to the current-mirror OTA in Section 3.3.3 makes the headroom-versus-performance trade-off of the final choice explicit. Each five-transistor OTA has no current amplification, so its effective transconductance is simply that of the input pair,

$$G_m = g_{m1} = g_{m2} = g_m, \tag{3.12}$$

the special case $K = 1$ of equation (3.4). The output node of the OTA again sees two device resistances in parallel, $r_{out} = r_{o2} \,\|\, r_{o4} \approx r_o/2$, which gives a low-frequency gain and a single-pole response of

$$A_0 = -\frac{1}{2} g_m r_o, \qquad \omega_{3\text{dB}} = \frac{1}{(r_o/2)\,C_L}, \qquad \omega_{GB} = \frac{g_m}{C_L}. \tag{3.13}$$

The slew rate is set by the tail current charging the load,

$$SR = \frac{I_{tail}}{C_L}, \tag{3.14}$$

without the factor K that the current-mirror OTA enjoyed. Comparing equations (3.13) and (3.14) with their current-mirror counterparts (3.6) and (3.7) shows what was given up, the gain, the gain-bandwidth product and the slew rate are all K times smaller. In return, the device stack is shorter. With V_{ov} the minimum drain-to-source voltage for saturation and V_{th} the threshold voltage, the input common-mode level of a five-transistor OTA must lie between $2V_{ov} + V_{th}$ and $V_{DD} - V_{ov}$, and the output between $2V_{ov}$ and $V_{DD} - V_{ov}$. These are the moderate swing ranges of the simplest single-stage OTA [29], and they fit the 0.8 V budget once the extra mirror loads of the second approach are removed, as the headroom budget of equation (3.11) showed.

The reduced performance is acceptable here because the OTA works as a unity-gain follower rather than as an open-loop high-gain amplifier. The slicer bank that follows sets the decision thresholds, so the OTA only has to reproduce the input voltage accurately and settle within the conversion window. The complementary structure places two such OTAs in parallel, an NMOS-input pair covering the upper part of the range and a PMOS-input pair covering the lower part, so that at least one of them keeps the transconductance of equation (3.12) available across almost the entire supply range.

3.6.3 Stability Without Frequency Compensation

An OTA placed in feedback is normally frequency-compensated to keep the loop stable. The follower here uses no compensation, for two reasons.

Compensation moves the dominant pole to a lower frequency, trading bandwidth for phase margin. The follower has to settle inside the conversion window before the next threshold is crossed, so a slower amplifier would lengthen the conversion delay τ_{conv} of Section 3.8.1 and tighten the slew-rate limit on the input frequency. Trading bandwidth for a phase margin the amplifier already has would only slow the converter.

A five-transistor OTA has a single high-impedance node, its output, so in unity-gain feedback the loop is dominated by one pole and stays stable by construction. Open-loop simulation of the complementary OTA confirmed this, as the magnitude and phase response of Figure 3.13 shows. The phase margin is 55.5° at the unity-gain frequency, just below the 60° usually taken as a comfortable target but still well clear of instability, and it narrows only for inputs under 0.1 V or above 0.7 V, where one of the input pairs runs out of overdrive near the rail.

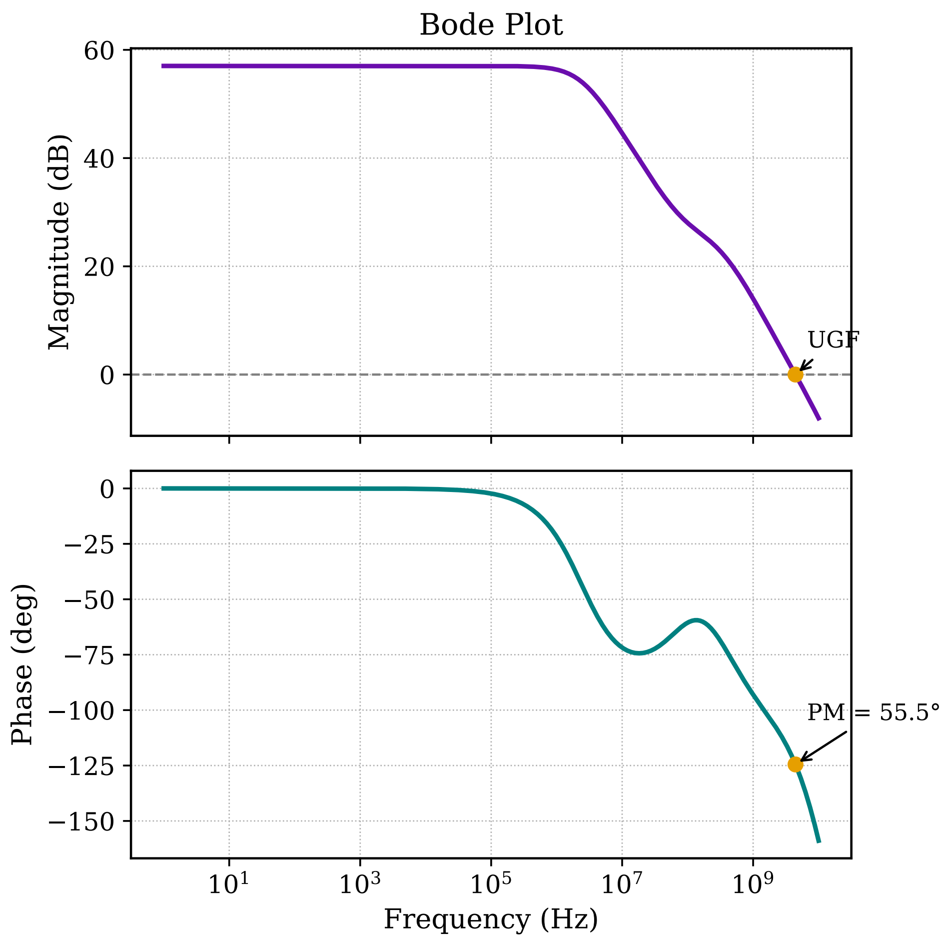


Figure 3.13: Open-loop magnitude and phase response of the complementary five-transistor OTA, showing the single dominant pole and the wide phase margin.

That high-margin band is the same 0.1–0.7 V window over which the input stage is most linear in Section 3.6.1. On either side of that band, one of the two complementary input pairs always keeps conducting, so the follower stays stable and the converter resolves crossings all the way to both rails. The cost is only a milder loss of follower linearity in the outer margins of the swing, not a loss of operation.

3.6.4 Ratioed Output Slicers Design

The quantisation stage of the converter is implemented through a bank of ratioed current-mode slicers directly connected to the output of the voltage follower. Each slicer is built from the current-mirror current comparator reviewed in Section 2.6 (Figure 2.12). That cell is normally a four-transistor structure in which a diode-connected transistor fixes the gate voltage of the output transistor. In the slicer this diode-connected device is replaced by the complementary OTA follower, so the follower output supplies that gate voltage and drives the current comparison directly, rather than the comparison being driven by an input current. The node then resolves high or low according to the resulting competition between the PMOS source and the NMOS sink, and the comparison level is set purely by the PMOS-to-NMOS width ratio of the slicer, not by current-mirror scaling as in the First Architecture. This ratio-defined thresholding is the same principle already used in the second architecture (Section 3.4), where the ratios lived in the PMOS current mirrors, here it is realised directly in the slicer transistors driven by the follower.

For reference, the balance point of the two competing output devices of an isolated such cell, the PMOS source and the NMOS sink driven by a common gate voltage, takes the same form as a CMOS inverter switching voltage,

$$V_M \approx \frac{V_{\mathrm{DD}}\sqrt{r}}{1+\sqrt{r}} \tag{3.15}$$

where V_M is the balance voltage of the cell, V_{DD} the supply voltage, and

$$r = \frac{(W/L)_p}{(W/L)_n}$$

is the ratio between the widths of the two competing output devices, the equal channel lengths cancelling.

The form follows from the balance condition itself. At V_M the gate and the output sit at the same voltage and both devices are in saturation carrying equal current, so the square-law current of the NMOS sink, with gate-source voltage V_M, equals that of the PMOS source, with source-gate voltage $V_{\mathrm{DD}} - V_M$,

$$\beta_n \,(V_M - V_{Tn})^2 = \beta_p \,(V_{\mathrm{DD}} - V_M - |V_{Tp}|)^2,$$

with $\beta = \mu C_{\mathrm{ox}}\,(W/L)$. Neglecting the threshold terms and the difference in carrier mobilities, so that the ratio of the two β collapses to the width ratio r, and taking the square root leaves $\sqrt{r}\,(V_{\mathrm{DD}} - V_M) = V_M$, which rearranges into Equation (3.15).

This bare expression rises with r and is shown only to identify the sizing parameter. It does not give the actual trip points.

The expression above, and the ratio assignment that goes with it, were only a starting point for the sizing. The trip point actually used for each slicer was then set on the simulated DC response of the follower and slicer together. The slicer comes right after the high-gain complementary follower, and its decision point, the input voltage at which the slicer output crosses the logic level, was read straight from that DC curve and the ratio adjusted until it landed where it was wanted.

The gain of the follower sets the wide span of the trip points and the shape of its transfer sets their direction, so that, as measured and tabulated below, the trip point moves to a lower input voltage as r increases, the opposite of what the plain expression suggests. Assigning a distinct ratio to each slicer then spreads the decision thresholds along the input voltage axis.

Each slicer therefore acts as an independent comparator operating at a predefined threshold voltage. Collectively, the outputs generate a thermometer coded representation of the input signal.

The sizing strategy is illustrated here with the reduced five-slicer bank introduced in Section 3.1. In that explanatory configuration, the topmost thermometer output, out_4, uses a ratio of $r = 1.5$ and trips at a low input voltage, while the bottommost output, out_0, adopts $r = 0.5$ and trips at a higher one. The intermediate slicers are distributed between these limits to generate approximately symmetric threshold spacing around the reference level.

As the input voltage increases from ground toward V_{DD}, the slicers cross their trip points one by one, the highest-ratio slicer first. At $V_{\mathrm{in}} = 0.4$ V the high-ratio slicers (out_4, out_3) have already switched while the low-ratio ones (out_1, out_0) have not, and the slicer with the central ratio $r = 1$ sits exactly at its trip point. The resulting thermometer code becomes 11?00, reproducing the same behaviour previously observed in the CMF ADC architecture.

The implemented converter applies the same strategy to its full bank of fifteen slicers. Each slicer receives a distinct ratio, listed in Table 3.3. The fifteen ratios increase monotonically from $3/5$ for out_0 to $23/14$ for out_{14}, and their numerators and denominators each sum to 166. Since the numerator of each ratio is the PMOS width of that slicer and the denominator its NMOS width, the equal channel lengths cancelling, this equality means the bank uses the same total PMOS and NMOS width, keeping its sizing balanced. The ratios are tuned on the DC response of the follower-plus-slicer chain, exercised across the full 0–0.8 V rail, so that each output crosses the 0.4 V logic level at a distinct input voltage, the fifteen crossings spreading from 0.75 V for out_0 down to 0.05 V for out_{14}, symmetric about the 0.4 V mid-level. In transient operation the usable input window narrows to the 0.1–0.7 V range of the follower (Section 3.6.1), within which the two extreme crossings move to 0.15 V and 0.65 V and the bank delivers the 4-bit quantisation used throughout the results. All the static-linearity and dynamic results reported in Chapter 4 use this 0.1–0.7 V transient window and the 37.5 mV LSB it implies, the 0.05–0.75 V figures describing only the DC sizing. Table 3.2 summarises the two conditions side by side.

Table 3.2: Decision thresholds of the slicer bank under the two operating conditions. The DC figures describe the sizing of the bank, while the transient figures describe operation and are used for all results.

Quantity	**DC sizing**	**Transient operation**
Usable input window	0–0.8 V	0.1–0.7 V
Lowest trip point (out_{14})	0.05 V	0.15 V
Highest trip point (out_0)	0.75 V	0.65 V
Logic reference level	0.4 V	0.4 V
LSB used for results	—	37.5 mV

Table 3.3: PMOS-to-NMOS sizing ratio r of each of the fifteen slicers. The numerators and the denominators each sum to 166.

Output	r	value	Output	r	value
out_0	$3/5$	0.600	out_8	$6/5$	1.200
out_1	$13/20$	0.650	out_9	$5/4$	1.250
out_2	$14/19$	0.737	out_{10}	$4/3$	1.333
out_3	$14/17$	0.824	out_{11}	$7/5$	1.400
out_4	$17/19$	0.895	out_{12}	$3/2$	1.500
out_5	$19/20$	0.950	out_{13}	$8/5$	1.600
out_6	$21/20$	1.050	out_{14}	$23/14$	1.643
out_7	$9/8$	1.125			

3.6.5 Asynchronous Control Logic and Clockless Event Generation Network

A key characteristic of the proposed architecture is the complete removal of the global clock from the conversion process. Instead of forcing periodic operation, event generation occurs exclusively as a consequence of changes in the analog input signal.

The OTA configured as a voltage follower continuously tracks the input signal and distributes this low impedance representation to all slicers simultaneously. Each slicer independently evaluates whether the input has crossed its corresponding threshold and immediately updates its output state.

The collection of slicer outputs forms an asynchronous thermometer code whose transitions directly represent input activity. Whenever one or more thresholds are crossed, a digital event is naturally generated without requiring any external timing reference.

This event driven operation establishes a direct relationship between signal activity and switching activity inside the converter. During intervals where the input remains constant, the slicers remain static and dynamic power consumption approaches zero. Conversely, periods of rapid input variation automatically increase event generation frequency and conversion activity.

Compared with previous current mirror based implementations, this architecture introduces a clear separation between input buffering and threshold generation. The OTA is responsible only for reproducing the input voltage, while threshold definition is achieved entirely through transistor sizing in the slicer stage.

This separation simplifies the design procedure, makes the circuit robust against saturation and removes the extra voltage headroom requirements. The result is a compact, scalable architecture suited to low-voltage asynchronous Flash conversion.

3.6.6 Results at the Typical Corner and Across Process Corners

The final architecture was characterised statically at the typical corner and at the four non-typical process corners, with the thresholds of the fifteen slicers extracted on rising and falling ramps as detailed in Chapter 4.

At the typical corner the converter needs no correction, both the DNL and the INL stay within approximately ±0.15 LSB across all codes, as Figure 3.14 shows.

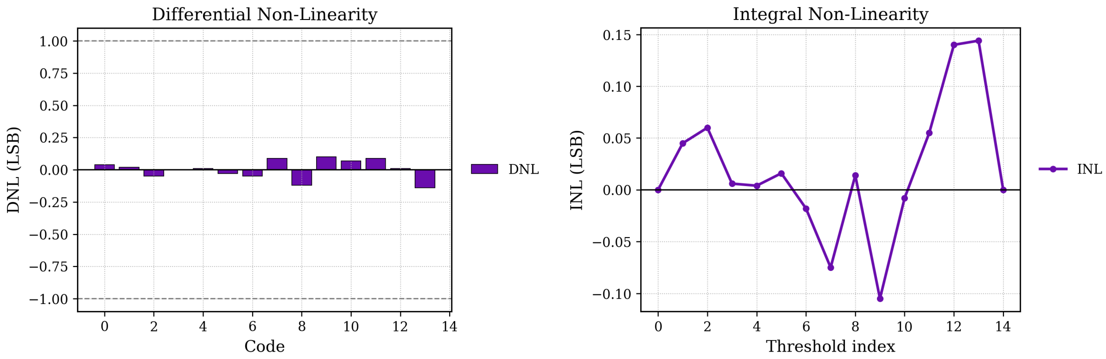


Figure 3.14: DNL per code (left) and INL per transition (right) of the final architecture at the typical (TT) corner.

The process corners are a different matter. Table 3.4 lists the worst-case errors at the typical corner and at SS, FF, FS and SF without any correction. The typical corner stays well within target, but every non-typical corner exceeds the 0.5 LSB DNL target on at least one ramp, and three of the four show worst-case INL above 1 LSB. The sizing-defined trip points of the slicer bank, accurate at TT, drift with the corner-dependent imbalance between NMOS and PMOS drive strengths, exactly as anticipated in the threshold model of equation (3.15).

Table 3.4: Worst-case static linearity of the final architecture at the typical and the non-typical process corners, without trimming (values in LSB, measured on rising and falling ramps). The TT figures are read from the per-code DNL and INL of Figure 3.14.

Corner	**DNL rise**	**DNL fall**	**INL rise**	**INL fall**
TT	0.1434	0.1421	0.1452	0.1446
SS	0.6958	0.6628	0.9489	1.0949
FF	0.3814	0.6372	0.6670	1.3724
FS	0.3814	0.6372	0.6670	1.3724
SF	0.5191	0.5142	0.5829	1.0787

The final architecture works, but its linearity does not survive process variation on its own. Some form of post-fabrication correction is required, the next section develops it.

3.7 Body Bias Trimming Strategy

The preceding section ended with a converter that is linear at the typical corner and out of specification at every non-typical one. This section develops the correction.

The slicer-based quantisation stage introduced earlier in this chapter defines each decision threshold exclusively through the PMOS-to-NMOS sizing ratio of the slicer, as expressed by equation (3.15). This makes the converter linearity directly dependent on the effective drive strengths of the two device types, and therefore directly exposed to fabrication process variability. This section describes the offline trimming strategy adopted to correct the resulting deterministic trip-point deviations across process corners.

The technique builds on the static, bulk-driven calibration approaches reviewed in Section 2.7: it adjusts the threshold voltages of the slicer transistors through the body effect, in the same spirit as the bulk-driven reference-ladder trimming of Yao [154] discussed in Section 2.7.3.2, but applied here to the bulk terminals of the slicer bank and indexed by the identified process corner rather than by a per-comparator offset measurement. The scope of this correction is deterministic inter-corner variation only: random intra-die mismatch, governed by Pelgrom's law [108], is not addressed by the corner-indexed table and is discussed as future work in Chapter 5.

3.7.1 Choice of Body Biasing Among the Reviewed Techniques

Among the trimming techniques surveyed in Section 2.7, several alternatives were considered before adopting body biasing. Internal resistive loading, in which a digitally switched resistor bank modulates the load of each comparator branch, is well suited to differential comparator stages but does not map naturally onto the slicers of the proposed architecture, which have no explicit load element to modulate. Introducing one would add capacitance to the only high-speed node of each slicer and degrade its switching delay. Dynamic offset-cancellation schemes such as auto-zeroing were excluded for the reasons detailed in Section 2.7.4. They require periodic clock phases, which would reintroduce into the converter precisely the clocked activity that the asynchronous architecture is designed to eliminate.

Bulk-driven adjustment, by contrast, satisfies all the constraints of the event-driven operation simultaneously. The bulk terminals of the slicer transistors are not part of the signal path, so driving them from a static bias source adds no capacitance to the switching nodes and consumes no dynamic power. Once the correction voltages are set, they are held by purely DC conduction paths, preserving the idle behaviour of the converter when the input is inactive. Finally, because the NMOS and PMOS bulk potentials act on the pull-down and pull-up networks independently, they provide exactly the two degrees of freedom needed to counteract both the symmetric (SS, FF) and the asymmetric (FS, SF) corner perturbations characterised in Section 3.7.3.

3.7.2 Function of NMOS and PMOS Transistors Under Different Body Bias

The threshold voltage of a MOSFET is not solely determined by the device geometry and doping profile but is also a function of the potential applied to its body terminal relative to the source. This dependence, commonly referred to as the body effect or substrate bias effect, is described for an NMOS transistor by

$$V_{tn} = V_{tn0} + \gamma_n \left(\sqrt{2\phi_{F,n} + V_{SB}} - \sqrt{2\phi_{F,n}} \right), \tag{3.16}$$

where V_{tn0} is the zero-bias threshold voltage, γ_n is the body effect coefficient, $\phi_{F,n}$ is the Fermi potential, and V_{SB} is the source-to-body voltage. The analogous expression for a PMOS transistor, accounting for the sign conventions appropriate to a p-channel device, takes the form

$$|V_{tp}| = |V_{tp0}| + \gamma_p \left(\sqrt{2\phi_{F,p} + |V_{SB}|} - \sqrt{2\phi_{F,p}} \right), \tag{3.17}$$

where the relevant source-to-body voltage for the PMOS is determined by the potential difference between the source and the N-well. In both cases, an increase in $|V_{SB}|$ raises the magnitude of the threshold voltage, a condition referred to as reverse body biasing (RBB). Conversely, reducing $|V_{SB}|$ below its nominal value or, in technologies that permit it, forward-biasing the body junction lowers $|V_t|$, a condition known as forward body biasing (FBB).

A direct consequence for the slicers in the proposed architecture is that each cell's trip point depends on the threshold voltages, and hence the drive strengths, of its PMOS source and NMOS sink. Forward-biasing the PMOS, by lowering its N-well below V_{DD}, reduces $|V_{tp}|$ and strengthens the source, which moves that slicer's trip to a lower input voltage, so the output crosses the logic level earlier. Forward-biasing the NMOS, by raising its bulk above ground, reduces V_{tn} and strengthens the sink, which moves the trip to a higher input voltage, so the crossing happens later. The two body terminals therefore act as two independent degrees of freedom, the PMOS well voltage pulling the trip point down and the NMOS bulk voltage pushing it up. Applying corrections to both allows independent tuning of the trip-point position and of the symmetry of the transition characteristic, a richer correction space than either terminal alone could offer.

3.7.3 Process Corner Dependence and Motivation for Trimming

In a deep-submicron CMOS process, fabrication variability causes the threshold voltages and carrier mobilities of the NMOS and PMOS devices to deviate from their nominal (typical-typical, TT) values. The standard process corners used to bound this variability, namely slow-slow (SS), fast-fast (FF), fast-NMOS slow-PMOS (FS) and slow-NMOS fast-PMOS (SF), capture the four extreme combinations of NMOS and PMOS behaviour and represent the conditions under which circuit performance is most likely to degrade.

For the slicer-based thermometer coder described in Chapter 3, the impact of process variation is particularly significant because the thermometer code relies on the precise relative ordering of the slicer trip points across the output bank. At the typical corner, the ratios r_k assigned to each slicer place the trip points at the intended positions relative to the reference level of 0.4 V, and the resulting differential non-linearity (DNL) and integral non-linearity (INL) of the converter remain within specification. At non-typical corners, however, the effective transistor strengths deviate from their nominal values in a manner that is not uniform across the bank. Since the PMOS and NMOS devices within each slicer are affected differently depending on the corner, the trip point of each slicer shifts by a cell-dependent amount, breaking the intended monotonic spacing and introducing non-linearities that manifest as elevated DNL and INL.

At the SS corner, both NMOS and PMOS threshold voltages increase, reducing the drive strength of both pull-down and pull-up networks. Although the symmetric nature of this degradation might suggest a self-compensating effect on the trip point, the body effect coefficients γ_n and γ_p and the nominal threshold values are in general not matched between the two device types, so the net shift in V_M is non-zero and, more importantly, cell-dependent due to the different sizing ratios across the slicer bank. The FF corner produces the complementary effect. Reduced threshold voltages increase the drive strength of both device types, again shifting the trip points by amounts that vary across the bank. The FS and SF corners are qualitatively different in that they introduce an asymmetric perturbation. At FS the NMOS is faster than nominal while the PMOS is slower, which biases all slicers toward a lower trip point and compresses the effective decision levels toward the bottom of the input range. At SF the complementary imbalance expands the decision levels toward the top. In all four non-typical corners, the loss of linearity was found to cause DNL excursions beyond 0.5 LSB, which takes the converter outside its linearity target under these conditions.

3.7.4 Offline Trimming Strategy via Body Bias Adjustment

To restore the linearity of the thermometer coder across process corners without modifying the core circuit topology, an offline trimming strategy based on body bias adjustment was adopted. The correction is applied per slicer rather than as a single global setting. Each of the fifteen slicers has its own PMOS well and NMOS bulk terminal, and these are driven individually so that each decision threshold can be nudged back toward its nominal position.

The default condition, used at the typical corner, ties the PMOS wells to $V_{DD} = 0.8$ V and the NMOS bulks to ground, so that the source-to-bulk voltage of every device is zero and no body effect is present. A correction is then a deliberate departure from this default on a chosen slicer. The trimming is offline in the sense that these bulk voltages are determined once, from the identified process corner, and held constant during normal operation. No continuous adaptive feedback is involved.

3.7.4.1 Determination of the Per-Slicer Correction Voltages

For each of the four non-typical corners, a set of per-slicer bulk voltages was determined with the objective of bringing the DNL of the converted thermometer code below 0.5 LSB across the full input range. The degree to which each corner meets that target is reported in Chapter 4.

The values were obtained by extracting the trip point of every slicer at the corner, identifying which slicers had moved above or below their nominal positions, and forward-biasing the appropriate device (PMOS to pull a high trip back down, NMOS to push a low trip back up) until the spacing was restored. Only the slicers that needed correction were touched. The rest stayed at the default of Section 3.7.4. Because the four corners perturb the bank differently, both the set of adjusted slicers and the applied voltages differ from corner to corner, as Table 3.5 shows. Each vector was verified in simulation to reduce the worst-case DNL at its corner under the trip-point direction established above, with the detailed outcome reported in Chapter 4.

Table 3.5: Per-slicer body bias trim voltages applied at each process corner. A value in the PMOS column is the N-well voltage of that slicer, which forward-biases its PMOS and lowers its trip point. A value in the NMOS column is the bulk voltage of that slicer, which forward-biases its NMOS and raises its trip point. Each trimmed slicer is adjusted on one device only, so a given slicer index appears in at most one column. Slicers not listed keep the default PMOS well voltage of 0.8 V and NMOS bulk voltage of 0 V. The typical corner uses no trimming. The notation $k\!:\!V$ denotes slicer index k set to voltage V.

Corner	**PMOS trim, N-well (V)** **(lowers trip point)**	**NMOS trim, bulk (V)** **(raises trip point)**
TT	default (all 0.8)	default (all 0)
FF	6:0.5, 7:0.5, 8:0.7	0:0.3, 1:0.3, 10:0.2, 11:0.5, 12:0.5, 13:0.1
FS	0:0.75, 7:0.7	1:0.23, 4:0.1, 12:0.1, 14:0.1
SF	1:0.75, 2:0.7	3:0.1, 4:0.2, 5:0.2, 11:0.1, 12:0.15, 13:0.1, 14:0.3
SS	0:0.6, 9:0.75, 10:0.72	2:0.2, 3:0.15, 4:0.2, 5:0.1

3.7.4.2 Verilog-A Behavioural Implementation

The trimming was implemented in simulation using a Verilog-A behavioural model, listed in full in Appendix A. The model takes the process corner as a string parameter and, for the selected corner, drives the thirty bulk terminals of the slicer bank, fifteen PMOS wells and fifteen NMOS bulks, to the voltages of Table 3.5, leaving the untouched slicers at the 0.8 V and 0 V defaults. Each output behaves as an ideal voltage source connected to the corresponding bulk node, so the correction acts on the transistor-level netlist without requiring any modification to it. This keeps all five corners within the same simulation framework as the nominal design and, because the corner is a single parameter, makes it straightforward to extend the table to intermediate corners or to temperature-corner combinations should a finer correction granularity be required in future iterations.

3.7.5 Summary

This section turned the corner problem of the preceding architecture into a correction mechanism. The body effect gives an independent handle on each slicer's trip point. Forward-biasing a slicer's PMOS well lowers its trip point, and forward-biasing its NMOS bulk raises it. Body biasing was preferred over the other techniques of Section 2.7 because it adds no capacitance to the signal path and no clocked activity to the converter.

A per-slicer set of bulk voltages was determined for each process corner (Table 3.5) and applied through the Verilog-A behavioural model of Appendix A, which drives the bulk terminals of the slicer bank without touching the transistor-level netlist. The correction targets deterministic inter-corner shifts only, leaving random mismatch outside its reach. The effect of these corrections on the DNL and INL of the converter, together with the rest of the final characterisation, is presented in Chapter 4.

3.8 Analytical Modelling and Design Constraints

This section establishes the theoretical limits and design trade-offs of the asynchronous Flash ADC. In a synchronous architecture the performance limits follow mostly from the clock frequency. In a level-crossing converter they depend instead on the dynamics of the input signal itself, on the circuit propagation delays and on the noise.

3.8.1 Maximum Input Wave Frequency and Slope Overload

In synchronous ADCs the maximum input frequency follows from the Nyquist-Shannon sampling theorem ($f_{in} < f_s/2$). An asynchronous Flash or level-crossing ADC has no uniform sampling grid. Its limit is the maximum tracking speed of its internal circuitry, expressed as a slew rate (SR) constraint. When the input changes faster than the converter can track, the result is a distortion known as slope overload.

Consider a full-scale sinusoidal input voltage $V_{in}(t)$:

$$V_{in}(t) = \frac{V_{FS}}{2} \sin(2\pi f_{in} t) \tag{3.18}$$

where V_{FS} represents the full-scale voltage range of the converter and f_{in} is the input frequency. The time derivative of this signal yields its instantaneous slope, which achieves its maximum absolute value at the zero-crossing points:

$$SR_{max} = \left.\frac{dV_{in}(t)}{dt}\right|_{max} = \pi f_{in} V_{FS} \tag{3.19}$$

The asynchronous Flash ADC resolves the input signal by triggering a new state or quantisation conversion whenever the input crosses a discrete threshold voltage step Δ, corresponding to the nominal resolution of N bits:

$$\Delta = \frac{V_{FS}}{2^N} \tag{3.20}$$

Even in an ideal, noiseless scenario, each conversion takes a finite minimum time. The input follower must settle, the slicer associated with the crossed threshold must switch, and the event-generation logic must propagate the result. The sum of these contributions is the total conversion propagation delay τ_{conv}. Consequently, the maximum rate at which the ADC can track voltage transitions (the converter's intrinsic Slew Rate, SR_{ADC}) is bounded by:

$$SR_{ADC} = \frac{\Delta}{\tau_{conv}} = \frac{V_{FS}}{2^N \tau_{conv}} \tag{3.21}$$

where $\Delta = V_{FS}/2^N$ is the quantisation step of one LSB, V_{FS} the full-scale input range, N the resolution and τ_{conv} the conversion delay defined above.

To surpass slope overload distortion, the maximum slope of the input signal must never exceed the tracking capability of the asynchronous circuitry ($SR_{max} \leq SR_{ADC}$):

$$\pi f_{in} V_{FS} \leq \frac{V_{FS}}{2^N \tau_{conv}} \tag{3.22}$$

where the left-hand side $\pi f_{in} V_{FS}$ is the peak slope of a full-scale sinusoid of frequency f_{in}, the most demanding input the converter has to track.

Rearranging this inequality yields the absolute maximum input frequency $f_{in,max}$ that the converter can digitise without undergoing slope overload under ideal conditions:

$$f_{in,max} = \frac{1}{\pi 2^N \tau_{conv}} \tag{3.23}$$

Transient noise lowers this limit considerably. Superimposed thermal and substrate noise produces rapid local voltage fluctuations (dV_{noise}/dt) that inflate the effective slope of the signal and make the voltage bounce back and forth across a quantisation threshold. These bounces consume the processing window τ_{conv} and appear at the output as thermometer-code metastability, the so-called bubble errors. The practical maximum slew rate under noise is therefore well below the ideal value. Slope overload appears at lower input frequencies and the measured bubble rate rises.

3.8.2 Resolution and Power Dissipation

The signal-dependent power profile is one of the main advantages of asynchronous data converters. In a standard synchronous Flash ADC the power consumption is roughly constant. The whole comparator bank ($2^N - 1$ devices) and the clock distribution network toggle at the sampling frequency f_s regardless of what the input is doing. In an asynchronous implementation, power dissipation follows the rate of change of the input.

When the input signal is entirely static ($f_{in} = 0$), no quantisation thresholds are crossed. Under this condition, the dynamic switching power ideally drops to zero, and the system only dissipates static power (P_{static}), which originates from the continuous bias currents in the analog references, pre-amplifiers, and comparators:

$$P_{total} = P_{static} + P_{dynamic} \tag{3.24}$$

In the implemented converter this zero-power condition is a theoretical limit rather than a measurable operating point. The complementary follower is continuously biased, so its input pairs always conduct a residual bias current, and the measured consumption at idle settles at a non-zero floor set by P_{static} instead of falling to the ideal zero.

For a dynamic input, such as a full-scale sinusoid at frequency f_{in}, the signal traverses the full-scale range V_{FS} twice per period $T = 1/f_{in}$ (one upward and one downward excursion across the 2^N levels in each half). The average number of quantisation level crossings per second, which can be read as an effective average sampling frequency $f_{s,avg}$, is therefore proportional to both the frequency and the resolution:

$$f_{s,avg} = 2 \cdot 2^N \cdot f_{in} = 2^{N+1} f_{in} \tag{3.25}$$

Assuming that each individual asynchronous conversion cycle draws an average switching energy E_{conv} from the power supply V_{DD}, the dynamic power consumption $P_{dynamic}$ can be modelled analytically as:

$$P_{dynamic} = E_{conv} \cdot f_{s,avg} = E_{conv} \cdot 2^{N+1} f_{in} \tag{3.26}$$

Substituting this back into the total power equation yields:

$$P_{total} = P_{static} + E_{conv} \cdot 2^{N+1} f_{in} \tag{3.27}$$

Equation (3.27) shows that $P_{dynamic}$ grows linearly with the signal frequency f_{in} but exponentially with the nominal resolution N.

In practical circuit implementations, the physical performance is constrained by noise, which splits the design analysis into nominal resolution (N) and Effective Number of Bits ($ENOB$), the latter derived directly from the Signal-to-Noise-and-Distortion Ratio ($SNDR$ or $SINAD$) as defined in Chapter 2.

Transient noise degrades the Signal-to-Noise Ratio (SNR) and the $SNDR$, and with them the $ENOB$. It also introduces spurious, non-monotonic threshold crossings (chatter), and each of these noise-driven crossings triggers a redundant conversion cycle that encodes no useful information. The bubble rate rises and $P_{dynamic}$ grows well beyond the ideal prediction. Power goes up while $ENOB$ goes down, so the Walden Figure of Merit (FoM_W) suffers twice over. Keeping the noise-induced bubble rate low is therefore a first-order design constraint for energy efficiency.

3.9 Chapter Summary

This chapter established the architecture of the proposed asynchronous Flash ADC and the theory that bounds its performance. The converter replaces clock-bound sampling with signal-dependent quantisation. With no global clock, it avoids the energy that synchronous architectures waste on sparse or bursty signals.

The structural development progressed through a sequence of architectures. A current-mirror Flash configuration adopted from the literature came first. A current-mirror implementation of the complementary dual-input extension followed, rejected for exceeding the voltage headroom available at 0.8 V. A bulk-driven amplifier was then tried for its rail-to-rail reach at low voltage but discarded as too sensitive to process corners. The final voltage-mode solution uses a complementary pair of five-transistor OTAs configured as a unity-gain follower, driving a bank of ratioed CMOS slicers whose switching thresholds are set purely by transistor sizing. The event-generation network described in Section 3.6.5 completes the conversion chain, detecting the transitions of the thermometer outputs and encoding them combinationally into the binary output word, with no clock involved at any point in the signal path.

The results presented along the way told a consistent story. The first approach validated the conversion principle but not the input range. The second could not be biased within the supply. The third recovers the headroom (equation (3.11)), is linear at the typical corner, and loses its linearity at all four non-typical process corners (Table 3.4). That last result is the brief for Section 3.7.

Finally, the analytical models bounded the design space. The maximum input frequency is set not by a Nyquist limit but by the slew rate constraint SR_{ADC}, in which the propagation delay τ_{conv} determines the onset of slope overload. The power analysis showed that dynamic consumption grows linearly with the input frequency f_{in} but exponentially with the nominal resolution N. The same models quantify the cost of transient noise, degraded SNR and $SNDR$, bubble errors, and redundant conversion cycles that raise the power while the effective resolution ($ENOB$) falls, to the detriment of the FoM_W. These results are the baseline for the simulation setup and the performance evaluation of the following chapters.

Chapter 4

Simulation Results and Performance Benchmarking

The previous chapters left two claims to settle. One is that the trimming of Section 3.7 restores the linearity lost at the process corners. The other is that removing the clock pays off in power against the synchronous reference of Section 2.4. This chapter settles both. It describes the simulation and reconstruction methodology, characterises the asynchronous converter statically and dynamically, measures the effect of the trimming, quantifies the impact of random mismatch, and closes with the power measurements and the comparison between the two converters under a matched workload.

4.1 Simulation Setup and Input Test Signals

To validate the theoretical framework of Chapter 3 and evaluate the physical performance of the proposed asynchronous Flash ADC, transistor-level simulations were run in a 16-nm CMOS process at a nominal supply of 0.8 V. This section describes the simulation environment, the reconstruction methodology required for non-uniform asynchronous data, and the input test stimuli.

Two families of stimuli were used. Static linearity was measured with a trapezoidal pulse spanning 0.1 V to 0.7 V, with equal rise and fall ramps swept from 35 ns down to 5 ns and a fixed 20 ns hold between them. Dynamic performance was measured with a sinusoid at $f_{in} = 9$ MHz, swinging rail to rail across the full 0–0.8 V supply (amplitude 0.4 V around a 0.4 V common mode). The converter quantises the central 0.1–0.7 V band defined by the fifteen slicer thresholds (Section 4.2.1), with the end codes absorbing the portion of the swing beyond it. The simulation ran for 200 μs with transient noise enabled.

In a synchronous converter the output samples sit on a periodic clock grid. The asynchronous level-crossing Flash ADC instead produces output transitions only when a quantisation threshold is crossed, so a standard Fast Fourier Transform (FFT), which requires uniformly sampled records, cannot be applied directly to the raw output. The reconstruction used here is deliberately simple. The fifteen digital outputs and the input are linearly interpolated onto a uniform analysis grid of 180 MHz, each interpolated output is compared against the logic threshold of 0.4 V to recover the thermometer bits, the output code is the count of asserted bits, and the equivalent analog value follows from mapping that code back onto the 0.1–0.7 V quantisation range of the fifteen slicer thresholds, the end codes absorbing the part of the rail-to-rail swing that falls outside that band.

The analysis grid is twenty times the input frequency and coherent with it ($N = 36000$ samples place the fundamental exactly on bin 1800), so the FFT uses a rectangular window and no spectral leakage correction is needed. The Signal-to-Noise Ratio (SNR), the Signal-to-Noise-and-Distortion Ratio (SINAD), the Effective Number of Bits (ENOB), the Total Harmonic Distortion (THD, first seven harmonics, aliased into the first Nyquist zone where necessary) and the Spurious-Free Dynamic Range (SFDR) are extracted from this spectrum.

The same resampled record defines the bubble error rate used throughout this chapter. A sample exhibits a bubble error when its fifteen-bit thermometer pattern contains more than one $0 \leftrightarrow 1$ transition, and the bubble rate is the fraction of analysis samples for which this happens.

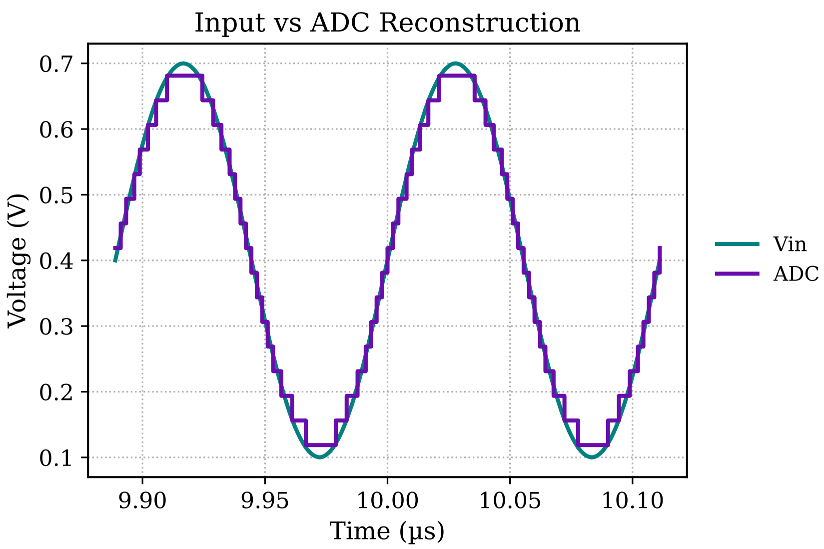


Figure 4.1: Input sinusoid (Vin) and reconstructed converter output (ADC) over two input periods of the dynamic test.

Figure 4.1 shows the input sinusoid against the reconstructed output over a representative interval of the test, and Figure 4.2 shows the spectrum of the reconstructed output, with the fundamental signal bin, the noise floor and the harmonic distortion components.

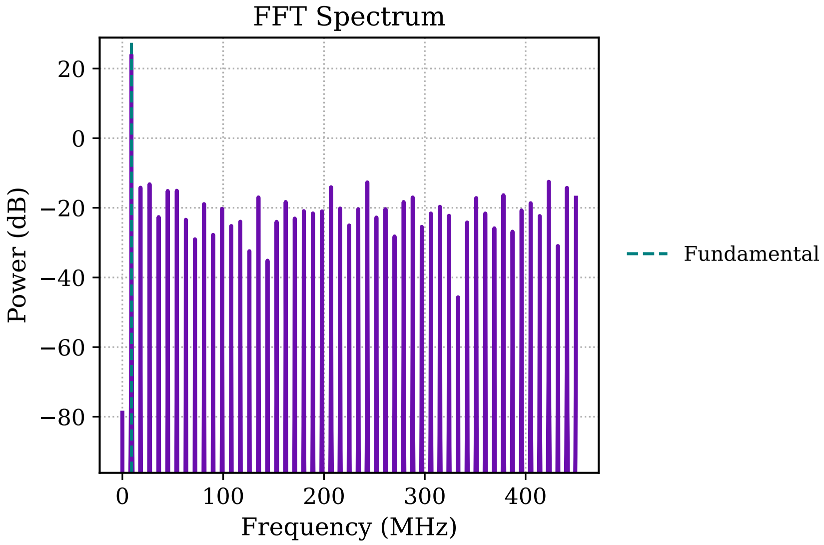


Figure 4.2: FFT spectrum of the reconstructed asynchronous Flash ADC output signal.

The choice of 9 MHz follows from the converter's measured tracking limit. Driven rail to rail, the 9 MHz sinusoid (amplitude 0.4 V) has a peak slew rate of $2\pi f_{in}A \approx 22.5$ V/µs at mid-scale, the value returned by the simulator, and the converter digitises the whole record without a single bubble error. That bubble-free result is the direct evidence that 9 MHz lies inside the tracking envelope. The hysteresis analysis of Section 4.2.1 provides a more conservative, static bound. The rising and falling thresholds separate by more than 1 LSB once the ramp slew passes roughly 20 V/µs, which at 9 MHz contributes the small amount of hysteresis-induced distortion already reflected in the ENOB sitting just above the nominal four bits, rather than producing bubble errors. A 20 MHz sinusoid would more than double the peak slew rate, to ≈ 50 V/µs, deep into slope overload, so 9 MHz is, in practice, the full-scale sinusoidal bandwidth of the converter.

Table 4.1 summarises the dynamic metrics extracted from the 9 MHz transient noise test. The SINAD of 26.50 dB corresponds to an ENOB of 4.11 bits, marginally above the nominal 4-bit resolution. This is consistent with the level-crossing property discussed in Chapter 2, whereby the absence of a uniform sampling clock removes the conventional quantisation noise floor and can push the SNDR past the limit of a clocked converter with the same nominal resolution. No bubble errors were observed over the 36000 analysis samples of this test.

This figure should be read with two qualifications. It is a single operating point, one input frequency and one transient-noise seed, so the 0.11 bit margin above four is an indicative result rather than a statistically averaged one, and a sweep over frequency, amplitude and noise seeds is left as a more complete dynamic characterisation. Part of that fractional bit may also originate in the reconstruction itself, since the linear interpolation onto the 180 MHz analysis grid (Section 4.1) smooths the level-crossing record before the FFT and can flatter the measured SINAD slightly. The conservative reading is therefore that the converter achieves essentially its nominal 4-bit dynamic resolution, with the level-crossing mechanism accounting for the small surplus.

Table 4.1: Dynamic performance of the asynchronous converter ($f_{in} = 9$ MHz, full-scale input, with transient noise).

Metric	**Value**	**Unit**
SINAD	26.50	dB
ENOB	4.11	bit
SNR	31.55	dB
THD	−28.13	dB
SFDR	30.63	dB
Bubble rate	0.00	%

4.2 Static Linearity Results

The static linearity metrics, Differential Non-Linearity (DNL) and Integral Non-Linearity (INL), quantify the deterministic mismatches of the architecture and the circuit. The proposed asynchronous Flash ADC contains no reference ladder, so its DNL and INL are set by the deviations of the slicer trip points from their nominal sizing-defined positions. Those deviations are the deterministic, corner-dependent shifts analysed in Section 3.7, plus the random threshold variations of the slicer bank.

4.2.1 Threshold Extraction and Linearity Versus Conversion Rate

The fifteen decision thresholds were measured with the trapezoidal pulse described in Section 4.1. For each slicer, the input voltage at which its digital output crosses the 0.4 V logic threshold is recorded by linear interpolation, once on the rising ramp and once on the falling ramp. The difference between the two readings is the input-referred hysteresis of that slicer, and their average is taken as its decision threshold. The DNL of each of the sixteen codes follows from the widths between consecutive sorted thresholds relative to the ideal LSB of 37.5 mV, the INL of each transition from the deviation of its threshold from the ideal uniform level, and the offset from the mean of those deviations. Since the widths of the two end codes depend directly on the chosen range limits, the DNL figures reported below exclude them and cover codes 1 to 14.

The hysteresis is not a designed property of the slicers. It is the input-referred signature of the conversion delay τ_{conv} modelled in Section 3.8.1. While the input ramps, every delay in the follower and the slicer translates into a voltage error proportional to the ramp rate, so the rising and falling readings separate as the ramp gets steeper. A measurement is considered valid when the worst-case hysteresis stays below 1 LSB. Table 4.2 reports the static metrics for ramp times from 35 ns down to 5 ns, with the equivalent conversion rate computed from the active ramp time (and, in parentheses in the discussion below, with the 20 ns hold included).

The data confirms the behaviour predicted by the slew rate model. The DNL, INL and offset stay essentially flat across the sweep (all fifteen thresholds were found in every run, with no

Table 4.2: Static linearity and hysteresis of the asynchronous converter versus input ramp time. DNL excludes the two end codes. Valid means worst-case hysteresis below 1 LSB.

Ramp (ns)	**Rate** (MHz)	**Offset** (LSB)	**Max |DNL|** (LSB)	**Max |INL|** (LSB)	**Max |hyst.|** (LSB)	**Valid**
35	14.3	−0.012	0.412	0.518	0.959	Yes
30	16.7	0.019	0.403	0.486	1.049	No
25	20.0	0.010	0.315	0.453	1.163	No
10	50.0	−0.046	0.456	0.351	2.494	No
5	100.0	−0.154	0.350	0.937	4.873	No

missing codes), but the hysteresis grows roughly in proportion to the ramp rate, from 0.96 LSB at 35 ns to 4.87 LSB at 5 ns. The hysteresis is therefore the binding constraint on speed. The fastest ramp that meets the 1 LSB criterion is 35 ns, which corresponds to a maximum conversion rate of 14.3 MHz over the active ramps.

4.2.2 Linearity Across Process Corners and Trimming

These non-idealities are corrected by the body bias trimming technique implemented through the Verilog-A behavioural framework of Section 3.7. To measure its effect, the static error profiles were extracted before and after applying the trimming vectors at each process corner, separately on the rising and on the falling ramp. The untrimmed behaviour was established in Section 3.6.6: linear at the typical corner, out of specification at all four non-typical corners. Table 4.3 reports the worst-case errors at those corners, before and after trimming.

Table 4.3: Worst-case static linearity per process corner, before and after body bias trimming, measured on the rising and falling ramps (values in LSB).

	DNL rise		**DNL fall**		**INL rise**		**INL fall**	
Corner	Before	After	Before	After	Before	After	Before	After
SS	0.6958	0.4882	0.6628	0.5043	0.9489	0.1905	1.0949	0.0213
FF	0.3814	0.4008	0.6372	0.3636	0.6670	0.7435	1.3724	0.3514
FS	0.3814	0.4979	0.6372	0.5859	0.6670	0.6614	1.3724	0.9807
SF	0.5191	0.3414	0.5142	0.4700	0.5829	1.0261	1.0787	1.4333

The correction is effective but not uniform across corners. The worst-case DNL improves at all four corners, falling below the 0.5 LSB target at SS, FF and SF and remaining marginally above it at FS (0.586 LSB on the falling ramp). The INL gains are largest where the untrimmed errors were worst. At SS the falling-ramp INL drops from 1.09 to 0.02 LSB, and at FF from 1.37 to 0.35 LSB. The SF corner is the exception. Its per-slicer trim, chosen to meet the DNL target, trades INL for DNL, so while the DNL improves, the worst-case INL degrades from 1.08 to 1.43 LSB. Even with per-slicer control, a correction indexed only to the process corner and limited to forward body bias cannot satisfy both metrics at once when the SF perturbation pulls them in opposite directions. The per-chip calibration discussed in Chapter 5, which measures and corrects each slicer individually, would relax that constraint.

4.3 Random Mismatch: Monte Carlo Analysis

The corner-indexed trimming of Section 3.7 corrects deterministic shifts only. Random intra-die mismatch, the device-to-device variation that remains even at a fixed process corner, is left uncorrected by design. To measure how much linearity this costs, a 300-run Monte Carlo simulation was run on two adjacent slicer outputs, out_0 and out_1.

Two quantities are central to what follows. The trip point, or threshold, of a slicer is the input voltage at which its output crosses the 0.4 V logic level. The threshold spacing is the gap between the trip points of the two adjacent slicers, which is exactly the width of the output code that lies between them. The nominal spacing is the ideal value of that gap, one LSB, equal to 37.5 mV in the transient window. In each run the two thresholds were extracted and their spacing recorded, and the deviation of that spacing from the nominal one LSB is the random variable studied below.

The spacing is used in place of a mean DNL because it maps directly onto the health of the code. A spacing near the nominal LSB gives a well-formed code, a spacing shrinking toward zero gives a missing code, and a spacing that turns negative means the two thresholds have swapped order, which is an arbitrarily large negative DNL. A mean DNL would be dominated by these inversions and lose physical meaning, so the spread is reported through the spacing distribution and the fraction of runs in each of these regimes.

Table 4.4: Monte Carlo statistics of the threshold spacing between two adjacent slicers (out_0, out_1), without trimming. The spacing is the code width separating the two thresholds, deviations are referred to the ideal LSB of 37.5 mV.

Quantity	**Value**	**Unit**
Monte Carlo runs	300	–
Threshold spacing standard deviation	215.6	mV
Threshold spacing standard deviation (in LSB)	5.75	LSB
Largest spacing deviation (ordering inversion)	15.18	LSB
Runs free of missing code (spacing > 0)	42.5	%
Runs within ± 0.5 LSB of nominal spacing	7.6	%
Runs within ± 1.0 LSB of nominal spacing	16.0	%

The standard deviation of the threshold spacing is 215.6 mV, or 5.75 LSB, almost six times the ideal LSB of 37.5 mV. Fewer than half of the runs (42.5%) keep the two thresholds in their nominal order with no missing code between them, and fewer than one run in ten holds the spacing within ± 0.5 LSB of the nominal value. The histogram of Figure 4.3 shows the distribution over the 300 runs, where the collapsed and inverted spacings form the strongly negative DNL tail.

The size of this spread follows from Pelgrom's law (Section 2.7). The slicers use the small integer width ratios of Table 3.3, with PMOS and NMOS widths from 2 to 23 unit transistors at equal channel length, so their device area is small and the threshold mismatch $\sigma_{V_{th}} = A_{V_{th}}/\sqrt{WL}$ is correspondingly large, where $\sigma_{V_{th}}$ is the standard deviation of the threshold voltage, $A_{V_{th}}$ the Pelgrom mismatch coefficient of the process and W and L the transistor width and length. The spacing between two adjacent slicers carries the mismatch of two independent thresholds, so its variance is the sum of the two device variances, which is how a nominally one-LSB code ends up with a multi-LSB standard deviation.

Random threshold mismatch, not corner variation, is therefore the dominant linearity hazard of the slicer bank. This confirms the limitation acknowledged in Chapter 5 and puts numbers behind the per-slicer foreground calibration proposed there as future work.

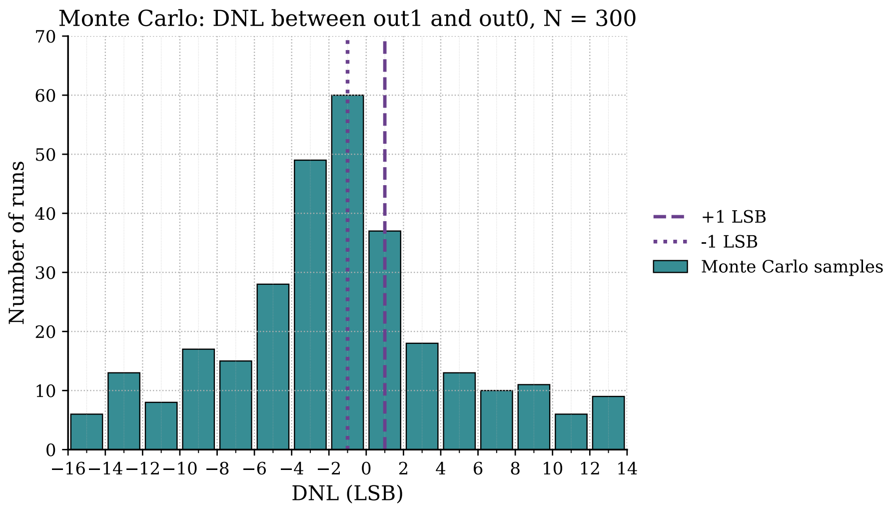


Figure 4.3: Monte Carlo distribution of the DNL of the code between two adjacent slicers (out_0 and out_1), with the ± 1 LSB limits marked.

Figure 4.4 checks whether that spread is Gaussian. The threshold spacing deviation is plotted against the quantiles of a normal distribution, and the points follow a straight line over almost the whole range, with a coefficient of determination of $R^2 = 0.982$ and a slope of about 5.7 LSB that recovers the 5.75 LSB standard deviation of Table 4.4. The spacing is therefore close to normally distributed, as expected for the sum of two independent and nominally Gaussian threshold mismatches. The target lines at $\pm 2.94\sigma$ mark the range a 300-run sample is expected to cover. For N runs the most extreme sample falls near the normal quantile of $p = 1 - 1/(2N)$, and for $N = 300$ this gives $p \approx 0.9983$ and a z-score of 2.94, so staying within specification across these runs asks the spacing to hold inside roughly three standard deviations, which the multi-LSB spread of Table 4.4 clearly does not.

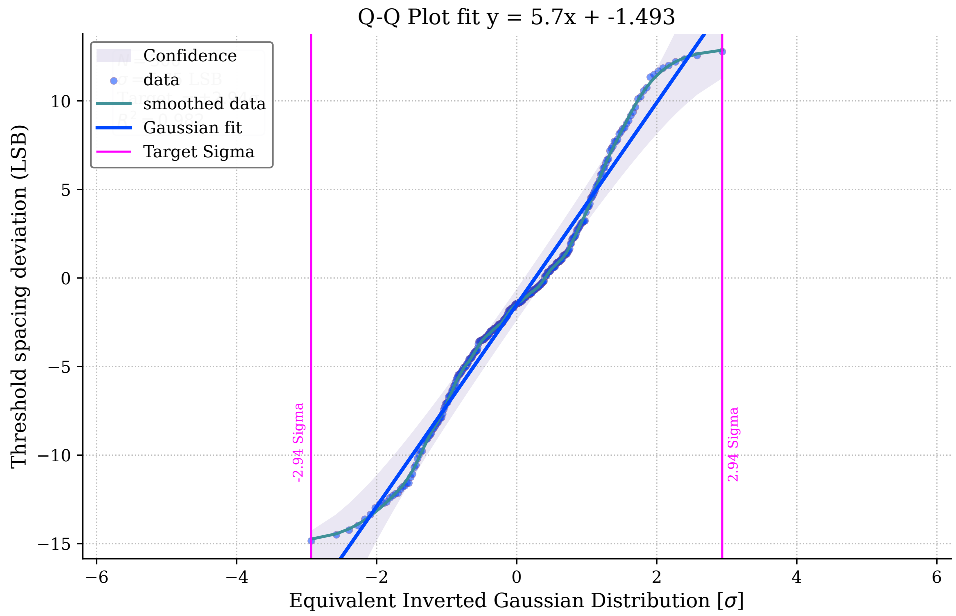


Figure 4.4: Quantile-quantile plot of the Monte Carlo threshold spacing deviation against a normal distribution, for the 300 runs of Table 4.4. The near-linear fit ($R^2 = 0.982$, slope ≈ 5.7 LSB) shows the spacing is close to Gaussian. The target lines at $\pm 2.94\sigma$ are the expected extreme of a 300-sample draw, the normal quantile of $p = 1 - 1/(2N)$ with $N = 300$.

4.4 Power Dissipation and Energy Efficiency

A defining feature of the asynchronous Flash ADC is its signal-dependent power profile. In direct contrast to traditional synchronous architectures that exhibit a static power baseline due to continuous clock toggling, the dynamic power of the proposed converter is governed strictly by the frequency of threshold-crossing events.

4.4.1 Power Measurement Methodology

To keep the comparison between the two converters fair, the power figures reported here cover only the core conversion path of each design. The encoder, that is, the digital logic that translates the thermometer code into the final binary word, was excluded from all power measurements on both sides of the comparison.

The reasoning is straightforward, both designs require some form of thermometer-to-binary conversion at their outputs, and this encoding logic is largely independent of whether the preceding conversion is synchronous or asynchronous. Including it on one side of the comparison but not the other would artificially inflate one power budget and lead to a conclusion that does not fairly reflect the underlying architectural trade-off. The most defensible approach is therefore to exclude it from both and compare only the converter cores.

Under this methodology, the power reported for the synchronous reference in Table 4.7 includes the resistor ladder, the 15 StrongARM latch comparators, the ring oscillator and the tapered clock buffer chain described in Section 2.4. The clock distribution path is precisely the overhead that the asynchronous design eliminates. The power reported for the proposed asynchronous converter includes the voltage follower, the slicer bank and the event-generation logic of Chapter 3.

4.4.2 Signal-Dependent Power Scaling

Driven by the full-scale 9 MHz sinusoid swinging rail to rail across the 0–0.8 V supply, the converter draws an average of 99.5 µW, the instantaneous consumption peaking at 108 µW around the mid-scale crossing, where the level-crossing rate is highest. With the input idle, it draws 94.8 µW. The idle figure is the static floor anticipated in Section 3.8.2: the complementary follower is continuously biased, so its input pairs conduct a residual class-A bias current even when no threshold is being crossed. The difference between the active average and the idle floor, 4.7 µW, is the activity-dependent component of the consumption.

Energy efficiency is assessed with the standard Walden Figure of Merit, applied as in the level-crossing literature of Chapter 2, with the rate taken as twice the signal bandwidth B:

$$\mathrm{FoM}_W = \frac{P}{2^{\mathrm{ENOB}} \cdot 2B} \tag{4.1}$$

where P is the average power drawn by the converter, ENOB its effective number of bits and B the signal bandwidth. The signal bandwidth of the proposed converter is its full-scale sinusoidal tracking limit, $B = 9$ MHz (Section 4.1). At that operating point, with $P = 99.5$ µW and an ENOB of 4.11, equation (4.1) gives $\mathrm{FoM}_W \approx 320$ fJ/conv.-step. As in the level-crossing references, the figure of merit improves for sparse inputs, not through the denominator but through the power term, as the converter approaches its 94.8 µW idle floor.

The measured power figures are summarised in Table 4.5.

Table 4.5: Measured power consumption and energy efficiency of the asynchronous converter.

Performance Parameter	**Value**	**Unit**
Power consumption, input active	99.5	μW
Power consumption, input idle	94.8	μW
Activity-dependent component	4.7	μW
FoM_W ($B = 9$ MHz)	320	fJ/conv.-step

4.4.3 Average Power Under Duty-Cycled Operation

The two power figures of Table 4.5 are the end points of a continuous relationship. The average power of the converter interpolates between the idle floor and the full-activity value through an activity factor α,

$$P_{avg}(\alpha) = P_{idle} + (P_{active} - P_{idle})\,\alpha = 94.8 + 4.7\,\alpha \quad [\mu\text{W}]. \tag{4.2}$$

where α is the activity factor, a dimensionless number between 0 and 1 that measures how busy the input keeps the converter, and P_{idle} and P_{active} are the idle and full-activity powers. The activity factor is the rate at which the input crosses quantisation thresholds, normalised to the rate reached at full activity,

$$\alpha = \frac{f_{cross}}{f_{cross,max}}, \tag{4.3}$$

where f_{cross} is the number of threshold crossings per second produced by the input, the effective sampling rate $f_{s,avg}$ defined in Section 3.8.2, and $f_{cross,max}$ is that same rate under the full-scale 9 MHz sinusoid that defines P_{active}, the 288 MHz reached in Section 4.5. Dividing one rate by the other cancels the units of hertz and fixes the scale, so $\alpha = 0$ is a quiet input with no crossings, where the converter rests at P_{idle}, and $\alpha = 1$ is the full-activity input, where it draws P_{active}. When the input simply switches between this full-activity state and silence, α is also the fraction of time spent active.

As a simple example, consider a window of 350 ns. At the highest input frequency the converter handles, 9 MHz, one cycle of the signal lasts about 111 ns, so at most three full cycles of activity fit inside that window. An input that goes through only one cycle in the same 350 ns is active at a third of the full rate, so $\alpha = 1/3$, and equation (4.2) gives an average power of $94.8 + 4.7/3 \approx 96.4$ µW.

The synchronous reference has no such dependence. Its clock and comparator bank switch every cycle regardless of the input, so its power stays at 886 µW for any α. The ratio between the two is $886/(94.8 + 4.7\alpha)$, which is 8.9 at full activity and rises to 9.3 as the input becomes sparse. The advantage of the asynchronous converter is therefore largest exactly where the target applications operate, at low activity. The asynchronous power rises gently with activity against the flat line of the clocked reference.

The nature of this idle figure should be stated plainly. At 94.8 µW the floor is 95% of the active power, so the converter is not a near-zero-power part when the input is quiet. What falls to zero is the dynamic, activity-dependent component, while the static bias of the complementary follower remains and dominates the budget. The benefit demonstrated here is therefore twofold and bounded. The clock-distribution network is removed entirely, and the remaining 4.7 µW of dynamic power scales down with input activity. Collapsing the floor itself would require duty-cycling the follower bias during idle intervals, a power-gating extension left for future work. Without it, the 94.8 µW bias current sets the practical lower bound on the converter's average power.

This activity-dependent power, rather than a per-conversion figure of merit, is the appropriate way to compare a clockless converter with a clocked one, as established by the continuous-time system of Schell and Tsividis, whose power fell from 800 µW at full activity to a 250 µW quiescent level driven entirely by the input [117]. The Walden FoM, normalised to a fixed bandwidth, is blind to this scaling by construction.

4.5 Synchronous Reference Flash ADC Results

The synchronous reference, described in Section 2.4, was simulated under the same 16-nm, 0.8 V conditions as the proposed design and characterised for both dynamic performance and power, following the power measurement methodology of Section 4.4.1. Driven by the same 9 MHz full-scale sinusoid sampled at 288 MS/s, it reaches an SNDR of 24.98 dB and an ENOB of 3.86 bits, exercising all sixteen output codes with no missing levels. It is therefore a competent 4-bit Flash converter, not a weakened reference chosen to flatter the comparison. Its main role here is to quantify the power that a clocked Flash spends on this conversion workload, where it draws 886 µW. Its static linearity is shown in Figure 4.5, with the measured transfer function against the ideal staircase, the DNL per code and the INL per transition.

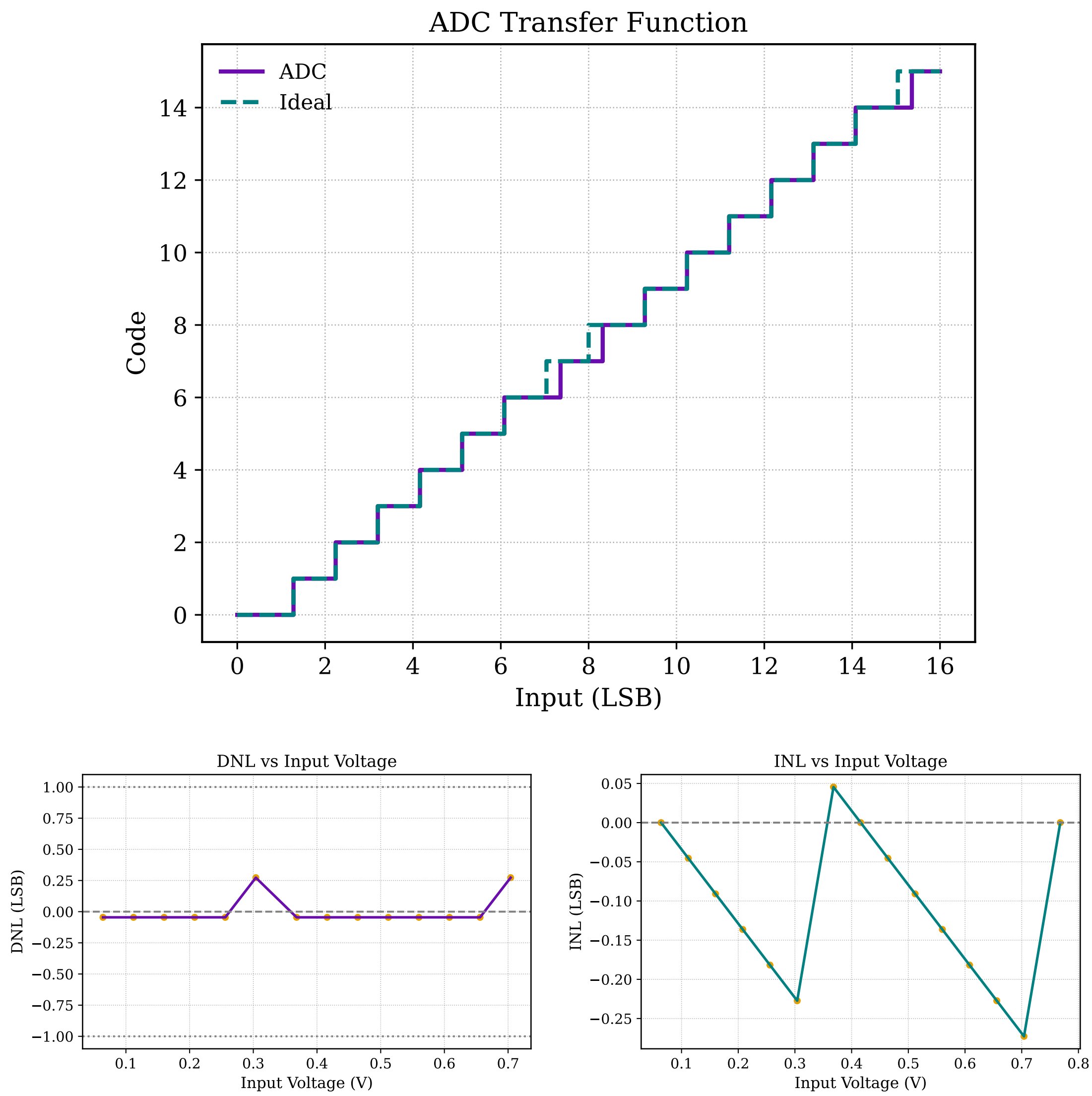


Figure 4.5: Static characterisation of the synchronous reference: (top) measured transfer function against the ideal staircase, (bottom left) DNL per code and (bottom right) INL per transition.

Neither operating point comes from the original publication. The design of [38] was demonstrated at 20 MS/s with a 332 kHz input in 65 nm, and reproducing that point here would compare the two converters in a regime unrelated to the one the asynchronous design targets. The matched point follows instead from the activity model of Section 3.8.2 and from the slew rate envelope of Section 4.1. The fastest full-scale sinusoid the asynchronous converter tracks cleanly is 9 MHz, and at that input a 4-bit level-crossing converter generates $f_{s,avg} = 2^{N+1} f_{in} = 32 \times 9$ MHz $= 288$ M crossings per second, which is the full-activity rate $f_{cross,max}$ of the activity factor in Section 4.4.3. Clocked at 288 MS/s with the same 9 MHz stimulus, the synchronous reference performs the same number of conversions per second, on the same input, as the asynchronous converter in the test of Section 4.1. The power figures of the two converters therefore describe the same workload. The performance summary is presented in Table 4.7, and the power is separated into its components in

Table 4.6. Together they form the baseline for the comparison of the following section.

The power consumption of the synchronous reference is not a single block. In any clocked Flash converter the clock must physically reach all fifteen comparators, so a converter of this kind always carries a clock generator and a distribution network between the generator and the array. That routing belongs to the architecture itself, and any honest power figure for a synchronous Flash has to include it. As described in Section 2.4, the distribution was modelled with a tapered buffer chain whose final $\times 24$ fanout stage represents the load of a normal, physically compact route to the comparator bank. Table 4.6 separates the measured consumption into the converter core, the ring oscillator and the clock routing. The complete set draws 886 µW, of which the ring oscillator and the clock routing account for 447 µW. That 447 µW is the clock overhead the asynchronous design removes entirely, and on its own it exceeds the full 99.5 µW of the asynchronous core by more than four times.

The routing term depends on the modelled load, so the headline ratio carries a corresponding sensitivity that should be stated openly. The 322 µW of clock routing follows from the $\times 24$ final buffer chosen in Section 2.4 to represent a compact on-chip clock route, this is a schematic-level estimate, not a value extracted from a physical layout. Since the dynamic power of that stage scales roughly with the load it drives, halving the final fanout to $\times 12$ lowers the routing to about 161 µW and the complete synchronous figure to roughly 725 µW, while doubling it to $\times 48$ raises them to about 644 and 1210 µW. The power ratio against the 99.5 µW asynchronous core therefore spans approximately $7\times$ to $12\times$ across this range, with the $8.9\times$ of the nominal $\times 24$ model sitting in the middle. The qualitative conclusion does not depend on the exact buffer size, since the entire clock-distribution term is removed by the asynchronous architecture regardless of how it is sized, only the precise multiplier moves with the assumption.

Table 4.6: Power breakdown of the synchronous reference Flash ADC.

Block	**Power**	**Unit**
ADC core (comparators and reference ladder)	439	μW
Ring oscillator	125	μW
Clock routing (buffer chain, $\times 24$ fanout)	322	μW
Complete converter	886	μW

4.6 Comparative Analysis and Discussion

Table 4.8 puts the developed design in context. It compares the core performance metrics of the synchronous reference Flash ADC of Section 2.4, the proposed asynchronous Flash ADC, and recent state-of-the-art architectures from the literature.

Table 4.7: Simulated performance summary of the synchronous reference Flash ADC in 16-nm CMOS.

Parameter	**Value**	**Unit**
Supply Voltage	0.8	V
Resolution	4	bit
Sampling Frequency	288	MS/s
Input Range	0–0.8	V
Power Consumption	886	μW
Dynamic Performance ($f_{in} = 9$ MHz, full-scale)		
SNR	25.60	dB
SINAD / SNDR	24.98	dB
ENOB	3.86	bit
THD	-30.05	dB
SFDR	33.07	dB

Table 4.8: Performance comparison with the synchronous baseline and state-of-the-art literature.

Parameter	**Sync. base.**	**This Work**	**[117]**	**[150]**	**[135]**
Technology (nm)	16	16	90	130	65
Supply (V)	0.8	0.8	1.0	0.8	1.2
Resolution (bit)	4	4	8	8	10
Signal BW (MHz)	9	9	0.02	0.02	0.02
f_s (MS/s)	288	event	event	event	event
SNDR (dB)	24.98	26.50	47–62	47–54	64
ENOB (bit)	3.86	4.11	7.51–10.01	7.5–8.7	10.3
Power (μW)	886	99.5	250–800	3–9	0.066–2.8
FoM$_W$ (fJ/c.s.)	3390	320	193986–6858200	180–1240[2]	1.8–65
Architecture	Sync. Flash	Async. LC Flash	CT system	LC-ADC	CT LC-ADC

[1] Reference [117] is a complete continuous-time ADC/DSP/DAC system. [2] Figure of merit not reported in the original work, computed here from the reported power, ENOB and signal bandwidth.

The comparison makes the trade-off concrete. On the same 9 MHz full-scale stimulus and at a matched conversions-per-second workload (Section 4.5), the complete synchronous converter draws 886 µW against 99.5 µW for the active asynchronous core, a factor of nearly nine. A substantial part of the synchronous budget goes to clock generation and distribution, blocks that do not exist in the asynchronous design.

The two converters are close on dynamic performance, the asynchronous reaching 26.50 dB SINAD against 24.98 dB for the synchronous reference, a difference of about 1.5 dB that is best read as near-parity at the 4-bit level rather than as a decisive advantage for either. One asymmetry deserves comment, since the two converters do not digitise the same band. The synchronous ladder spans the full 0–0.8 V rail to rail, whereas the asynchronous converter quantises the central 0.1–0.7 V band, with the two end codes absorbing the portion of the rail-to-rail swing that falls outside it (Section 4.1). Those end codes are wide quantisation bins of about 0.1 V each, mapped on reconstruction to a representative level, so the part of the sinusoid beyond the active band is coarsely quantised rather than hard-clipped against a rail, the resulting distortion is bounded and shows up in the THD and SFDR of Table 4.1 without collapsing the SINAD. Inside the active band the asynchronous LSB is finer than the synchronous one (37.5 mV against 50 mV), which lowers the quantisation error over the central part of the swing and offsets the coarser end regions. The net effect is the rough dynamic parity observed, and the case for the asynchronous design rests on power rather than on any dynamic-range superiority.

Two cautions keep this comparison honest. First, two efficiency figures tell the same story for the synchronous baseline. On the standard Walden FoM with the rate at twice the bandwidth, the reference reaches 3390 fJ/conv.-step against 320 fJ for the proposed converter, around ten times worse. A per-conversion FoM referenced instead to the actual 288 MS/s clock would reward the oversampling rather than penalise the wasted clock energy, which is why the matched-workload power, 886 against 99.5 µW, remains the most direct efficiency figure for the comparison. Second, the 320 fJ/conv.-step of the proposed converter is well above the best dedicated level-crossing ADCs in the literature (single-digit to tens of fJ for the dedicated low-bandwidth designs of Chapter 2). This is expected, since those are high-resolution, kilohertz-bandwidth converters optimised for energy, whereas this work is a medium-resolution, megahertz-range Flash converter benchmarked against a synchronous Flash of the same family. The energy argument of this work rests on the matched-workload power saving against that synchronous baseline and on the drop towards the 94.8 µW idle floor for sparse inputs, not on a record FoM.

A note on the fairness of the power comparison is warranted, since it could be suspected of favouring the asynchronous design by construction. Three observations bound that concern. First, both converters have the same 4-bit resolution and their cores are measured under the same exclusions (Section 4.4.1), so the structural terms are symmetric. Second, the comparison does not depend on choosing a favourable, sparse operating point. The matched-workload point used here is already full activity, the fastest full-scale input the converter tracks, and even there the asynchronous core draws 99.5 µW against 886 µW, a factor of 8.9. As the input becomes sparse the ratio only widens, to 9.3. The advantage therefore holds across the whole activity range.

Because the dynamic component of the asynchronous power is small next to its static floor, this saving comes mainly from the lower absolute consumption of the clockless core, with the activity-dependent scaling adding only a secondary contribution. Third, the one structural asymmetry that remains, the reduced linearity of the complementary follower near the supply rails, runs against the asynchronous design rather than in its favour.

4.7 Chapter Summary

The measurements of this chapter close the loop opened in Chapter 1. Two distinct speed limits bound the converter and should not be conflated. The first is a ramp-rate limit of 14.3 MHz equivalent conversion rate, the fastest trapezoidal ramp whose input-referred hysteresis stays within 1 LSB, bounded by hysteresis rather than by linearity. The second is the full-scale sinusoidal bandwidth of 9 MHz, set by the slew-rate envelope, which the converter digitises with an ENOB of 4.11 bits and no bubble errors. The sinusoidal bandwidth is the lower of the two because a sinusoid reaches a higher peak slope than the linear ramp of the same period. The body bias trimming improves the worst-case DNL at all four process corners and the worst-case INL at three of them, with the SF corner exposing the limit of a corner-indexed, forward-bias-only per-slicer correction. The Monte Carlo analysis confirms that random mismatch, outside the scope of the corner-indexed trimming, remains the dominant linearity hazard. On power, the comparison under a matched workload gives 99.5 µW against 886 µW for the synchronous reference, and the gap widens for sparse inputs, where the asynchronous converter falls to its 94.8 µW idle floor while the clocked reference keeps paying for every cycle. The final chapter weighs these results against the objectives of the dissertation.

Chapter 5

Conclusions

5.1 Summary of the Work Developed

This dissertation addressed the design and validation of an asynchronous Flash ADC intended for low-power, event-driven applications such as Internet of Things sensor nodes and other sources of non-periodic signals. The central motivation for this work stems from the fundamental inefficiency of conventional synchronous ADC architectures when dealing with time-sparse signals. A global clock forces continuous switching activity across the comparator bank and clock distribution network, dissipating dynamic power regardless of whether the input signal is actually carrying new information. Removing that clock and replacing it with a level-crossing, event-driven mechanism was the core hypothesis explored throughout the dissertation.

The work began with a review of the state of the art in data conversion. The classical ADC architectures (Flash, SAR, Pipeline and Sigma-Delta) were examined with particular attention to the trade-offs between speed, resolution and power consumption. This analysis was grounded in performance data from 140 publications drawn from ISSCC and JSSC, from which the Walden figure of merit was extracted and compared across a wide range of sampling frequencies and technology nodes. The review made clear that synchronous Flash converters occupy the high-speed frontier but at the cost of a poor energy figure of merit, and that asynchronous, event-driven operation is one of the most promising paths toward breaking this constraint for sparse-signal applications. A dedicated discussion on the adaptation of standard figures of merit to asynchronous systems was also included, since applying a fixed Nyquist sampling frequency to a level-crossing converter is inherently inappropriate and leads to misleading efficiency estimates.

The design of the asynchronous ADC itself evolved through a sequence of architectures. The first candidate was a current-mirror flash configuration in which a transconductance cell converts the differential input voltage into an error current, and a current-to-code converter maps that current to a thermometer-coded output by exploiting asymmetric PMOS and NMOS mirror ratios at each comparison node. This topology operates in continuous time and requires no external clock, which made it a natural fit for the event-driven approach. The decision element of this stage, the current comparator, was not assumed. Six published current comparator topologies (current-mirror, Traff, Lu Chen, hysteresis, Min and Kim and CC-II) were redrawn at transistor level and simulated in a common testbench, and the current-mirror cell was selected for having the lowest propagation delay and the second-lowest average power of the six. The power and delay figures reported for these comparators therefore come from direct simulation, not from estimates. The second architecture investigated a complementary dual-input topology aimed at extending the input common-mode range toward rail-to-rail operation by combining an NMOS and a PMOS differential pair with cross-coupled output branches. Although conceptually attractive, this implementation could not be made to work within the 0.8 V supply budget of the target 16 nm process. The mirror loads and the cross-coupling stacked additional transistors in every signal path, the accumulated overdrive requirements exceeded the available voltage headroom, and no feasible bias point could be found that kept all devices in saturation simultaneously. A bulk-driven amplifier was then evaluated as a low-voltage route to the same rail-to-rail input, but its positive-feedback transconductance boost proved too sensitive to process corners and was discarded. The final architecture kept the complementary rail-to-rail input but rebuilt it with two five-transistor OTAs, one with an NMOS and one with a PMOS input pair, each loaded by a single active current mirror and biased through the low-voltage tail mirror retained from the second architecture. With far fewer stacked devices, this arrangement fits the available headroom, and the complementary OTA, configured as a unity-gain voltage follower, drives a bank of ratioed current-mode slicers through its low-impedance output. Each slicer is sized with a distinct PMOS-to-NMOS transistor ratio, which sets its switching threshold at a specific voltage level within the supply range. The ensemble of slicers produces an asynchronous thermometer code whose transitions directly encode level-crossing events, with no clock required at any point in the signal path.

Alongside the asynchronous core, a synchronous reference Flash ADC was implemented in the same 16 nm technology and under the same supply conditions to serve as a fair benchmark. This reference uses fifteen StrongARM latch comparators driven by a ring oscillator with a tapered buffer chain, and its power budget explicitly accounts for the clock distribution network, which is precisely the overhead that the asynchronous design aims to eliminate. The comparison between the two converters therefore isolates the effect of removing the global clock from an otherwise structurally similar Flash architecture.

A critical practical challenge of the slicer-based thermometer coder is its sensitivity to process variation. Since the threshold of each slicer is determined by the ratio of PMOS to NMOS drive strength, any departure from the nominal device parameters shifts the trip points in a bit-dependent manner, degrading the differential and integral non-linearity of the converter. To address this, an offline body bias trimming strategy was designed and implemented using a Verilog-A behavioural model. By applying distinct bulk voltages to the NMOS and PMOS transistors of the slicer stage, the threshold voltage of each device type can be shifted through the body effect, so the trip points can be steered back toward their nominal positions after fabrication. Correction vectors were determined for the four standard process corners (SS, FF, FS and SF) and were shown in simulation to improve the worst-case DNL at all four corners and the worst-case INL at three of them, with the largest gains at the symmetric corners, the SF corner proved the hardest to correct with a single global bias pair.

Finally, the complete system was validated through transistor-level analog and mixed-signal co-simulation. The dynamic performance of the asynchronous converter was characterised using a non-uniform to uniform resampling algorithm applied to the raw level-crossing output, enabling standard spectral metrics such as SNDR and ENOB to be extracted and compared against both the synchronous reference and representative state-of-the-art designs. The power measurements confirmed the key theoretical prediction. Dynamic power consumption scales linearly with input signal frequency, and the converter enters a near-zero dynamic consumption state when the input is idle. Measured at a matched conversions-per-second workload, the asynchronous core draws 99.5 µW against the 886 µW of the complete synchronous reference, and the gap widens for sparse inputs as the converter falls towards its idle floor. This power saving, rather than a record Walden figure of merit, is where the energy advantage of the proposed converter lies in the sparse-signal regime targeted by this work.

The contribution of this work lies less in any single block than in their combination, ratioed-slicer quantisation, corner-indexed body bias trimming and clockless event generation, brought together at 0.8 V in a 16 nm process and benchmarked against a synchronous reference of equal resolution and technology.

5.2 Concluded Objectives

The three specific objectives defined at the outset of this dissertation were all successfully accomplished.

The first objective was to implement a valid asynchronous Flash ADC circuit proposal and validate it under typical operating conditions. This was achieved through the final topology, a complementary five-transistor OTA driving the ratioed-slicer bank, which operates in continuous time without any global clock signal. At the typical corner the converter is linear, with the DNL and INL of the slicer bank staying within about ± 0.15 LSB, and the theoretical analysis confirmed that its dynamic power scales with the input activity and approaches zero during idle intervals, validating the intended activity-dependent power profile.

The second objective was to implement a trimming approach for the proposed schematic. This was fulfilled through the body bias trimming strategy of Section 3.7, implemented as a hardware-description-language behavioural model in Verilog-A that applies pre-computed NMOS and PMOS bulk voltages to the slicer bank to restore the intended trip-point positions. The approach was validated across the SS, FF, FS and SF process corners, reducing the worst-case DNL at all corners and the worst-case INL at three of the four, as reported in Table 4.3.

The third objective was to validate the complete ADC system, the schematic together with the trimming, through analog and mixed-signal co-simulation. The transistor-level netlist was simulated in a 16 nm CMOS process under a 0.8 V supply, and both static linearity and dynamic performance metrics were extracted. The FFT-based spectral analysis, performed on the uniformly resampled level-crossing output, confirmed correct operation of the asynchronous core together with the trimming logic. As part of this validation the converter was benchmarked against a synchronous Flash reference implemented under identical technology and supply conditions. It does not set a record figure of merit, which is expected for a medium-resolution, megahertz-range Flash converter, but at a matched conversions-per-second workload it draws about a ninth of the power of the synchronous reference, and the saving grows further when the input is mostly idle, which is the operating regime targeted by this work.

Although process corner variation was successfully addressed by the offline trimming strategy, Monte Carlo statistical variation was not corrected within the scope of this work. The body bias trimming vectors were determined for the four deterministic process corners and are therefore not adapted to the continuous distribution of device mismatch that Monte Carlo analysis exposes. The Monte Carlo results of Chapter 4 quantify the cost, the spread of a single code width reaches several LSB. This limitation is acknowledged as an important gap between the presented prototype and a fully robust production-ready implementation, and it is the primary motivation for the first suggestion for future work described in the following section.

5.3 Future Work

Two directions for future development are identified as the most impactful extensions of the work presented in this dissertation.

5.3.1 Analogue Process Corner Detection Approach

The trimming strategy validated in this work relies on the correct identification of the active process corner, which in the current implementation is supplied externally to the Verilog-A behavioural model and used to select the appropriate pre-computed correction vector. In a physical integrated circuit this external identification is not available, the fabricated die has no direct means of knowing which combination of NMOS and PMOS device parameters it has been assigned by the foundry.

A natural extension would therefore be a dedicated on-chip analogue process detection circuit that senses the effective transistor characteristics of the fabricated device and communicates the inferred process corner directly to the trimming logic. Such a circuit could be implemented as a small reference cell, for example, a pair of matched inverter-based ring oscillators, one biased toward NMOS-dominated operation and the other toward PMOS-dominated operation, whose relative oscillation frequencies or propagation delays encode the current process corner. The ratio between the two delay measurements maps unambiguously onto the NMOS/PMOS balance of the process and distinguishes SS from FF and FS from SF corners. The resulting analogue measurement would be digitised by a simple comparator and used to address a lookup table of pre-computed body bias correction values, replacing the external corner identification currently provided by the Verilog-A model. This approach would make the trimming procedure fully self-contained and autonomous, removing the need for any external process characterisation while preserving the static, low-overhead nature of the offline correction strategy.

5.3.2 Monte Carlo Trimming

A more fundamental limitation of the current trimming approach is that it corrects deterministic inter-corner shifts but leaves intra-die stochastic mismatch unaddressed. In practice, even at the nominal process corner, individual transistors within the slicer bank will exhibit random threshold voltage variations whose statistical distribution is characterised by Pelgrom's law [108]. These variations are not captured by the four-corner trimming table, and their cumulative effect on DNL and INL can be significant in a Flash converter where each comparator threshold must be individually accurate.

Extending the calibration to handle Monte Carlo variation would require moving from a corner-indexed lookup table to a converter-specific foreground calibration routine capable of measuring the actual trip point of each slicer independently and computing an individual correction for each one. One viable approach would be to inject a known ramp stimulus at the converter input during a dedicated startup calibration phase and record the digital code transitions to infer the actual trip point of each slicer. The deviation of each measured trip point from its nominal target could then be mapped to the body bias correction required to bring it back into specification, using the sensitivity relationship established in Section 3.7. Since the NMOS and PMOS bulk voltages of each slicer would need to be individually controlled, this approach requires the addition of a per-cell bias generator, which increases the area overhead relative to a global corner correction. Nevertheless, the achievable linearity improvement and the corresponding reduction in post-fabrication yield loss would likely justify this added complexity in any application where the converter is produced at volume and must meet tight DNL specifications across the full statistical distribution of process variability.

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Appendix A

Verilog-A Body Bias Trimming Model

This appendix lists the Verilog-A behavioural model used to apply the body bias trimming described in Section 3.7. The module exposes, for each of the fifteen slicers, one PMOS bulk output (`pbulk0`–`pbulk14`) and one NMOS bulk output (`nbulk0`–`nbulk14`). Based on the `process` parameter, which selects the active process corner, it drives each bulk terminal to the corresponding correction voltage. Bulks that are not adjusted for the selected corner remain at their default values, `VP_DEFAULT` $= 0.8$ V for the PMOS wells and `VN_DEFAULT` $= 0.0$ V for the NMOS bulks, which corresponds to no body effect and is the condition used at the typical corner.

```
`include "constants.vams"
`include "disciplines.vams"

module adc_bulk_trim_ctrl (
    pbulk0,  pbulk1,  pbulk2,  pbulk3,  pbulk4,
    pbulk5,  pbulk6,  pbulk7,  pbulk8,  pbulk9,
    pbulk10, pbulk11, pbulk12, pbulk13, pbulk14,

    nbulk0,  nbulk1,  nbulk2,  nbulk3,  nbulk4,
    nbulk5,  nbulk6,  nbulk7,  nbulk8,  nbulk9,
    nbulk10, nbulk11, nbulk12, nbulk13, nbulk14
);

    // Output ports
    output pbulk0,  pbulk1,  pbulk2,  pbulk3,  pbulk4;
    output pbulk5,  pbulk6,  pbulk7,  pbulk8,  pbulk9;
    output pbulk10, pbulk11, pbulk12, pbulk13, pbulk14;

    output nbulk0,  nbulk1,  nbulk2,  nbulk3,  nbulk4;
    output nbulk5,  nbulk6,  nbulk7,  nbulk8,  nbulk9;
    output nbulk10, nbulk11, nbulk12, nbulk13, nbulk14;

    electrical pbulk0,  pbulk1,  pbulk2,  pbulk3,  pbulk4;
    electrical pbulk5,  pbulk6,  pbulk7,  pbulk8,  pbulk9;
    electrical pbulk10, pbulk11, pbulk12, pbulk13, pbulk14;
```

```
    electrical nbulk0,  nbulk1,  nbulk2,  nbulk3,  nbulk4;
    electrical nbulk5,  nbulk6,  nbulk7,  nbulk8,  nbulk9;
    electrical nbulk10, nbulk11, nbulk12, nbulk13, nbulk14;

    // Process corner selector: "tt", "ff", "ss", "sf", or "fs"
    parameter string process = "tt";

    // Default PMOS and NMOS bulk voltages
    parameter real VP_DEFAULT = 0.8;
    parameter real VN_DEFAULT = 0.0;

    // Internal voltage variables
    real vp0,  vp1,  vp2,  vp3,  vp4;
    real vp5,  vp6,  vp7,  vp8,  vp9;
    real vp10, vp11, vp12, vp13, vp14;

    real vn0,  vn1,  vn2,  vn3,  vn4;
    real vn5,  vn6,  vn7,  vn8,  vn9;
    real vn10, vn11, vn12, vn13, vn14;

    analog begin

        // Default bulk voltage assignment
        vp0  = VP_DEFAULT; vp1  = VP_DEFAULT; vp2  = VP_DEFAULT;
        vp3  = VP_DEFAULT; vp4  = VP_DEFAULT; vp5  = VP_DEFAULT;
        vp6  = VP_DEFAULT; vp7  = VP_DEFAULT; vp8  = VP_DEFAULT;
        vp9  = VP_DEFAULT; vp10 = VP_DEFAULT; vp11 = VP_DEFAULT;
        vp12 = VP_DEFAULT; vp13 = VP_DEFAULT; vp14 = VP_DEFAULT;

        vn0  = VN_DEFAULT; vn1  = VN_DEFAULT; vn2  = VN_DEFAULT;
        vn3  = VN_DEFAULT; vn4  = VN_DEFAULT; vn5  = VN_DEFAULT;
        vn6  = VN_DEFAULT; vn7  = VN_DEFAULT; vn8  = VN_DEFAULT;
        vn9  = VN_DEFAULT; vn10 = VN_DEFAULT; vn11 = VN_DEFAULT;
        vn12 = VN_DEFAULT; vn13 = VN_DEFAULT; vn14 = VN_DEFAULT;

        // TT corner: no trimming (all bulks at default)
        if (process == "tt") begin
            // PMOS bulks remain at 0.8 V, NMOS bulks remain at 0.0 V
        end

        // FF corner trimming
        else if (process == "ff") begin
            vp8 = 0.7;  vp7 = 0.5;  vp6 = 0.5;
            vn13 = 0.1; vn12 = 0.5; vn11 = 0.5;
            vn10 = 0.2; vn1  = 0.3; vn0  = 0.3;
        end

        // FS corner trimming
        else if (process == "fs") begin
```

```
            vp7 = 0.7;  vp0 = 0.75;
            vn1  = 0.23; vn4  = 0.1; vn14 = 0.1; vn12 = 0.1;
        end

        // SF corner trimming
        else if (process == "sf") begin
            vp2 = 0.7;  vp1 = 0.75;
            vn3  = 0.1;  vn4  = 0.2; vn5  = 0.2; vn11 = 0.1;
            vn12 = 0.15; vn13 = 0.1; vn14 = 0.3;
        end

        // SS corner trimming
        else if (process == "ss") begin
            vp10 = 0.72; vp9 = 0.75; vp0 = 0.6;
            vn2 = 0.2; vn3 = 0.15; vn4 = 0.2; vn5 = 0.1;
        end

        // Drive PMOS bulk voltage outputs (ideal voltage sources)
        V(pbulk0)  <+ vp0;  V(pbulk1)  <+ vp1;  V(pbulk2)  <+ vp2;
        V(pbulk3)  <+ vp3;  V(pbulk4)  <+ vp4;  V(pbulk5)  <+ vp5;
        V(pbulk6)  <+ vp6;  V(pbulk7)  <+ vp7;  V(pbulk8)  <+ vp8;
        V(pbulk9)  <+ vp9;  V(pbulk10) <+ vp10; V(pbulk11) <+ vp11;
        V(pbulk12) <+ vp12; V(pbulk13) <+ vp13; V(pbulk14) <+ vp14;

        // Drive NMOS bulk voltage outputs (ideal voltage sources)
        V(nbulk0)  <+ vn0;  V(nbulk1)  <+ vn1;  V(nbulk2)  <+ vn2;
        V(nbulk3)  <+ vn3;  V(nbulk4)  <+ vn4;  V(nbulk5)  <+ vn5;
        V(nbulk6)  <+ vn6;  V(nbulk7)  <+ vn7;  V(nbulk8)  <+ vn8;
        V(nbulk9)  <+ vn9;  V(nbulk10) <+ vn10; V(nbulk11) <+ vn11;
        V(nbulk12) <+ vn12; V(nbulk13) <+ vn13; V(nbulk14) <+ vn14;

    end

endmodule
```

Listing A.1: Verilog-A model of the per-slicer body bias trimming controller.